BTC Vol Desk PDF Handoff Packet - README

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/pdf-packet/README.md`

BTC Vol Desk PDF Handoff Packet

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Investor / Diligence PDFs

These are the cleaner documents to show an investor for diligence context. They remain internal-diligence-only unless counsel approves external distribution.

- `01-investor-and-diligence-pdfs/00-packet-index.pdf` — Packet Index
- `01-investor-and-diligence-pdfs/01-one-page-tear-sheet.pdf` — One Page Tear Sheet
- `01-investor-and-diligence-pdfs/02-investor-memo.pdf` — Investor Memo
- `01-investor-and-diligence-pdfs/03-static-site-export.pdf` — Static Site Text Export
- `01-investor-and-diligence-pdfs/04-latest-monitor-report.pdf` — Latest Monitor Report
- `01-investor-and-diligence-pdfs/05-backtest-research-evidence.pdf` — Backtest Research Evidence
- `01-investor-and-diligence-pdfs/06-legal-wrapper-draft.pdf` — Legal Wrapper Draft

Internal Team PDFs

These are for internal execution, legal, source acquisition, RFQ workflow, and implementation planning.

- `02-internal-team-pdfs/00-master-plan-v2.pdf` — Master Plan v2 Evidence-First Hardening
- `02-internal-team-pdfs/01-licensed-source-acquisition-package.pdf` — Licensed Source Acquisition Package
- `02-internal-team-pdfs/02-source-intake-validation.pdf` — Source Intake Skeleton Validation
- `02-internal-team-pdfs/03-source-blocker-audit.pdf` — Source Blocker Audit
- `02-internal-team-pdfs/04-rfq-verification-plan.pdf` — RFQ Verification Plan
- `02-internal-team-pdfs/05-rfq-verification-board.pdf` — RFQ Verification Board
- `02-internal-team-pdfs/06-launch-model-decision-matrix.pdf` — Launch Model Decision Matrix
- `02-internal-team-pdfs/07-investor-deep-dive.pdf` — Investor Deep Dive
- `02-internal-team-pdfs/08-data-evidence-appendix.pdf` — Data Evidence Appendix
- `02-internal-team-pdfs/09-client-map-commercial-wedge.pdf` — Client Map / Commercial Wedge
- `02-internal-team-pdfs/10-treasury-case-study.pdf` — Treasury Case Study
- `02-internal-team-pdfs/11-miner-production-hedge-case-study.pdf` — Miner Production Hedge Case Study

- `02-internal-team-pdfs/12-board-policy-template.pdf` — BTC Treasury Board Policy Template
- `02-internal-team-pdfs/13-miner-hedge-calculator-spec.pdf` — Miner Hedge Calculator Spec
- `02-internal-team-pdfs/14-data-pipeline-hardening-spec.pdf` — Data Pipeline Hardening Spec
- `02-internal-team-pdfs/15-legal-perimeter-memo.pdf` — Legal Perimeter Memo
- `02-internal-team-pdfs/16-investor-deck-outline.pdf` — Investor Deck Outline
- `02-internal-team-pdfs/17-30-day-execution-sprint.pdf` — 30-Day Execution Sprint

Non-negotiable gates

- Screen-only, not executable.
- Not a quote, not a fund offering, not client-facing advice.
- External use remains blocked until counsel-approved wrapper.
- Quote-verified status requires real two-counterparty evidence and attestations.
- Licensed historical source coverage remains incomplete; fixtures do not count as covered.

Packet Index

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/packet_index.md`

BTC Vol Desk Investor Packet

Run ID: `btcvol-20260518-163034`

Status: SCREEN-ONLY · NOT EXECUTABLE

Publishability: `internal-diligence-only`

Packet SHA-256: see `packet_manifest.json`

> Not a client portal. Not an execution venue. Not a fund offering. This packet is for internal/investor diligence review only.

Contents

- `btc-vol-desk-investor-memo.md` — investor_memo — `09463acc6b6bc4f8a538a4632d5799780a79013e3933391fd559a19d5248a8f0`
- `one-page-tear-sheet.md` — investor_tearsheet — `084d7cb6f1d759c1c29679958b5759f22dc288961e529bfd5580a06f0abc6b8a`
- `site/index.html` — site_index — `3a9fa72fde60c7103d5bd1c99bb037de4613d22a4056108cce29b264c86193a7`
- `site/site-data.json` — site_data — `0a3ce0e7baf5f604f55f3fb984205e392371bead401407c43664752729c454f2`
- `evidence/latest-report.md` — latest_report — `7cd4fea856c5c859c730d951252ad378085144ebdfd6912a58f8193f5fa2d645`
- `evidence/latest-evidence-bundle.zip` — latest_evidence_bundle — `6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b`
- `evidence/evidence_manifest.json` — evidence_manifest — `d72d97a8d980159a7aa796d362f8a09325152cd9b606721693c9d4650eabdc6d`
- `evidence/candidate_triage.jsonl` — candidate_triage — `9d63bd946b784f0054b574fd83a78c775fa86e2a9a746068720be8916500ac46`
- `evidence/quote_evidence.jsonl` — quote_evidence — `e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
- `evidence/backtest-report.md` — backtest_report — `ac18310dddc991c9fd22bfbce98bfbcc10279ac8eed99613e27f36cea6cbdb48`
- `evidence/legal-wrapper-package-v1.json` — legal_wrapper_json — `6f58bce9ad0542df36048fd3758c80715038376422add873a9e5745ed8ff2e50`
- `evidence/legal-wrapper-package-v1.md` — legal_wrapper_md — `ba150abff9ba812ccf52ea1f74a8ce0695cabb041d72465675beac7bbfb70c89`

Required Verification

1. Review `packet_manifest.json`.
2. Confirm artifact SHA-256 values match local files.

3. Confirm the latest evidence bundle remains independently verifiable.
4. Treat all economics as SCREEN-ONLY · NOT EXECUTABLE until quote/execution evidence and legal wrapper are approved.

One Page Tear Sheet

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/one-page-tear-sheet.md`

BTC Treasury & Miner Hedging Desk — One-Page Tear Sheet

Positioning: BTC Treasury & Miner Hedging Desk powered by a purpose-built cross-venue volatility evidence engine

Evidence status: SCREEN-ONLY · NOT EXECUTABLE

Control: Not a client portal. Not an execution venue. Not a fund offering.

Legal gate: Counsel-approved wrapper required before any external client, fund, RFQ, or execution workflow.

> Internal evidence prototype. Public screen/model data only. No RFQ sent. No executable quote.

> This is an internal/investor diligence summary. Public screen marks, model-estimated IVs, and manual indicative quote records are evidence inputs only — not executable economics.

Evidence Snapshot

- Run ID: `btcvol-20260518-163034`
- As-of CST: `2026-05-18 16:30:34 CDT`
- BTC Reference: `$77,457.06`
- Configured-source quality: `GREEN`
- Current screen-source availability: `Current screen/vendor captures available: Deribit + IBIT + CME Databento`
- Overall evidence readiness: `YELLOW until real quote evidence exists`
- Freshness: `GREEN`
- Screen-only dislocations: `6`
- Quote-review flags: `3`
- Evidence bundle SHA-256: `6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b`

Why Now

- Corporate BTC treasuries and miners need risk-management infrastructure, not retail crypto trading screens.
- ETF options, Deribit, CME, and OTC liquidity fragment BTC volatility into inconsistent surfaces.
- A desk/platform can normalize exposures, preserve evidence, and separate screen-only signals from quote-verified diligence.

Initial ICP

- BTC treasury holders needing board-safe covered-call, collar, or put-spread collar programs.
- Miners needing runway protection without overhedging production or creating collateral stress.

- Strategic investors/OTC partners who can provide broker, legal, or counterparty workflow support.

Current Evidence

- Top screen-only candidate: IBIT 0D ATM vs Deribit 1D ATM (-16.49 vol pts, SCREEN-ONLY · NOT EXECUTABLE)
- Quote evidence ledger: 0 demo/manual indicative quote records, 0 real quote-verified records, 0 quote-verified candidates, 0 trade-verified candidates.
- Real counterparty quote evidence captured: `0` records / `0` candidates.
- Publishability: `not-investor-publishable` until legal wrapper, counterparty permissions, and external records support wider use.

Evidence Operations

1. Monitor public Deribit and IBIT option screens.
2. Normalize ETF exposure into BTC-equivalent terms.
3. Triage dislocations as `SCREEN-ONLY · NOT EXECUTABLE`.
4. Capture manual indicative quote evidence in JSONL when available.
5. Require two real counterparty indicative quote records before internal quote-verified status.
6. Require external execution record before post-trade verification.

Diligence Ask

Use this packet to evaluate seed/build capital, strategic OTC/broker introductions, legal structuring support, or a pilot treasury/miner mandate. The fund/risk sleeve should remain a later-stage extension after evidence history, quote workflow, and the sponsor’s counsel-approved legal wrapper are stronger.

Hard Boundaries

- Not a client portal. Not an execution venue. Not a fund offering.
- No RFQ is sent by this packet.
- No executable quote, counterparty commitment, advisory service, or securities offering is implied.
- Counsel-approved wrapper required before any external client, fund, RFQ, or execution workflow.

Investor Memo

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/btc-vol-desk-investor-memo.md`

BTC Treasury & Miner Hedging Desk Investor Memo

Status: SCREEN-ONLY · NOT EXECUTABLE

Positioning: BTC Treasury & Miner Hedging Desk powered by a purpose-built cross-venue volatility evidence engine

> Internal evidence prototype. Public screen/model data only. No RFQ sent. No executable quote.

> Not a client portal. Not an execution venue. Not a fund offering. This memo summarizes an internal proof-of-concept and evidence workflow.

Thesis

BTC volatility markets are fragmented across ETF-listed options, offshore crypto options venues, CME-linked institutional channels, and OTC counterparties. Corporate BTC treasuries and miners need hedge design, evidence integrity, and quote-verification discipline more than another generic crypto trading screen.

The opportunity is a desk/platform: use a purpose-built cross-venue volatility evidence engine to identify screen-only dislocations, convert them into treasury/miner hedge structures, and promote only independently documented opportunities through a manual quote-verification workflow.

Evidence Snapshot

- Run ID: `btcvol-20260518-163034`
- As-of CST: `2026-05-18 16:30:34 CDT`
- BTC reference: `$77,457.06`
- IBIT BTC/share: `0.0005679118586151414`
- Configured-source quality: `GREEN` / `80`
- Current screen-source availability: `Current screen/vendor captures available: Deribit + IBIT + CME Databento`
- Overall evidence readiness: `YELLOW until real quote evidence exists`
- Freshness: `GREEN`
- Screen-only dislocations: `6`
- Quote-review flags: `3`
- Bundle SHA-256: `6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b`

All displayed economics are public screen marks or model-estimated values. They are evidence inputs, not executable quotes.

Screen-Only Candidate Examples

- #1 IBIT 0D ATM vs Deribit 1D ATM — -16.49 vol pts; priority `high`; SCREEN-ONLY · NOT EXECUTABLE.
- #2 IBIT 2D ATM vs Deribit 2D ATM — +8.39 vol pts; priority `high`; SCREEN-ONLY · NOT EXECUTABLE.
- #3 IBIT 4D ATM vs Deribit 4D ATM — +5.46 vol pts; priority `high`; SCREEN-ONLY · NOT EXECUTABLE.
- #4 IBIT 8D ATM vs Deribit 11D ATM — -1.51 vol pts; priority `low`; SCREEN-ONLY · NOT EXECUTABLE.
- #5 IBIT 11D ATM vs Deribit 11D ATM — +1.27 vol pts; priority `low`; SCREEN-ONLY · NOT EXECUTABLE.

Treasury Case Study

- Structure: corporate BTC treasury hedge policy sleeve.
- Hedged BTC: 350 BTC hedged.
- Caveat: hypothetical scenario only; no suitability, premium, margin, tax, accounting, liquidity, or counterparty commitment is implied.
- Illustrative model output — protected value: $20,332,477.31 illustrative floor-protected sleeve.
- Illustrative model output — floor / cap: $58,092.79 / $96,821.32.
- Control: Premium and executable levels require two-counterparty quote verification.

Miner Case Study

- Structure: runway-protection hedge for conservative monthly production.
- Hedged monthly production: 60 BTC/month hedged.
- Caveat: hypothetical scenario only; no hedge recommendation, suitability review, margin model, tax/accounting treatment, or executable quote is implied.
- Illustrative model output — monthly floor revenue: $3,253,196.37.
- Pre-hedge cash runway: 3.75 months.
- Control: Indicative economics require quote verification before investor/client use.

Quote Verification Workflow

Manual demo workflow only; no RFQ is sent and no executable quote is implied.

Current stage counts:

- Screen-only: 5
- Reviewed: 0
- Internal RFQ draft: 0
- Indicative quote 1: 0
- Indicative quote 2: 0
- Quote verified: 0
- Post-trade record verified: 0

Workflow gates:

1. Screen-only dislocation detected by the monitor.
2. Candidate triaged for materiality and structure fit.
3. RFQ package generated for review, not auto-sent.
4. After approval and manual outreach, counterparties may provide indicative quotes.
5. Evidence ledger may mark candidate quote-verified after required indicative quote records are captured.
6. Post-trade verification only occurs after an actual external execution record exists; this memo does not imply execution.

Business Model Sequence

1. Research/evidence engine
2. Treasury and miner hedge structuring
3. Partner-led RFQ and execution support
4. Principal risk sleeve only after chosen legal wrapper

The recommended route remains desk/platform-first. The fund sleeve is an expansion path after evidence, legal wrapper, counterparty access, and investor demand are established.

Investor Ask

- Seed build capital for data licensing, counterparty connectivity, compliance review, and proof-of-market pilots.
- Introductions to BTC treasury decision-makers, miners, OTC desks, derivatives counsel, and execution partners.
- Approval to harden the evidence engine into a controlled diligence workflow without launching a client portal or executable RFQ system prematurely.

Evidence Room

- Report: `artifacts/institutional/data/reports/btc-vol-desk-monitor-2026-05-18-163034.md`
- Evidence bundle: `artifacts/institutional/data/normalized/btcvol-20260518-163034/btcvol-20260518-163034-evidence-bundle.zip`
- Evidence manifest: `artifacts/institutional/data/normalized/btcvol-20260518-163034/evidence_manifest.json`
- Candidate ledger: `artifacts/institutional/data/normalized/btcvol-20260518-163034/candidate_triage.jsonl`
- Bundle SHA-256: `6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b`

Limitations / Gating Items

- SCREEN-ONLY · NOT EXECUTABLE: all current market economics are public screen marks, model-estimated IVs, or internal review outputs.
- CME remains unavailable until licensed/vendor/broker feed is configured.
- Quote-verified economics require two independently captured indicative quote records.
- Trade/post-trade verification requires an actual external execution record; this POC has no trading capability.
- Screen-only dislocation differences are gross IV observations only; they exclude bid/ask width, slippage, financing, borrow, margin, collateral, tax/accounting, settlement, exercise style, venue hours, operational constraints, and counterparty risk.
- Tenor matching between IBIT ETF options and Deribit BTC options is approximate and not an arbitrage claim or trading recommendation.

• Counsel-approved wrapper required before any external client, fund, RFQ, or execution workflow.

Static Site Text Export

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/site/index.html`

BTC Vol Desk Evidence Legal Case Studies Quote Verification Backtest Sources Evidence Room SCREEN-ONLY · NOT EXECUTABLE

BTC Vol Desk

Evidence Legal Case Studies Quote Verification Backtest Sources Evidence Room

SCREEN-ONLY · NOT EXECUTABLE

Desk / Platform First · Fund sleeve later BTC Treasury & Miner Hedging Desk powered by a purpose-built cross-venue volatility evidence engine An institutional proof-of-concept for converting fragmented BTC volatility markets into treasury and miner hedging intelligence, then into quote-verified opportunities through a controlled RFQ evidence process. Internal evidence prototype. Public screen/model data only. No RFQ sent. No executable quote. Counsel-approved wrapper required before any external client, fund, RFQ, or execution workflow.

Desk / Platform First · Fund sleeve later

BTC Treasury & Miner Hedging Desk powered by a purpose-built cross-venue volatility evidence engine

An institutional proof-of-concept for converting fragmented BTC volatility markets into treasury and miner hedging intelligence, then into quote-verified opportunities through a controlled RFQ evidence process.

Internal evidence prototype. Public screen/model data only. No RFQ sent. No executable quote. Counsel-approved wrapper required before any external client, fund, RFQ, or execution workflow.

Latest evidence run btcvol-20260518-163034 2026-05-18 16:30:34 CDT

Latest evidence run

btcvol-20260518-163034

2026-05-18 16:30:34 CDT

BTC reference $77,457.06 Dislocations 6 Configured-source quality GREEN Freshness GREEN

BTC reference $77,457.06

BTC reference

$77,457.06

Dislocations 6

Dislocations

6

Configured-source quality GREEN

Configured-source quality

GREEN

Freshness GREEN

Freshness

GREEN

Static screen evidence only; not live market data, not an executable quote, and not client-facing advice.

Internal packet integrity

Verifier PASS

Hashes, controls, credential scan, and CTA scan pass.

External use

BLOCKED

Legal, quote, and licensed-source gates are not cleared.

Current source availability

Current screen/vendor captures available: Deribit + IBIT + CME Databento

Current captures support internal review only.

Licensed readiness

0/6 groups ready

Required before external use.

Problem

BTC volatility is fragmented across ETF options, Deribit, CME, and OTC desks. Treasuries and miners need hedging programs, not retail crypto trading screens.

Solution

A cross-venue evidence engine normalizes exposures, labels source confidence, and identifies review candidates before any quote or execution claim is made.

Business

Lead with hedge structuring and evidence workflow. Add RFQ partnerships and a fund/risk sleeve only after the chosen wrapper supports it.

Latest Static Evidence Snapshot

IBIT BTC/share 0.0005679118586151414 Current screen-source availability Current screen/vendor captures available: Deribit + IBIT + CME Databento Bundle hash 6504204d3adf…dd7724210b

IBIT BTC/share 0.0005679118586151414

IBIT BTC/share

0.0005679118586151414

Current screen-source availability Current screen/vendor captures available: Deribit + IBIT + CME Databento

Current screen-source availability

Current screen/vendor captures available: Deribit + IBIT + CME Databento

Bundle hash 6504204d3adf…dd7724210b

Bundle hash

6504204d3adf…dd7724210b

Static as of run timestamp; not real-time market data. All displayed economics are public screen marks or model-estimated values. They are evidence inputs, not executable quotes. Overall evidence readiness: YELLOW until real quote evidence exists.

#1 IBIT 0D ATM vs Deribit 1D ATM IBIT cheap vs Deribit -16.49 vol pts SCREEN-ONLY · NOT EXECUTABLE draft two-counterparty indicative RFQ review (internal only) #2 IBIT 2D ATM vs Deribit 2D ATM IBIT rich vs Deribit +8.39 vol pts SCREEN-ONLY · NOT EXECUTABLE draft two-counterparty indicative RFQ review (internal only) #3 IBIT 4D ATM vs Deribit 4D ATM IBIT rich vs Deribit +5.46 vol pts SCREEN-ONLY · NOT EXECUTABLE draft two-counterparty indicative RFQ review (internal only) #4 IBIT 8D ATM vs Deribit 11D ATM IBIT cheap vs Deribit -1.51 vol pts SCREEN-ONLY · NOT EXECUTABLE watch only

#1

IBIT 0D ATM vs Deribit 1D ATM

IBIT cheap vs Deribit

-16.49 vol pts

SCREEN-ONLY · NOT EXECUTABLE

draft two-counterparty indicative RFQ review (internal only)

#2

IBIT 2D ATM vs Deribit 2D ATM

IBIT rich vs Deribit

+8.39 vol pts

SCREEN-ONLY · NOT EXECUTABLE

draft two-counterparty indicative RFQ review (internal only)

#3

IBIT 4D ATM vs Deribit 4D ATM

IBIT rich vs Deribit

+5.46 vol pts

SCREEN-ONLY · NOT EXECUTABLE

draft two-counterparty indicative RFQ review (internal only)

#4

IBIT 8D ATM vs Deribit 11D ATM

IBIT cheap vs Deribit

-1.51 vol pts

SCREEN-ONLY · NOT EXECUTABLE

watch only

Institutional Readiness Gate Readiness gates are diligence controls, not approval to trade, advise, market, or raise capital. NOT READY FOR INVESTMENT/CLIENT USE

Institutional Readiness Gate Readiness gates are diligence controls, not approval to trade, advise, market, or raise capital.

Institutional Readiness Gate

Readiness gates are diligence controls, not approval to trade, advise, market, or raise capital.

NOT READY FOR INVESTMENT/CLIENT USE

Gate status 0/4 readiness gates passed Evidence status SCREEN-ONLY · NOT EXECUTABLE External use BLOCKED

Gate status 0/4 readiness gates passed

Gate status

0/4 readiness gates passed

Evidence status SCREEN-ONLY · NOT EXECUTABLE

Evidence status

SCREEN-ONLY · NOT EXECUTABLE

External use BLOCKED

External use

BLOCKED

BLOCKED Licensed historical source coverage not ready Required: All required source groups covered by non-fixture, replay-ready sources Blocker: licensed historical source coverage incomplete BLOCKED Backtest sample gate not ready Required: At least one scenario passes minimum snapshot and synthetic-trade thresholds Blocker: backtest sample gate insufficient BLOCKED Two-counterparty quote evidence not ready Required: At least one candidate has quote-verified evidence from two distinct counterparties Blocker: two-counterparty quote evidence missing BLOCKED Counsel-approved legal wrapper not ready Required: Approved legal/business wrapper before external investor/client use Blocker: counsel-approved wrapper missing

BLOCKED

Licensed historical source coverage

not ready

Required: All required source groups covered by non-fixture, replay-ready sources

Blocker: licensed historical source coverage incomplete

BLOCKED

Backtest sample gate

not ready

Required: At least one scenario passes minimum snapshot and synthetic-trade thresholds

Blocker: backtest sample gate insufficient

BLOCKED

Two-counterparty quote evidence

not ready

Required: At least one candidate has quote-verified evidence from two distinct counterparties

Blocker: two-counterparty quote evidence missing

BLOCKED

Counsel-approved legal wrapper

not ready

Required: Approved legal/business wrapper before external investor/client use

Blocker: counsel-approved wrapper missing

- Blocker licensed historical source coverage incomplete

Blocker

licensed historical source coverage incomplete

- Blocker backtest sample gate insufficient

Blocker

backtest sample gate insufficient

- Blocker two-counterparty quote evidence missing

Blocker

two-counterparty quote evidence missing

- Blocker counsel-approved wrapper missing

Blocker

counsel-approved wrapper missing

Readiness Remediation Plan

- Next action Replace IBIT/Deribit fixtures and missing groups with licensed/replay-ready source manifests.

Next action

Replace IBIT/Deribit fixtures and missing groups with licensed/replay-ready source manifests.

- Next action Accumulate enough point-in-time replay observations for sample-gate pass thresholds.

Next action

Accumulate enough point-in-time replay observations for sample-gate pass thresholds.

- Next action Capture two distinct external indicative quote records for at least one candidate.

Next action

Capture two distinct external indicative quote records for at least one candidate.

- Next action Obtain counsel-approved business/legal wrapper before external investor/client use.

Next action

Obtain counsel-approved business/legal wrapper before external investor/client use.

Legal / Business Wrapper Legal/business wrapper draft only. This artifact is not legal advice and does not authorize external client, investor, RFQ, execution, advisory, or fund activity. DRAFT — NOT APPROVED FOR EXTERNAL USE

Legal / Business Wrapper Legal/business wrapper draft only. This artifact is not legal advice and does not authorize external client, investor, RFQ, execution, advisory, or fund activity.

Legal / Business Wrapper

Legal/business wrapper draft only. This artifact is not legal advice and does not authorize external client, investor, RFQ, execution, advisory, or fund activity.

DRAFT — NOT APPROVED FOR EXTERNAL USE

Counsel approval FALSE Wrapper status draft-blocked External use BLOCKED

Counsel approval FALSE

Counsel approval

FALSE

Wrapper status draft-blocked

Wrapper status

draft-blocked

External use BLOCKED

External use

BLOCKED

Analytics / research internal-only Allowed now: True · External requires: counsel-approved disclaimer and distribution controls Advisory / structuring blocked-pre-wrapper Allowed now: False · External requires: RIA/CTA/CPO/BD perimeter analysis and written engagement terms Execution / RFQ support blocked-pre-wrapper Allowed now: False · External requires: broker/FCM/OTC partner model, no agency ambiguity, counterparty onboarding controls Principal risk-taking blocked-pre-wrapper Allowed now: False · External requires: entity/risk capital approval, mandate limits, conflicts policy, trade surveillance Structured products / fund sleeve blocked-pre-wrapper Allowed now: False · External requires: securities/private fund counsel, offering docs, investor eligibility, marketing review

Analytics / research

internal-only

Allowed now: True · External requires: counsel-approved disclaimer and distribution controls

Advisory / structuring

blocked-pre-wrapper

Allowed now: False · External requires: RIA/CTA/CPO/BD perimeter analysis and written engagement terms

Execution / RFQ support

blocked-pre-wrapper

Allowed now: False · External requires: broker/FCM/OTC partner model, no agency ambiguity, counterparty onboarding controls

Principal risk-taking

blocked-pre-wrapper

Allowed now: False · External requires: entity/risk capital approval, mandate limits, conflicts policy, trade surveillance

Structured products / fund sleeve

blocked-pre-wrapper

Allowed now: False · External requires: securities/private fund counsel, offering docs, investor eligibility, marketing review

- Legal blocker counsel approval missing

Legal blocker

counsel approval missing

- Legal blocker external advisory/client/RFQ/fund use prohibited until wrapper approved

Legal blocker

external advisory/client/RFQ/fund use prohibited until wrapper approved

- Legal blocker quote evidence remains internal diligence only

Legal blocker

quote evidence remains internal diligence only

- Hard gate No external investor/client use until counsel-approved wrapper exists

Hard gate

No external investor/client use until counsel-approved wrapper exists

- Hard gate No RFQ submission by automation

Hard gate

No RFQ submission by automation

- Hard gate No executable quote labels without two distinct external indicative quote records

Hard gate

No executable quote labels without two distinct external indicative quote records

- Hard gate No trade-verified label without execution record and post-trade evidence

Hard gate

No trade-verified label without execution record and post-trade evidence

- Hard gate No offering/fund/product language without securities/private-fund counsel approval

Hard gate

No offering/fund/product language without securities/private-fund counsel approval

Current Source Availability

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Current captured sources can support internal diligence, but they do not clear licensed historical readiness, quote verification, legal approval, or executable-economics gates.

Current availability 6/6 current source groups available Internal diligence READY Investment/client use BLOCKED

Current availability 6/6 current source groups available

Current availability

6/6 current source groups available

Internal diligence READY

Internal diligence

READY

Investment/client use BLOCKED

Investment/client use

BLOCKED

IBIT options · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: Nasdaq public option-chain JSON · License: public_screen_source · Rows: 6 SHA-256 aa95e3aba2f5ab93a4564bd33293d31d1dca2f6361383349ab2df6bce0dc4e03 Status note Available for internal diligence only; does not clear licensed historical readiness. Deribit options · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: Deribit public API · License: public_api_screen_data · Rows: 11 SHA-256 2d9e10f7fa683946da696f09ad6bdd10e358727042dea357ad3cb20ee25284dd Status note Available for internal diligence only; does not clear licensed historical readiness. CME Bitcoin options · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: Databento CME · License: licensed_vendor_api_databento · Rows: 1708 SHA-256 654789e32f58dee8bab1428e0ad41780776e7b1750374f76ff54bde2af69d702 Status note Available for internal diligence only; does not clear licensed historical readiness. BTC reference · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: Monitor run manifest · License: run_derived_reference · Rows: 1 SHA-256 missing Status note Available for internal diligence only; does not clear licensed historical readiness. IBIT holdings · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: iShares holdings capture/cache · License: public_fund_holdings_or_valid_cache · Rows: 1570 SHA-256 1e1291ea3cfe0ad1b0aae1ff609059a5fe2eeb19c585c7b33ab77f00d560272b Status note Available for internal diligence only;

does not clear licensed historical readiness. Rates/fee curves · available SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE Provider: FRED rates CSV · License: public_reference_rates · Rows: 16794 SHA-256 f863aa00472ca8378e06a3f8dc080606e2240c529ad2c968ba3b648e90b410b2 Status note Available for internal diligence only; does not clear licensed historical readiness.

IBIT options · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: Nasdaq public option-chain JSON · License: public_screen_source · Rows: 6

- SHA-256 aa95e3aba2f5ab93a4564bd33293d31d1dca2f6361383349ab2df6bce0dc4e03

SHA-256

aa95e3aba2f5ab93a4564bd33293d31d1dca2f6361383349ab2df6bce0dc4e03

- Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

Deribit options · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: Deribit public API · License: public_api_screen_data · Rows: 11

- SHA-256 2d9e10f7fa683946da696f09ad6bdd10e358727042dea357ad3cb20ee25284dd

SHA-256

2d9e10f7fa683946da696f09ad6bdd10e358727042dea357ad3cb20ee25284dd

- Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

CME Bitcoin options · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: Databento CME · License: licensed_vendor_api_databento · Rows: 1708

- SHA-256 654789e32f58dee8bab1428e0ad41780776e7b1750374f76ff54bde2af69d702

SHA-256

654789e32f58dee8bab1428e0ad41780776e7b1750374f76ff54bde2af69d702

- Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

BTC reference · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: Monitor run manifest · License: run_derived_reference · Rows: 1

- SHA-256 missing

SHA-256

missing

• Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

IBIT holdings · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: iShares holdings capture/cache · License: public_fund_holdings_or_valid_cache · Rows: 1570

• SHA-256 1e1291ea3cfe0ad1b0aae1ff609059a5fe2eeb19c585c7b33ab77f00d560272b

SHA-256

1e1291ea3cfe0ad1b0aae1ff609059a5fe2eeb19c585c7b33ab77f00d560272b

• Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

Rates/fee curves · available

SCREEN-ONLY · CURRENT SOURCE AVAILABILITY · NOT EXECUTABLE

Provider: FRED rates CSV · License: public_reference_rates · Rows: 16794

• SHA-256 f863aa00472ca8378e06a3f8dc080606e2240c529ad2c968ba3b648e90b410b2

SHA-256

f863aa00472ca8378e06a3f8dc080606e2240c529ad2c968ba3b648e90b410b2

• Status note Available for internal diligence only; does not clear licensed historical readiness.

Status note

Available for internal diligence only; does not clear licensed historical readiness.

Licensed Source Intake Contract Defines the licensed/replay-ready source package required before historical coverage can pass readiness gates. It does not create live data access or executable economics. SCREEN-ONLY · INTAKE CONTRACT · NOT EXECUTABLE

Licensed Source Intake Contract Defines the licensed/replay-ready source package required before historical coverage can pass readiness gates. It does not create live data access or executable economics.

Licensed Source Intake Contract

Defines the licensed/replay-ready source package required before historical coverage can pass readiness gates. It does not create live data access or executable economics.

SCREEN-ONLY · INTAKE CONTRACT · NOT EXECUTABLE

Source Intake Validation

Validation status NOT READY Coverage 0/6 source groups structurally valid Evidence status SCREEN-ONLY · SOURCE INTAKE VALIDATION · NOT EXECUTABLE

Validation status NOT READY

Validation status

NOT READY

Coverage 0/6 source groups structurally valid

Coverage

0/6 source groups structurally valid

Evidence status SCREEN-ONLY · SOURCE INTAKE VALIDATION · NOT EXECUTABLE

Evidence status

SCREEN-ONLY · SOURCE INTAKE VALIDATION · NOT EXECUTABLE

- Intake blocker IBIT options history: fixture/manual_fixture sources cannot satisfy readiness

Intake blocker

IBIT options history: fixture/manual_fixture sources cannot satisfy readiness

- Intake blocker Deribit options history: fixture/manual_fixture sources cannot satisfy readiness

Intake blocker

Deribit options history: fixture/manual_fixture sources cannot satisfy readiness

- Intake blocker CME Bitcoin options history: source group missing

Intake blocker

CME Bitcoin options history: source group missing

- Intake blocker BTC reference history: source group missing

Intake blocker

BTC reference history: source group missing

- Intake blocker IBIT holdings history: source group missing

Intake blocker

IBIT holdings history: source group missing

- Intake blocker Rates and fee curves: source group missing

Intake blocker

Rates and fee curves: source group missing

IBIT options history IBIT options history accepted formats: csv, jsonl, parquet required fields: available_ts, expiration, strike, option_type, bid, ask, volume, open_interest, source_ref provider examples: OPRA/vendor export, broker historical option chain Deribit options history Deribit options history accepted formats: csv, jsonl, parquet required fields: available_ts, instrument_name, underlying_price, bid_iv, ask_iv, mark_iv, open_interest, source_ref provider examples: Deribit historical export, licensed vendor replay CME Bitcoin options history CME Bitcoin options history accepted formats: csv, jsonl, parquet required fields: available_ts, symbol, expiration, strike, option_type, bid, ask, settlement, source_ref provider examples: Databento licensed history, CME/broker historical export BTC reference history BTC reference history accepted formats: csv, jsonl, parquet required fields: available_ts, btc_usd, venue_or_index, source_ref provider examples: CF Benchmarks, vendor BTC index, broker reference IBIT holdings history IBIT holdings history accepted formats: csv, jsonl, parquet required fields: available_ts, btc_per_share, shares_outstanding, fund_assets, source_ref provider examples: iShares daily holdings archive, fund administrator export Rates and fee curves Rates and fee curves accepted formats: csv, jsonl, parquet required fields: available_ts, tenor, rate, borrow_or_fee, source_ref provider examples: SOFR/rate vendor, broker borrow/fee export

IBIT options history

IBIT options history

accepted formats: csv, jsonl, parquet

required fields: available_ts, expiration, strike, option_type, bid, ask, volume, open_interest, source_ref

provider examples: OPRA/vendor export, broker historical option chain

Deribit options history

Deribit options history

accepted formats: csv, jsonl, parquet

required fields: available_ts, instrument_name, underlying_price, bid_iv, ask_iv, mark_iv, open_interest, source_ref

provider examples: Deribit historical export, licensed vendor replay

CME Bitcoin options history

CME Bitcoin options history

accepted formats: csv, jsonl, parquet

required fields: available_ts, symbol, expiration, strike, option_type, bid, ask, settlement, source_ref

provider examples: Databento licensed history, CME/broker historical export

BTC reference history

BTC reference history

accepted formats: csv, jsonl, parquet

required fields: available_ts, btc_usd, venue_or_index, source_ref

provider examples: CF Benchmarks, vendor BTC index, broker reference

IBIT holdings history

IBIT holdings history

accepted formats: csv, jsonl, parquet

required fields: available_ts, btc_per_share, shares_outstanding, fund_assets, source_ref

provider examples: iShares daily holdings archive, fund administrator export

Rates and fee curves

Rates and fee curves

accepted formats: csv, jsonl, parquet

required fields: available_ts, tenor, rate, borrow_or_fee, source_ref

provider examples: SOFR/rate vendor, broker borrow/fee export

- Validation gate No future available_ts relative to decision_ts

Validation gate

No future available_ts relative to decision_ts

- Validation gate Raw file SHA-256 captured before normalization

Validation gate

Raw file SHA-256 captured before normalization

- Validation gate Provider/license label present for every source group

Validation gate

Provider/license label present for every source group

- Validation gate Fixture/manual_fixture sources cannot satisfy readiness

Validation gate

Fixture/manual_fixture sources cannot satisfy readiness

- Validation gate Rows with crossed/wide/missing markets remain diagnostic, not executable

Validation gate

Rows with crossed/wide/missing markets remain diagnostic, not executable

Scenario outputs are model-estimated and screen-only; no executable quote, client suitability review, or counterparty commitment is implied.

Case study · screen-only

Treasury Hedge Case Study

For corporate BTC holders: covered-call income, put-spread collars, and board-approved hedge policies that convert volatility into treasury risk management.

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 350 BTC hedged Conservative policy sleeve ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $20,332,477.31 illustrative floor-protected sleeve Illustrative protected value ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $58,092.79 illustrative floor / $96,821.32 cap Structure preview

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 350 BTC hedged Conservative policy sleeve

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

350 BTC hedged

Conservative policy sleeve

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $20,332,477.31 illustrative floor-protected sleeve Illustrative protected value

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$20,332,477.31 illustrative floor-protected sleeve

Illustrative protected value

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $58,092.79 illustrative floor / $96,821.32 cap Structure preview

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$58,092.79 illustrative floor / $96,821.32 cap

Structure preview

Premium and executable levels require two-counterparty quote verification.

Case study · screen-only

Miner Runway Protection

For miners: protect production runway without overhedging or creating collateral stress. The goal is survival and financing credibility, not speculative upside chasing.

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 60 BTC/month hedged Conservative production hedge ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $3,253,196.37 model-estimated monthly floor revenue Monthly floor revenue on hedged BTC ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 3.75 months pre-hedge cash runway Baseline operating runway

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 60 BTC/month hedged Conservative production hedge

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

60 BTC/month hedged

Conservative production hedge

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $3,253,196.37 model-estimated monthly floor revenue Monthly floor revenue on hedged BTC

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$3,253,196.37 model-estimated monthly floor revenue

Monthly floor revenue on hedged BTC

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE 3.75 months pre-hedge cash runway Baseline operating runway

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

3.75 months pre-hedge cash runway

Baseline operating runway

Indicative economics require quote verification before investor/client use.

Quote Verification Demo Board

Manual demo workflow only; no RFQ is sent and no executable quote is implied.

Screen-only 5 Reviewed 0 Internal RFQ draft 0 Indicative quote 1 0 Indicative quote 2 0 Quote verified 0 Post-trade record verified 0

Screen-only 5

Screen-only

5

Reviewed 0

Reviewed

0

Internal RFQ draft 0

Internal RFQ draft

0

Indicative quote 1 0

Indicative quote 1

0

Indicative quote 2 0

Indicative quote 2

0

Quote verified 0

Quote verified

0

Post-trade record verified 0

Post-trade record verified

0

#1 IBIT 0D ATM vs Deribit 1D ATM Screen-only Internal candidate review; draft RFQ package only after approval. Quotes captured: 0 / 2 · Publishability: internal-only SCREEN-ONLY · NOT EXECUTABLE #2 IBIT 2D ATM vs Deribit 2D ATM Screen-only Internal candidate review; draft RFQ package only after approval. Quotes captured: 0 / 2 · Publishability: internal-only SCREEN-ONLY · NOT EXECUTABLE #3 IBIT 4D ATM vs Deribit 4D ATM Screen-only Internal candidate review; draft RFQ package only after approval. Quotes captured: 0 / 2 · Publishability: internal-only SCREEN-ONLY · NOT EXECUTABLE #4 IBIT 8D ATM vs Deribit 11D ATM Screen-only Internal candidate review; draft RFQ package only after approval. Quotes captured: 0 / 2 · Publishability: internal-only SCREEN-ONLY · NOT EXECUTABLE

#1

IBIT 0D ATM vs Deribit 1D ATM

Screen-only

Internal candidate review; draft RFQ package only after approval.

Quotes captured: 0 / 2 · Publishability: internal-only

SCREEN-ONLY · NOT EXECUTABLE

#2

IBIT 2D ATM vs Deribit 2D ATM

Screen-only

Internal candidate review; draft RFQ package only after approval.

Quotes captured: 0 / 2 · Publishability: internal-only

SCREEN-ONLY · NOT EXECUTABLE

#3

IBIT 4D ATM vs Deribit 4D ATM

Screen-only

Internal candidate review; draft RFQ package only after approval.

Quotes captured: 0 / 2 · Publishability: internal-only

SCREEN-ONLY · NOT EXECUTABLE

#4

IBIT 8D ATM vs Deribit 11D ATM

Screen-only

Internal candidate review; draft RFQ package only after approval.

Quotes captured: 0 / 2 · Publishability: internal-only

SCREEN-ONLY · NOT EXECUTABLE

Screen-only dislocation detected by the monitor. Candidate triaged for materiality and structure fit. RFQ package generated for review, not submission. After approval and manual outreach, counterparties may provide indicative quotes. Evidence ledger may mark candidate quote-verified after required indicative quote records are captured. Post-trade verification only occurs after an actual external execution record exists; this demo has no trading capability.

Screen-only dislocation detected by the monitor.

Candidate triaged for materiality and structure fit.

RFQ package generated for review, not submission.

After approval and manual outreach, counterparties may provide indicative quotes.

Evidence ledger may mark candidate quote-verified after required indicative quote records are captured.

Post-trade verification only occurs after an actual external execution record exists; this demo has no trading capability.

Backtest / Research Evidence

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

Point-in-time research only using synthetic fills; not executable economics and not an investment conclusion.

1d scenario SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE History 19 snapshots Synthetic trades 17 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $743,350.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $27,650.00 Win rate 64.7% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00 Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only. 7d scenario SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE History 19 snapshots Synthetic trades 2 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $54,000.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $0.00 Win rate 100.0% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00 Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only. 30d scenario SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE History 19 snapshots Synthetic trades 2 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $48,300.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $150.00 Win rate 50.0% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00 Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only.

1d scenario

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

History 19 snapshots Synthetic trades 17 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $743,350.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $27,650.00 Win rate 64.7% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

History 19 snapshots

History

19 snapshots

Synthetic trades 17 synthetic trades

Synthetic trades

17 synthetic trades

Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $743,350.00

Gross PnL

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$743,350.00

Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $27,650.00

Max drawdown

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$27,650.00

Win rate 64.7%

Win rate

64.7%

Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

Cost/trade

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$250.00

Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only.

7d scenario

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

History 19 snapshots Synthetic trades 2 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $54,000.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $0.00 Win rate 100.0% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

History 19 snapshots

History

19 snapshots

Synthetic trades 2 synthetic trades

Synthetic trades

2 synthetic trades

Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $54,000.00

Gross PnL

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$54,000.00

Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $0.00

Max drawdown

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$0.00

Win rate 100.0%

Win rate

100.0%

Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

Cost/trade

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$250.00

Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only.

30d scenario

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

History 19 snapshots Synthetic trades 2 synthetic trades Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $48,300.00 Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $150.00 Win rate 50.0% Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

History 19 snapshots

History

19 snapshots

Synthetic trades 2 synthetic trades

Synthetic trades

2 synthetic trades

Gross PnL ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $48,300.00

Gross PnL

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$48,300.00

Max drawdown ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $150.00

Max drawdown

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$150.00

Win rate 50.0%

Win rate

50.0%

Cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

Cost/trade

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$250.00

Sample gate: insufficient-history · 0.0 vol pts slippage · synthetic fills only.

Backtest Robustness Sample gate pass count 0 / 3 Minimum gates 30 snapshots / 20 synthetic trades Effective cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00 Tenors with synthetic trades: 1d, 7d, 30d. Sample gate ready: False. Synthetic fills only; cost sensitivity is illustrative and not executable economics.

Backtest Robustness

Sample gate pass count 0 / 3 Minimum gates 30 snapshots / 20 synthetic trades Effective cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

Sample gate pass count 0 / 3

Sample gate pass count

0 / 3

Minimum gates 30 snapshots / 20 synthetic trades

Minimum gates

30 snapshots / 20 synthetic trades

Effective cost/trade ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE $250.00

Effective cost/trade

ILLUSTRATIVE MODEL OUTPUT · NOT A QUOTE · NOT EXECUTABLE · NOT INVESTOR-PUBLISHABLE

$250.00

Tenors with synthetic trades: 1d, 7d, 30d. Sample gate ready: False. Synthetic fills only; cost sensitivity is illustrative and not executable economics.

Sample gate: insufficient-history. Backtest PnL is a scaffold for research hygiene, not an investment conclusion.

- Backtest report artifacts/institutional/backtests/vol-spread-backtest-multitenor-v1.md

Backtest report

artifacts/institutional/backtests/vol-spread-backtest-multitenor-v1.md

- Backtest SHA-256 ac18310dddc991c9fd22bfbce98bfbcc10279ac8eed99613e27f36cea6cbdb48

Backtest SHA-256

ac18310dddc991c9fd22bfbce98bfbcc10279ac8eed99613e27f36cea6cbdb48

- Backtest summary JSON artifacts/institutional/backtests/vol-spread-backtest-multitenor-v1.json

Backtest summary JSON

artifacts/institutional/backtests/vol-spread-backtest-multitenor-v1.json

- Summary JSON SHA-256 233e874dbd528f954a69dfd51e730b98821ec64b10d2c0f7b3d345bf30f4d176

Summary JSON SHA-256

233e874dbd528f954a69dfd51e730b98821ec64b10d2c0f7b3d345bf30f4d176

- Control strict chronology enforced

Control

strict chronology enforced

- Control strategy sees prior history only

Control

strategy sees prior history only

- Control synthetic fills; not executable economics

Control

synthetic fills; not executable economics

Source Diagnostics Historical backtests use hashed point-in-time sources; synthetic fills only; not executable economics. SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

Source Diagnostics

Historical backtests use hashed point-in-time sources; synthetic fills only; not executable economics.

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

Historical source manifest 2 sources Required-source coverage 0/6 licensed source groups covered · 2 fixture-only · 4 missing Manifest hash 238ace299e2a…052bf3d939

Historical source manifest 2 sources

Historical source manifest

2 sources

Required-source coverage 0/6 licensed source groups covered · 2 fixture-only · 4 missing

Required-source coverage

0/6 licensed source groups covered · 2 fixture-only · 4 missing

Manifest hash 238ace299e2a…052bf3d939

Manifest hash

238ace299e2a…052bf3d939

Required universe IBIT Deribit CME BTC reference IBIT holdings Rates/fees Coverage gaps IBIT, Deribit, CME, BTC reference, IBIT holdings, Rates/fees Coverage matrix IBIT fixture-only Deribit fixture-only CME missing BTC reference missing IBIT holdings missing Rates/fees missing

Required universe

IBIT Deribit CME BTC reference IBIT holdings Rates/fees

IBIT

Deribit

CME

BTC reference

IBIT holdings

Rates/fees

Coverage gaps

IBIT, Deribit, CME, BTC reference, IBIT holdings, Rates/fees

Coverage matrix

IBIT fixture-only Deribit fixture-only CME missing BTC reference missing IBIT holdings missing Rates/fees missing

IBIT fixture-only

Deribit fixture-only

CME missing

BTC reference missing

IBIT holdings missing

Rates/fees missing

0 covered · 4 missing · 2 fixture-only. This is an operational ledger for source acquisition and replay readiness, not a trading signal.

IBIT · fixture-only screen_only_not_executable Provider: manual_fixture · License: manual_fixture · Rows: n/a Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z · Sources: 1 SHA-256 a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430 Blocker Fixture source; replace with licensed historical export before investment use. Next action Expand IBIT historical coverage window and retain manifest hashes for replay. Deribit · fixture-only screen_only_not_executable Provider: manual_fixture · License: manual_fixture · Rows: n/a Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z · Sources: 1 SHA-256 8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76 Blocker Fixture source; replace with licensed historical export before investment use. Next action Load Deribit historical options replay source with raw-file hash and coverage dates. CME · missing screen_only_not_executable Provider: missing · License: missing · Rows: n/a Coverage: n/a → n/a · Sources: 0 SHA-256 missing Blocker No CME source captured in latest historical manifest. Next action Configure licensed CME/Databento historical source; keep screen-only until quote/trade evidence exists. BTC reference · missing screen_only_not_executable Provider: missing · License: missing · Rows: n/a Coverage: n/a → n/a · Sources: 0 SHA-256 missing Blocker No BTC reference source captured in latest historical manifest. Next action Add point-in-time BTC reference marks used for every decision frame. IBIT holdings · missing screen_only_not_executable Provider: missing · License: missing · Rows: n/a Coverage: n/a → n/a · Sources: 0 SHA-256 missing Blocker No IBIT holdings source captured in latest historical manifest. Next action Add point-in-time IBIT holdings/BTC-per-share files for every decision date. Rates/fees · missing screen_only_not_executable Provider: missing · License: missing · Rows: n/a Coverage: n/a → n/a · Sources: 0 SHA-256 missing Blocker No Rates/fees source captured in latest historical manifest. Next action Add rates, borrow, fees, and transaction-cost assumptions with source hashes.

IBIT · fixture-only

screen_only_not_executable

Provider: manual_fixture · License: manual_fixture · Rows: n/a

Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z · Sources: 1

- SHA-256 a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430

SHA-256

a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430

- Blocker Fixture source; replace with licensed historical export before investment use.

Blocker

Fixture source; replace with licensed historical export before investment use.

- Next action Expand IBIT historical coverage window and retain manifest hashes for replay.

Next action

Expand IBIT historical coverage window and retain manifest hashes for replay.

Deribit · fixture-only

screen_only_not_executable

Provider: manual_fixture · License: manual_fixture · Rows: n/a

Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z · Sources: 1

- SHA-256 8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76

SHA-256

8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76

- Blocker Fixture source; replace with licensed historical export before investment use.

Blocker

Fixture source; replace with licensed historical export before investment use.

- Next action Load Deribit historical options replay source with raw-file hash and coverage dates.

Next action

Load Deribit historical options replay source with raw-file hash and coverage dates.

CME · missing

screen_only_not_executable

Provider: missing · License: missing · Rows: n/a

Coverage: n/a → n/a · Sources: 0

- SHA-256 missing

SHA-256

missing

- Blocker No CME source captured in latest historical manifest.

Blocker

No CME source captured in latest historical manifest.

- Next action Configure licensed CME/Databento historical source; keep screen-only until quote/trade evidence exists.

Next action

Configure licensed CME/Databento historical source; keep screen-only until quote/trade evidence exists.

BTC reference · missing

screen_only_not_executable

Provider: missing · License: missing · Rows: n/a

Coverage: n/a → n/a · Sources: 0

- SHA-256 missing

SHA-256

missing

- Blocker No BTC reference source captured in latest historical manifest.

Blocker

No BTC reference source captured in latest historical manifest.

- Next action Add point-in-time BTC reference marks used for every decision frame.

Next action

Add point-in-time BTC reference marks used for every decision frame.

IBIT holdings · missing

screen_only_not_executable

Provider: missing · License: missing · Rows: n/a

Coverage: n/a → n/a · Sources: 0

- SHA-256 missing

SHA-256

missing

- Blocker No IBIT holdings source captured in latest historical manifest.

Blocker

No IBIT holdings source captured in latest historical manifest.

- Next action Add point-in-time IBIT holdings/BTC-per-share files for every decision date.

Next action

Add point-in-time IBIT holdings/BTC-per-share files for every decision date.

Rates/fees · missing

screen_only_not_executable

Provider: missing · License: missing · Rows: n/a

Coverage: n/a → n/a · Sources: 0

- SHA-256 missing

SHA-256

missing

- Blocker No Rates/fees source captured in latest historical manifest.

Blocker

No Rates/fees source captured in latest historical manifest.

- Next action Add rates, borrow, fees, and transaction-cost assumptions with source hashes.

Next action

Add rates, borrow, fees, and transaction-cost assumptions with source hashes.

Deribit BTC historical options fixture SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE Deribit · manual_fixture · BTC options License: manual_fixture · Redistribution: internal_only · Execution: screen_only_not_executable Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z Raw source artifacts/institutional/historical/raw/fixtures/deribit_fixture_2025-01-02.jsonl SHA-256 8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76 IBIT OPRA historical options fixture SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE OPRA · manual_fixture · IBIT options License: manual_fixture · Redistribution: internal_only · Execution: screen_only_not_executable Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z Raw source artifacts/institutional/historical/raw/fixtures/ibit_opra_fixture_2025-01-02.csv SHA-256 a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430

Deribit BTC historical options fixture

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

Deribit · manual_fixture · BTC options

License: manual_fixture · Redistribution: internal_only · Execution: screen_only_not_executable

Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z

- Raw source artifacts/institutional/historical/raw/fixtures/deribit_fixture_2025-01-02.jsonl

Raw source

artifacts/institutional/historical/raw/fixtures/deribit_fixture_2025-01-02.jsonl

- SHA-256 8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76

SHA-256

8d2552db0baaf54cc177ca6ddc6aece0863ad374f3baddfcf7c860b3677baa76

IBIT OPRA historical options fixture

SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE

OPRA · manual_fixture · IBIT options

License: manual_fixture · Redistribution: internal_only · Execution: screen_only_not_executable

Coverage: 2025-01-02T15:30:00Z → 2025-01-02T15:30:00Z

- Raw source artifacts/institutional/historical/raw/fixtures/ibit_opra_fixture_2025-01-02.csv

Raw source

artifacts/institutional/historical/raw/fixtures/ibit_opra_fixture_2025-01-02.csv

- SHA-256 a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430

SHA-256

a803cdd9ba7804417eeb2a97af06064db7757b0ec919fb9acc4bfc0b89acd430

- Historical source manifest artifacts/institutional/historical/manifests/historical-source-manifest-fixtures-v1.json

Historical source manifest

artifacts/institutional/historical/manifests/historical-source-manifest-fixtures-v1.json

- Manifest SHA-256 238ace299e2a2d6af4f8ecc223149346a876d0184df7fca40fb0ae052bf3d939

Manifest SHA-256

238ace299e2a2d6af4f8ecc223149346a876d0184df7fca40fb0ae052bf3d939

- Replay controls PIT hashed · synthetic fills · not executable economics

Replay controls

PIT hashed · synthetic fills · not executable economics

Quote Evidence Ledger

Manual evidence capture for diligence only. Demo/manual indicative rows do not count as real quote-verified evidence. Real quote-verified evidence remains internal diligence only until trade records and the sponsor's counsel-approved business/legal wrapper permit external use.

Demo/manual indicative records 0 demo records Real quote-verified records 0 real records Trade-verified candidates 0 trade-verified candidates

Demo/manual indicative records 0 demo records

Demo/manual indicative records

0 demo records

Real quote-verified records 0 real records

Real quote-verified records

0 real records

Trade-verified candidates 0 trade-verified candidates

Trade-verified candidates

0 trade-verified candidates

Not investor-publishable: indicative quotes are not executable and do not represent a client-facing RFQ, trade, or fund offering. Quote-verified candidates: 0. Invalid rows: 0.

No manual quote evidence captured yet. Screen-only candidates remain internal review items.

- Quote control Two distinct counterparties required for quote-verified candidate status

Quote control

Two distinct counterparties required for quote-verified candidate status

- Quote control same-dealer duplicate quotes cannot promote a candidate

Quote control

same-dealer duplicate quotes cannot promote a candidate

- Quote control Quote evidence remains not investor-publishable until legal wrapper approval

Quote control

Quote evidence remains not investor-publishable until legal wrapper approval

- Quote records none captured

Quote records

none captured

Desk / Platform First

- Research/evidence engine
- Treasury and miner hedge structuring
- Partner-led RFQ and execution support
- Principal risk sleeve only after chosen legal wrapper

This is the recommended Option B route: prove the desk and platform first; use the fund sleeve as an expansion path, not the opening claim.

Investor Diligence Ask

Use the site to ask for seed/build capital, strategic OTC/broker introductions, legal structuring support, or a pilot treasury/miner mandate.

Fund sleeve later once data history, quote evidence, and wrapper are strong enough.

Fund sleeve later

Evidence Room

Artifacts below are generated by the current evidence monitor and should travel with the investor memo.

- Report artifacts/institutional/data/reports/btc-vol-desk-monitor-2026-05-18-163034.md

Report

artifacts/institutional/data/reports/btc-vol-desk-monitor-2026-05-18-163034.md

- Internal dashboard artifacts/institutional/dashboard/index.html

Internal dashboard

artifacts/institutional/dashboard/index.html

- Evidence bundle artifacts/institutional/data/normalized/btcvol-20260518-163034/btcvol-20260518-163034-evidence-bundle.zip

Evidence bundle

artifacts/institutional/data/normalized/btcvol-20260518-163034/btcvol-20260518-163034-evidence-bundle.zip

- Evidence manifest artifacts/institutional/data/normalized/btcvol-20260518-163034/evidence_manifest.json

Evidence manifest

artifacts/institutional/data/normalized/btcvol-20260518-163034/evidence_manifest.json

- Candidate ledger artifacts/institutional/data/normalized/btcvol-20260518-163034/candidate_triage.jsonl

Candidate ledger

artifacts/institutional/data/normalized/btcvol-20260518-163034/candidate_triage.jsonl

- Quote evidence ledger artifacts/institutional/data/quote_evidence.jsonl

Quote evidence ledger

artifacts/institutional/data/quote_evidence.jsonl

- Bundle SHA-256 6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b

Bundle SHA-256

6504204d3adf8ecd1d5bac94775f2639e08baec722a06f41623346dd7724210b

- Evidence status SCREEN-ONLY · NOT EXECUTABLE

Evidence status

SCREEN-ONLY · NOT EXECUTABLE

Latest Monitor Report

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/evidence/latest-report.md`

BTC Vol Desk Monitor — 2026-05-18 16:30:34 CDT

Run ID: `btcvol-20260518-163034`

Evidence standard: all public/API screen outputs are `screen-only` unless linked to quote or trade records.

BTC / ETF Reference

- BTC spot/reference: 77,457.06
- BTC per ETF share: 0.000567911859

Deribit ATM IV Term Structure

Expiry	DTE	Symbol	IV mark
2026-05-19	1	BTC-19MAY26-77000-P	27.64%
2026-05-20	2	BTC-20MAY26-77000-P	32.42%
2026-05-21	3	BTC-21MAY26-77000-P	34.06%
2026-05-22	4	BTC-22MAY26-77000-C	35.22%
2026-05-29	11	BTC-29MAY26-77000-P	37.25%
2026-06-05	18	BTC-5JUN26-77000-C	37.46%
2026-06-26	39	BTC-26JUN26-78000-P	37.67%
2026-07-31	74	BTC-31JUL26-77000-C	38.72%
2026-09-25	130	BTC-25SEP26-78000-C	40.42%
2026-12-25	221	BTC-25DEC26-78000-P	43.79%
2027-03-26	312	BTC-26MAR27-78000-P	44.92%

IBIT / ETF ATM IV Term Structure

Expiry	DTE	Symbol	IV mark
2026-05-18	0	IBIT-2026-05-18-43.5-C	11.15%
2026-05-20	2	IBIT-2026-05-20-43.5-C	40.81%
2026-05-22	4	IBIT-2026-05-22-43.5-C	40.68%
2026-05-26	8	IBIT-2026-05-26-43.5-C	35.74%
2026-05-27	9	IBIT-2026-05-27-43.5-C	36.36%

Expiry	DTE	Symbol	IV mark
2026-05-29	11	IBIT-2026-05-29-43.5-C	38.52%

CME / Databento BTC Options Status

Evidence label: Databento CME rows are `screen_only_not_executable`; they are licensed vendor marks, not executable quotes.

Expiry	DTE	Symbol	Bid	Ask
2026-05-22	4	UD:B4: VT 2614591	9.00	45.00
2026-05-29	11	BTCK6 P55000	2.00	70.00
2026-05-29	11	BTCK6 P55500	4.00	70.00
2026-05-29	11	BTCK6 P56000	6.00	75.00
2026-05-29	11	BTCK6 P56500	8.00	75.00
2026-05-29	11	BTCK6 P57000	11.00	80.00
2026-05-29	11	BTCK6 P57500	13.00	80.00
2026-05-29	11	BTCK6 P58000	16.00	85.00
2026-05-29	11	BTCK6 P58500	20.00	90.00
2026-05-29	11	BTCK6 P59000	23.00	90.00
2026-05-29	11	BTCK6 P59500	25.00	95.00
2026-05-29	11	BTCK6 P60000	30.00	100.00
2026-05-29	11	BTCK6 P60500	35.00	105.00
2026-05-29	11	BTCK6 P61000	40.00	115.00
2026-05-29	11	BTCK6 P61500	45.00	120.00
2026-05-29	11	BTCK6 P62000	55.00	125.00
2026-05-29	11	BTCK6 P62500	60.00	135.00
2026-05-29	11	BTCK6 P63000	70.00	140.00
2026-05-29	11	BTCK6 P63500	80.00	150.00
2026-05-29	11	BTCK6 P64000	85.00	160.00

Dislocation Board

Candidate	Gross IV diff	Confidence	Next action
IBIT 0D ATM vs Deribit 1D ATM	-16.49 vol pts	screen-only	quote review
IBIT 2D ATM vs Deribit 2D ATM	8.39 vol pts	screen-only	quote review
IBIT 4D ATM vs Deribit 4D ATM	5.46 vol pts	screen-only	quote review

Candidate	Gross IV diff	Confidence	Next action
IBIT 8D ATM vs Deribit 11D ATM	-1.51 vol pts	screen-only	watch
IBIT 9D ATM vs Deribit 11D ATM	-0.89 vol pts	screen-only	watch
IBIT 11D ATM vs Deribit 11D ATM	1.27 vol pts	screen-only	watch

Quality Warnings

- Deribit normalized rows: 888
- iShares BTC/share current fetch or parse failed: iShares holdings endpoint returned HTML instead of holdings CSV
- iShares BTC/share fallback used latest cached official CSV: artifacts/institutional/data/raw/btcvol-20260515-183316/ishares_ibit_holdings.csv
- IBIT holdings source is stale-cache; use for continuity only until current iShares CSV recovers
- IBIT BTC/share independent market-implied cross-check diff: 1.04%
- IBIT normalized rows: 402
- CME Databento normalized rows: 1708
- CME Databento BBO rows: 5000

Backtest Research Evidence

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/evidence/backtest-report.md`

BTC Vol Spread Backtest — Multi-Tenor Research Pack

Evidence status: `SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE`

Research evidence only. All modeled results use screen-only historical marks and synthetic fills; not executable economics.

BTC Vol Spread Backtest — 1d-spread-threshold-5

Evidence status: `SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE`

This is research evidence, not executable economics. The engine uses historical monitor snapshots, synthetic fills, and fixed cost assumptions; it does not prove that any spread was tradable.

Controls

- No future leakage: the strategy receives only prior snapshots at each decision point.
- Chronology is enforced; out-of-order snapshots fail instead of being silently sorted.
- Synthetic fills are explicitly labeled and are not quotes, RFQs, or executions.

Summary

- Tenor: `1d`
- Snapshot count: 19
- Trade count: 17

- Gross PnL: $743,350.00
- Max drawdown: $27,650.00
- Win rate: 64.7%
- Minimum evidence gate: insufficient-history

Assumptions

- execution: screen-only synthetic fills; not executable economics
- anti_lookahead: DecisionFrame excludes next_* future fields; ExecutionResolver resolves synthetic exits after decision time
- source: point-in-time historical spread snapshots
- spread_field: spread_1d_vol_pts
- threshold_vol_pts: 5.0
- notional_vega: 10000.0
- cost_per_trade: 250.0

Trades

#	Side	Entry	Exit	Net PnL
1	short_spread	5.71	-11.97	$176,550.00
2	long_spread	-11.97	-14.71	$-27,650.00
3	long_spread	-14.71	-11.52	$31,650.00
4	long_spread	-11.52	-12.99	$-14,950.00
5	long_spread	-12.99	-13.13	$-1,650.00
6	long_spread	-13.13	-12.86	$2,450.00
7	long_spread	-12.86	-7.86	$49,750.00
8	long_spread	-7.86	13.83	$216,650.00
9	short_spread	13.83	13.15	$6,550.00
10	short_spread	13.15	6.43	$66,950.00
11	short_spread	6.43	6.43	$-250.00
12	short_spread	6.43	6.36	$450.00
13	short_spread	6.36	8.96	$-26,250.00
14	short_spread	8.96	-16.97	$259,050.00
15	long_spread	-16.97	-15.16	$17,850.00
16	long_spread	-15.16	-17.69	$-25,550.00
17	long_spread	-17.69	-16.49	$11,750.00

BTC Vol Spread Backtest — 7d-spread-threshold-5

Evidence status: `SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE`

This is research evidence, not executable economics. The engine uses historical monitor snapshots, synthetic fills, and fixed cost assumptions; it does not prove that any spread was tradable.

Controls

- No future leakage: the strategy receives only prior snapshots at each decision point.
- Chronology is enforced; out-of-order snapshots fail instead of being silently sorted.
- Synthetic fills are explicitly labeled and are not quotes, RFQs, or executions.

Summary

- Tenor: `7d`
- Snapshot count: 19
- Trade count: 2
- Gross PnL: $54,000.00
- Max drawdown: $0.00
- Win rate: 100.0%
- Minimum evidence gate: insufficient-history

Assumptions

- execution: screen-only synthetic fills; not executable economics
- anti_lookahead: DecisionFrame excludes next_* future fields; ExecutionResolver resolves synthetic exits after decision time
- source: point-in-time historical spread snapshots
- spread_field: spread_7d_vol_pts
- threshold_vol_pts: 5.0
- notional_vega: 10000.0
- cost_per_trade: 250.0

Trades

#	Side	Entry	Exit	Net PnL
1	short_spread	7.81	7.74	$450.00
2	short_spread	7.74	2.36	$53,550.00

BTC Vol Spread Backtest — 30d-spread-threshold-5

Evidence status: `SCREEN-ONLY · BACKTEST ONLY · NOT EXECUTABLE`

This is research evidence, not executable economics. The engine uses historical monitor snapshots, synthetic fills, and fixed cost assumptions; it does not prove that any spread was tradable.

Controls

- No future leakage: the strategy receives only prior snapshots at each decision point.
- Chronology is enforced; out-of-order snapshots fail instead of being silently sorted.
- Synthetic fills are explicitly labeled and are not quotes, RFQs, or executions.

Summary

- Tenor: `30d`
- Snapshot count: 19
- Trade count: 2
- Gross PnL: $48,300.00
- Max drawdown: $150.00
- Win rate: 50.0%
- Minimum evidence gate: insufficient-history

Assumptions

- execution: screen-only synthetic fills; not executable economics
- anti_lookahead: DecisionFrame excludes next_* future fields; ExecutionResolver resolves synthetic exits after decision time
- source: point-in-time historical spread snapshots
- spread_field: spread_30d_vol_pts
- threshold_vol_pts: 5.0
- notional_vega: 10000.0
- cost_per_trade: 250.0

Trades

#	Side	Entry	Exit	Net PnL
1	short_spread	5.87	5.86	$-150.00
2	short_spread	5.86	0.99	$48,450.00

Robustness Metrics

- Sample gate pass count: 0 / 3

- Sample gate ready: False
- Minimum gates: 30 snapshots and 20 synthetic trades per tenor
- Max observed sample: 19 snapshots / 17 synthetic trades
- Tenors with synthetic trades: 1d, 7d, 30d
- Best gross PnL tenor: 1d
- Worst gross PnL tenor: 30d
- Cost sensitivity: $250.00 explicit cost + 0.00 vol pts slippage = $250.00 effective cost/trade
- Control: Synthetic fills only; cost sensitivity is illustrative and not executable economics.

Legal Wrapper Draft

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/investor-packet/evidence/legal-wrapper-package-v1.md`

BTC Vol Desk Legal / Business Wrapper Draft v1

Evidence status: `SCREEN-ONLY · LEGAL WRAPPER DRAFT · NOT EXECUTABLE`

Status: `DRAFT — NOT APPROVED FOR EXTERNAL USE`

> Legal/business wrapper draft only. This artifact is not legal advice and does not authorize external client, investor, RFQ, execution, advisory, or fund activity.

Blockers

- counsel approval missing
- external advisory/client/RFQ/fund use prohibited until wrapper approved
- quote evidence remains internal diligence only

Business Boundary Matrix

- **Analytics / research** — internal-only — external requires: counsel-approved disclaimer and distribution controls
- **Advisory / structuring** — blocked-pre-wrapper — external requires: RIA/CTA/CPO/BD perimeter analysis and written engagement terms
- **Execution / RFQ support** — blocked-pre-wrapper — external requires: broker/FCM/OTC partner model, no agency ambiguity, counterparty onboarding controls
- **Principal risk-taking** — blocked-pre-wrapper — external requires: entity/risk capital approval, mandate limits, conflicts policy, trade surveillance
- **Structured products / fund sleeve** — blocked-pre-wrapper — external requires: securities/private fund counsel, offering docs, investor eligibility, marketing review

Counsel Questions

- Can analytics/research be distributed externally, and under what disclaimers?
- Does treasury/miner hedge structuring create investment adviser, CTA, CPO, broker-dealer, FCM, or swap-adviser issues?
- What engagement letter, suitability, KYC/KYB, and conflicts controls are required before any client mandate?
- Can RFQ support be handled through a partner broker/FCM/OTC desk without agency/execution ambiguity?
- What marketing/investor deck language is prohibited before a fund or structured-product wrapper exists?
- What records, surveillance, and approvals are required before principal risk-taking?

Hard Gates

- No external investor/client use until counsel-approved wrapper exists
- No RFQ submission by automation
- No executable quote labels without two distinct external indicative quote records
- No trade-verified label without execution record and post-trade evidence
- No offering/fund/product language without securities/private-fund counsel approval

Master Plan v2 Evidence-First Hardening

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-master-plan-v2-evidence-first-hardening.md`

BTC Vol Desk Master Plan v2 — Evidence-First Hardening

> **For Hermes:** Use subagent-driven-development / TDD for implementation slices. Every item requires tests, artifact rebuild, packet verification, browser-visible QA, and an explicit done/not-done/blocker ledger.

Date: 2026-05-18 CST

Scope: Frontend/static investor packet and dashboard, backend monitor/pipeline/CLI, strategy/backtest/signals/candidate triage, quote evidence, legal/evidence gates, packet/docs/artifacts, cron automation, tests, security/secrets, and commercial positioning.

Evidence status: `SCREEN-ONLY · NOT EXECUTABLE`

External use: `BLOCKED` until licensed historical sources, two-counterparty real quote evidence, and counsel-approved legal/business wrapper are all verified.

---

1. Audit inputs and verification proof

My audit lane

Verified locally:

- Focused BTC Vol test suite: `135 passed in 2.06s`.
- Investor packet verifier: `ok=true`, 13 artifacts checked, packet SHA matched.
- Source-intake skeleton validator: `ready=false`, `covered_source_groups=0/6`, correctly blocks placeholder licensed-source package.
- Cron job read-back: `BTC Vol daily point-in-time snapshot`, job ID `9a4c6cc52559`, enabled, script-only, next run `2026-05-18T14:30:00-07:00` = `4:30 PM CST`.
- Browser/visual audit of packet site: page renders cleanly, no executable controls visible, but dollar-value disclaimer adjacency and dense readability need improvement.

Key current packet state from `site-data.json`:

- Institutional readiness gate: `0/4 readiness gates passed`, `not-ready`.
- Quote evidence ledger: `0` valid records, `0` quote-verified candidates.
- Legal wrapper: `approved_by_counsel=false`, `draft-blocked`.
- Source-intake validation: `ready=false`, `0/6` licensed historical source groups ready.
- Current source availability: `6/6 current source groups available`, but this is current/screen/cached availability, not licensed historical readiness.

Claude Opus 4.7 separate audit lane

Verified execution:

- Claude auth: Claude Max account.
- Model usage included `claude-opus-4-7`.
- Separate read-only adversarial audit produced P0/P1/P2 findings.
- No files modified by Claude.

Claude's strongest verified findings were independently spot-checked in source:

- `packet_verify.py` verifies hashes/control language but does not enforce legal-wrapper approval semantics.
- `quote_evidence.py` lacks operator/counsel attestation fields and exposes raw counterparty strings in summaries.
- `site_data.py` has a fixture-vs-covered inconsistency between the source coverage matrix and tracker.
- `site_data.py` silently falls back to `btc_spot=80000` for case studies if spot is missing.
- `history.py` appends JSONL without file locking.
- `quality.py` penalizes zero-dislocation runs, which is backwards for quiet markets.
- `backtest_report.py` displays PnL numbers even when the sample gate is insufficient.
- Browser visual audit confirmed disclaimer-adjacency weakness around large case-study dollar values and backtest metrics.

---

2. Accepted audit findings

P0 — Boundary, legal, and evidence-integrity fixes

P0.1 — Legal-wrapper enforcement is advisory, not a hard verifier gate

Evidence: `packet_verify.py` checks packet hash, publishability string, artifacts, secret scan, CTA scan, and control language. It does not parse `legal_wrapper_json` or enforce `approved_by_counsel` status.

Important nuance: For the current internal packet, `approved_by_counsel=false` should not make internal verification fail. Instead the verifier must explicitly output an external-use gate status:

- internal packet integrity: can pass;
- external readiness: must fail until counsel approval exists.

Required fix: Add structured verifier fields:

- `legal_wrapper_present`
- `legal_wrapper_approved`
- `external_use_ok`
- `external_use_blockers`

`external_use_ok` must be false unless all required external gates pass.

P0.2 — Quote evidence promotion lacks operator/counsel attestation

Evidence: `quote_evidence.REQUIRED_FIELDS` requires RFQ/evidence basics but no `promoted_by_operator`, `promotion_timestamp`, `legal_review_ref`, or `promotion_basis`.

Required fix: Add promotion lifecycle controls:

- no skip from `screen-only` to `trade-verified`;
- `quote-verified` requires two distinct real external counterparties plus operator attestation;
- `trade-verified` requires execution record, operator attestation, and legal/compliance reference;
- demo/manual placeholders can never become quote-verified.

P0.3 — Raw counterparty names can flow to publishable surfaces

Evidence: `build_quote_evidence_summary()` exposes `counterparties` from raw records.

Required fix: Introduce counterparty pseudonymization:

- raw ledger can store actual counterparty name internally;
- rendered site/memo/tearsheet/packet can only expose `counterparty_pseudonym` or deterministic labels like `Counterparty A/B`;
- verifier scans rendered artifacts for raw counterparty names supplied in the ledger test fixture.

P0.4 — Fixture-only historical sources are counted as covered in one matrix

Evidence: `_source_coverage_matrix()` says `2/6 required source groups covered`; `_source_coverage_tracker()` correctly says `0 covered · 4 missing · 2 fixture-only`.

Required fix: Coverage matrix must support `covered | fixture-only | missing`; summary must read like `0/6 licensed source groups covered · 2 fixture-only · 4 missing`.

P0.5 — Large dollar outputs need claim-local disclaimers

Evidence: Browser visual audit showed case-study and backtest dollar values visually dominate while disclaimers are section-level or small.

Required fix: Every large monetary output in case studies and backtest cards must include an adjacent inline pill:

- `ILLUSTRATIVE MODEL OUTPUT`
- `NOT A QUOTE`
- `NOT EXECUTABLE`
- `NOT INVESTOR-PUBLISHABLE`

The warning must live in the same DOM/card/container as the monetary value.

P1 — Reliability, reproducibility, and model-honesty fixes

P1.1 — Run manifest append is not locked

Evidence: `history.append_run_manifest()` opens JSONL in append mode with no lock.

Fix: Add `fcntl.flock` around append on macOS/Linux. Add concurrency regression.

P1.2 — Site JSON is not deterministic

Evidence: `write_site_data()` uses `json.dumps(..., indent=2)` without `sort_keys=True`, while generated packets hash `site-data.json`.

Fix: Use sorted deterministic JSON for all hashed/generated JSON artifacts. Add reproducibility test: build same packet twice on same inputs, assert hashes match.

P1.3 — Multiple `now()` calls create timestamp drift

Evidence: `run_monitor.py` captures timestamps at separate stages.

Fix: Capture one run-level `as_of` at start, thread through source captures, report, freshness, manifest, and site.

P1.4 — Missing BTC spot silently becomes `$80,000`

Evidence: `site_data.py:694` uses `float(latest.get("btc_spot") or 80000)`.

Fix: Remove fallback. If spot missing, suppress case studies and render `spot unavailable — case studies suppressed`.

P1.5 — Quote-evidence dummy path masks unavailable state

Evidence: `site_data.py` passes `__no_quote_evidence_for_latest_run__.jsonl` when no quote path exists.

Fix: Let `load_quote_evidence_ledger(None)` return a typed `unavailable` result. Render unavailable distinctly from `empty but available`.

P1.6 — Backtest sample gate still displays performance-like PnL

Evidence: Backtest reports mark sample gate insufficient but still display gross PnL and trade PnL.

Fix: When `sample_gate_ready=false`, suppress headline PnL and scenario PnL with –; keep diagnostics and controls only.

P1.7 — Effective cost is computed but not reflected per scenario

Evidence: `effective_cost` appears in robustness metrics, while scenario dict records only input `cost_per_trade`.

Fix: Add `effective_cost_per_trade` to scenario outputs and UI labels.

P1.8 — Quality score penalizes quiet markets

Evidence: `quality.py` subtracts points when `dislocations == 0`.

Fix: Remove that penalty. Add penalty only for suspicious candidate explosions or source/parsing degradation.

P1.9 — Fetchers need bounded retry/backoff and source-level failure classification

Evidence: Current fetch paths are mostly single-attempt calls.

Fix: Add retry helper: 3 attempts, exponential backoff with jitter, 4xx fail-fast, 5xx/network retry. Record retry counts and final failure class in run manifest.

P1.10 — Packet verifier trusts builder-provided missing-artifacts metadata

Evidence: Verifier errors if `missing_artifacts` is non-empty, but does not independently assert required labels are represented.

Fix: Add `EXPECTED_PACKET_LABELS`; verifier fails if each required label is not either present in artifacts or explicitly listed as missing.

P2 — Polish and maintainability

- Replace hardcoded wording constants with central control-language constants where copy is reused.
- Add explicit parse-rejection logs in historical ingest.
- Align UTC/CST timestamp fields across Databento/current sources.
- Add units to backtest tables: vol pts, USD, synthetic PnL.
- Make vendor endpoints/config overridable while preserving safe defaults.
- Add dashboard/site HTML tests for forbidden CTAs, raw counterparty leaks, and required control labels.
- Move the investor ask higher in the packet narrative, above deep operational evidence sections.
- Increase font size/contrast for dense tables/hashes/source diagnostics.

---

3. Disputed / narrowed Claude findings

These are useful but should not be implemented literally without nuance:

1. **"Fail packet verifier if counsel not approved."**

Narrowed: internal packet integrity should still pass; external-use readiness must fail explicitly.

2. **"Fixture coverage is a P0."**

Accepted as P0 for investor/commercial interpretation, not because the backtester currently treats fixtures as live. The product already has a readiness gate, but the matrix headline is misleading.

3. **"Raw zip CTA scan skip may hide forbidden language."**

Keep skip for third-party raw captures, but add a separate contamination report for raw captures so vendor marketing text does not fail packet integrity while still being visible.

---

4. New master implementation plan

Phase A — Hard evidence/legal gates first

Exit criterion: Internal packet verification still passes, but external-use readiness is a first-class machine-readable failure until legal, quote, and licensed-source gates are complete.

1. Add external-use gate fields to `packet_verify.py`.
2. Add tests for missing legal wrapper, draft wrapper, approved wrapper, and external-use blocked state.
3. Add `EXPECTED_PACKET_LABELS` enforcement.
4. Rebuild packet and verify `internal_integrity_ok=true`, `external_use_ok=false`.
5. Browser QA Legal / Readiness sections.

Phase B — Quote evidence lifecycle and pseudonymization

Exit criterion: No raw counterparty name appears in rendered site/packet/memo/tearsheet; promotion requires operator/legal attestation.

1. Extend quote schema with promotion attestation and pseudonym fields.
2. Add state-machine validation tests.
3. Add same-counterparty duplicate tests and two-real-counterparty promotion tests.
4. Update site data and renderers to expose pseudonyms only.
5. Add packet verifier raw-name leak regression.

Phase C — Source coverage truth and case-study safety

Exit criterion: Fixture-only never reads as licensed coverage, and no case-study dollar value can appear without adjacent model-output/non-executable warning.

1. Fix `_source_coverage_matrix()` fixture classification.
2. Update source-diagnostics UI badges and copy.
3. Remove `$80,000` spot fallback.
4. Suppress case studies when spot unavailable.
5. Add inline control pills to every large case-study dollar value.
6. Visual/browser QA.

Phase D — Backtest honesty hardening

Exit criterion: Insufficient sample gates cannot produce headline performance-like PnL in packet/site.

1. Suppress PnL when sample gate fails.
2. Add `effective_cost_per_trade` per scenario.
3. Add units and synthetic-fill labels in every table/card.
4. Bind input manifest hash into backtest JSON.
5. Add reproducibility and zero/single/all-win/all-loss tests.

Phase E — Reproducibility and audit trail integrity

Exit criterion: Two builds on identical inputs are byte-identical for hashed artifacts; concurrent manifest appends are safe.

1. Deterministic JSON dump everywhere hashed.
2. Atomic/locked JSONL manifest append.
3. One run-level timestamp threaded through all outputs.
4. Add packet rebuild reproducibility test.
5. Add concurrent append regression.

Phase F — Operational resilience

Exit criterion: Transient source failures are retried and classified without leaking secrets or producing ambiguous failures.

1. Add retry/backoff helper for external fetches.
2. Store per-source retry count/failure class in run manifest.
3. Add secret-redacted error wrapper around credential-bearing fetchers.
4. Add tests for retryable 5xx, fail-fast 4xx, and redacted exception output.
5. Manual live run and evidence bundle verification.

Phase G — Frontend/packet polish pass

Exit criterion: Packet reads like an institutional evidence memo, not a dense internal dashboard, while preserving controls.

1. Move investor ask and executive narrative higher.
2. Increase dense-section font size and contrast.
3. Rename generic green statuses to precise labels like `source file present` or `configured-source healthy`.
4. Add a sticky/persistent top warning banner.
5. Add browser visual QA screenshots to completion proof.

---

5. Verification plan for every implementation slice

Each slice must run:

1. Focused RED/GREEN pytest for the changed behavior.
2. Full BTC Vol focused suite: `python -m pytest --tb=short -q tests/test_btc_vol*`.
3. Live/internal monitor run where applicable: `python -m institutional_btc_vol.cli run artifacts/institutional/data`.
4. Packet rebuild: `python -m institutional_btc_vol.cli build-packet artifacts/institutional/data artifacts/institutional/investor-packet`.
5. Packet verification: `python -m institutional_btc_vol.cli verify-packet artifacts/institutional/investor-packet`.
6. Static site browser QA: open `artifacts/institutional/investor-packet/site/index.html`, inspect console, visual screenshot/vision check.
7. Final ledger: Done / Not done / Blocked / Proof.

---

6. Current status after audit

Done:

- Complete dual audit run: my audit + separate Claude Opus 4.7 audit.
- Test baseline verified green: `135 passed`.
- Packet verifier baseline verified green for internal integrity.
- Browser visual audit completed.
- New master plan written here.

Implementation progress after plan creation:

- Phase A started and first verifier slice completed:
- `packet_verify.py` now reports legal-wrapper presence/approval and a separate `external_use_gate`.
- Internal packet integrity can pass while external-use readiness remains blocked.
- Verifier now enforces required packet artifact labels including legal wrapper artifacts and quote evidence.
- `investor_packet.py` now treats legal wrapper JSON/Markdown as required packet artifacts.
- Phase B started and first quote-evidence lifecycle slice completed:
- `quote_evidence.py` now requires operator/legal promotion attestation for `quote-verified` and `trade-verified` evidence.
- Quote evidence summaries now pseudonymize counterparties for rendered surfaces and set `raw_counterparties_redacted=true`.
- `quote_templates.py` now includes promotion/legal-review fields.
- `investor_site.py` renders pseudonymized counterparty labels when available.

Not done:

- Remaining Phase B state-machine depth and packet-level raw-counterparty leak verifier are not complete.
- P0.4/P0.5 and all P1/P2 fixes remain to be implemented.

Blocked / external gates:

- Licensed historical source readiness remains blocked until real licensed/replay-ready files replace skeleton placeholders.
- Quote verification remains blocked until real two-counterparty indicative quotes are captured.
- External use remains blocked until counsel approves the legal/business wrapper.
- Everything remains `SCREEN-ONLY · NOT EXECUTABLE`.

Licensed Source Acquisition Package

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/historical/licensed_source_package/licensed-source-acquisition-package-v1.md`

Licensed Historical Source Acquisition Package v1

Evidence status: `SCREEN-ONLY · LICENSED SOURCE ACQUISITION PACKAGE · NOT EXECUTABLE`

This package removes ambiguity about what is still needed. It does not promote current public/screen captures to licensed historical readiness.

Required source groups

IBIT options history

- Required fields: `available_ts, expiration, strike, option_type, bid, ask, volume, open_interest, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`
- Provider targets: OPRA historical options via broker/vendor export; ORATS/OptionMetrics/Polygon paid historical options if license permits replay/diligence use
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

Deribit options history

- Required fields: `available_ts, instrument_name, underlying_price, bid_iv, ask_iv, mark_iv, open_interest, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`
- Provider targets: Deribit historical/replay export with timestamped option book/IV fields; Kaiko/Amberdata/Tardis licensed Deribit options history
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

CME Bitcoin options history

- Required fields: `available_ts, symbol, expiration, strike, option_type, bid, ask, settlement, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`
- Provider targets: Databento CME options history export; CME DataMine/broker historical CME Bitcoin options export
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

BTC reference history

- Required fields: `available_ts, btc_usd, venue_or_index, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`

- Provider targets: CF Benchmarks/CME CF BRR/BRTI licensed reference history; Kaiko/Coin Metrics/Bloomberg BTC reference index export
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

IBIT holdings history

- Required fields: `available_ts, btc_per_share, shares_outstanding, fund_assets, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`
- Provider targets: iShares daily holdings/NAV archive under approved use; Bloomberg/refinitiv/FactSet ETF holdings history export
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

Rates and fee curves

- Required fields: `available_ts, tenor, rate, borrow_or_fee, source_ref`
- Accepted formats: `csv`, `jsonl`, `parquet`
- Provider targets: FRED/SOFR/Treasury time series with usage terms archived; Broker borrow/financing/fee curve export when available
- Status: `not-procured-for-licensed-historical-readiness`
- Next action: Request/export licensed historical file, then run build-source-intake-entry and validate-source-intake.

Operator commands after a licensed file is obtained

```
PYTHONPATH=. python3 -m institutional_btc_vol.cli build-source-intake-entry <raw-file> --
source-group "<source group>" --provenance "<provider/license>" --license-label "<license
label>" --source-ref "<vendor export id>" --output artifacts/institutional/historical/
source_entries/<entry>.json
PYTHONPATH=. python3 -m institutional_btc_vol.cli validate-source-intake artifacts/
institutional/historical/source_intake_manifest.json --output artifacts/institutional/
historical/source-intake-validation-current.md
```

Current truth: current captures are available for internal diligence, but licensed historical readiness stays blocked until these files are procured and validated.

Source Intake Skeleton Validation

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/historical/licensed_source_package/source-intake-skeleton-validation-v1.md`

Source Intake Validation Report

Evidence status: `SCREEN-ONLY · SOURCE INTAKE VALIDATION · NOT EXECUTABLE`

No executable quote, RFQ, advice, or investment readiness is implied by this report. It only validates whether historical source packages are structurally replay-ready.

- Ready: `False`
- Coverage: `0/6 source groups structurally valid`
- Blocker count: `18`

Blockers

- IBIT options history: raw_sha256 missing or invalid
- IBIT options history: row_count must be positive
- IBIT options history: available_end missing or invalid
- Deribit options history: raw_sha256 missing or invalid
- Deribit options history: row_count must be positive
- Deribit options history: available_end missing or invalid
- CME Bitcoin options history: raw_sha256 missing or invalid
- CME Bitcoin options history: row_count must be positive
- CME Bitcoin options history: available_end missing or invalid
- BTC reference history: raw_sha256 missing or invalid
- BTC reference history: row_count must be positive
- BTC reference history: available_end missing or invalid
- IBIT holdings history: raw_sha256 missing or invalid
- IBIT holdings history: row_count must be positive
- IBIT holdings history: available_end missing or invalid
- Rates and fee curves: raw_sha256 missing or invalid
- Rates and fee curves: row_count must be positive
- Rates and fee curves: available_end missing or invalid

Source Results

- IBIT options history: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label
- Deribit options history: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label
- CME Bitcoin options history: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label
- BTC reference history: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label
- IBIT holdings history: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label
- Rates and fee curves: blocked; rows=0; format=csv; provenance=replace-with-licensed-provider-or-broker-export; license=replace-with-license-or-contract-label

Source Blocker Audit

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/historical/source_audit/source_blocker_audit_current.md`

Source Intake Blocker Audit — Current Captures

- Run audited: `btcvol-20260517-162152`
- Audited at UTC: `2026-05-18T17:10:12.616640+00:00`
- Control: `SCREEN-ONLY · SOURCE AVAILABILITY AUDIT · NOT EXECUTABLE`

Finding

The production/source-intake blocker list is stale relative to current monitor artifacts. It is reading `artifacts/institutional/historical/source_intake_manifest.json`, which intentionally contains only two fixture historical sources. The repo actually has current live/source captures for most groups under `artifacts/institutional/data/raw/` and `normalized/`, but those captures are not yet converted into the strict historical licensed-source intake manifest.

Current local availability

- **IBIT options history**: found — `artifacts/institutional/data/raw/btcvol-20260517-162152/nasdaq_ibit_option_chain.json`
- sha256: `fdff135345c245b05548dc0ad44ca56183a3636d3615e647e40ccbfaa241b83c`; bytes: `186563`
- **Deribit options history**: found — `artifacts/institutional/data/raw/btcvol-20260517-162152/deribit_book_summary_btc_options.json`
- sha256: `fbb4621fb847f8e84c898d6ccd77e3a044fae9141e2b6ad118ff1b080833f738`; bytes: `533720`
- **CME Bitcoin options history raw BBO**: found — `artifacts/institutional/data/raw/btcvol-20260517-162152/databento_cme_btc_options_bbo_1m.csv`
- sha256: `4a57b6cf64294627276c09abcb8cf9785039235de6c8d9bf7900f96ff69d618e`; bytes: `646595`
- **CME Bitcoin options history normalized**: found — `artifacts/institutional/data/normalized/btcvol-20260517-162152/databento_cme_btc_options_bbo.csv`
- sha256: `c09e7e8df608abb8bbf76fc6fe6197ae8a0fdc68039cbfca782c0fc2e8515f6c`; bytes: `274518`
- **BTC reference history**: found — `artifacts/institutional/historical/source_audit/btc_reference_from_run_manifest.jsonl`
- sha256: `b46634d69256a4a6a313d564241603cb42c85e9ec237d95e825e766666e5273e`; bytes: `321`
- **IBIT holdings history official attempt**: found — `artifacts/institutional/data/raw/btcvol-20260517-162152/ishares_ibit_holdings.csv`
- sha256: `1b838d2c121f21f5a2ed8c43dab0841bc45a486a2bb8f756cdc27c637a2f5169`; bytes: `8539634`
- **IBIT holdings history cached valid CSV**: found — `artifacts/institutional/data/raw/btcvol-20260517-162152/ishares_ibit_holdings_fallback_cached.csv`
- sha256: `f7ba3f217ef58e76cd72e248c61d25c9646325252663b9f9eaeee392188ffeef`; bytes: `4749`
- **Rates and fee curves**: found — `artifacts/institutional/historical/source_audit/fred_rates_sofr_dgs1_dgs3mo_current.csv`
- sha256: `f863aa00472ca8378e06a3f8dc080606e2240c529ad2c968ba3b648e90b410b2`; bytes: `353760`

Fixability classification

- **Can fix now locally:** BTC reference file, rates/fee public reference file, manifest generation from existing captures, better report language that separates “data exists” from “readiness gate passed”.
- **Can partially fix from existing artifacts:** CME has Databento captured files in the latest run, but current environment has no `DATABENTO_API_KEY`, so it can be documented/replayed from existing raw files but not refreshed right now.
- **Cannot honestly clear as investment/client readiness yet:** IBIT and Deribit are public/screen sources, and the source-intake gate says licensed/replay-ready historical coverage. Public screen captures can support internal diligence but should not be promoted to licensed historical readiness.
- **Still externally gated:** license scope approval for redistribution/use, enough history for sample gates, and two-counterparty quote evidence.

Rates proof

Fetched FRED public CSV `SOFR,DGS1,DGS3MO` into the audit folder. Latest non-empty tail:

- `2026-05-11,3.60,3.79,3.70`
- `2026-05-12,3.60,3.80,3.70`
- `2026-05-13,3.59,3.79,3.69`
- `2026-05-14,3.56,3.79,3.69`
- `2026-05-15,3.55,,`

RFQ Verification Plan

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-rfq-verification-plan-v0.md`

Idea 1 — RFQ Verification Plan v0

INTERNAL RESEARCH DRAFT ONLY · NOT FOR EXTERNAL DISTRIBUTION · NO RFQ SUBMISSION · NO EXECUTION CAPACITY

Concept: BTC volatility intermediation desk — ETF / Deribit / CME / OTC pricing verification for treasury collars, miner production hedges, and cross-venue BTC vol spreads.

Prepared: 2026-05-14 CDT

Status: Internal verification design. Not client execution.

Legal status: Research workflow only until counsel approves RFQ/execution role. Do not send client RFQs, introduce client orders, solicit transactions, or accept transaction compensation without legal wrapper.

---

1. Purpose

The RFQ plan converts the current workstream from **screen-only thesis** into **quote-verified evidence**.

The goal is not to execute trades immediately. The goal is to determine whether the economics shown by the BTC Vol Desk Monitor survive real-world quote frictions:

- bid/ask width;
- size;
- counterparty willingness;
- margin/collateral terms;
- execution venue restrictions;
- legal onboarding;
- slippage;
- quote freshness;
- basis/wrapper mismatch;
- settlement and custody constraints.

The rule:

> No screen-only dislocation becomes an investor-facing return claim until at least two executable or indicative quotes confirm the structure after fees, collateral, and basis adjustments.

---

2. Confidence ladder

Every opportunity and product structure gets one of four execution labels.

Level 0 — Hypothesis

Definition: Economic idea only. No live data.

Example:

- “ETF puts may be richer than Deribit puts because ETF holders demand protection.”

Use:

- brainstorming;
- market-structure memo;
- target research.

Do not use:

- investor return claims;
- client term sheets;
- revenue forecasts.

Level 1 — Screen-only

Definition: Visible from public/vendor screens but not confirmed executable.

Example:

- Deribit mark IV vs Nasdaq IBIT mid-derived IV shows a 2–5 vol point difference.

Requirements:

- timestamp;
- source;
- bid/ask/mid/mark;
- size/OI/volume if available;
- calculation method;
- source-confidence label.

Use:

- research appendix;
- internal dislocation board;
- illustrative case studies.

Do not use:

- “tradable edge” language;
- expected-return claims;
- client execution recommendation.

Level 2 — Quote-verified

Definition: Counterparty or venue quote received for a defined structure, with quote timestamp, size, expiry, strike, settlement, collateral/margin terms, and quote status.

Quote types:

- indicative quote;
- firm quote with expiry window;
- broker market color;
- executable screen quote from approved account;
- dealer RFQ response.

Requirements:

- at least one quote for a defined structure;
- ideally two independent quotes;
- exact quote fields stored;
- quote age tracked;
- legal role clarified before any client-facing use.

Use:

- quote-verification appendix;
- investor evidence, carefully labeled;
- pricing sanity checks.

Do not use:

- “trade-verified” or realized economics language.

Level 3 — Trade-verified

Definition: Executed trade or firm executable quote in size within quoted window.

Requirements:

- execution confirmation or firm quote proof;
- size;
- all-in fees;
- margin/collateral impact;
- post-trade mark;

- hedge/recycling result;
- realized slippage vs model;
- compliance/approval record.

Use:

- investor performance evidence;
- operating model calibration;
- capacity estimates.

Do not use:

- broad scaling claims unless repeated across conditions.

---

3. What must be quote-verified first

Priority 1 — BTC treasury covered-call package

Why first: Most commercially intuitive product. Quotes should be easy to compare.

Standard RFQ package:

- Underlying: BTC or IBIT-equivalent BTC exposure.
- Notional: 500 BTC and 1,000 BTC scenarios.
- Tenors: 30D, 45D, 90D, 135D.
- Strikes: 15%, 20%, 25% OTM calls.
- Settlement: cash-settled BTC/USD or ETF shares depending wrapper.
- Quote: premium received, bid/ask, collateral/margin, fees.

Verification objective:

- Can a treasury monetize a 5–10% sleeve at attractive premium without unacceptable collateral or upside cap?

Priority 2 — BTC treasury put-spread collar

Why second: Better governance than tight full collars.

Standard RFQ package:

- Notional: 500 BTC and 1,000 BTC.
- Tenors: 90D and 135D.
- Put spread: buy 90% put / sell 75–80% put.
- Call-spread financing: sell 115–120% call / buy 125–140% call.
- Goal: define drawdown band, avoid unlimited upside give-up.

Verification objective:

- Can the desk create a board-friendly floor at modest cost while retaining upside above the call spread?

Priority 3 — Miner production collar

Why third: Natural recurring product with clear pain.

Standard RFQ package:

- Client exposure: 3-month conservative production.
- Notional: 100 BTC, 225 BTC, 500 BTC.
- Tenors: 45D, 90D, 135D.
- Structure: buy 80–90% put, sell 115–125% call.
- Variants: put-spread collar and outright disaster put.

Verification objective:

- Can miners protect runway at a cost that is cheaper than emergency equity/dilution or forced spot liquidation?

Priority 4 — ETF vs Deribit / CME vol spread package

Why fourth: This is the intellectual alpha, but it needs stronger proof.

Standard RFQ package:

- Compare equivalent expiry/strike/moneyness across:
- IBIT options;
- Deribit BTC options;
- CME BTC options when data/access available;
- OTC dealer quote.
- Structures:
- ATM straddle spread;
- 25-delta risk reversal;
- 90/110 collar package;
- calendar spread.

Verification objective:

- Does the screen-only IV gap survive all-in execution, hedging, margin, collateral, fees, and wrapper/basis adjustments?

---

4. Counterparty categories

A. Listed ETF options route

Use for:

- IBIT/spot BTC ETF covered calls;
- ETF collars;
- ETF-option wealth overlays;
- comparison against Deribit/CME.

Possible sources:

- listed options broker;
- prime broker;
- institutional options desk;
- OPRA/vendor data;
- OCC series validation.

Required fields:

- ETF ticker;
- expiry;
- strike;
- bid/ask;
- size at bid/ask;
- implied vol;
- delta/gamma/vega/theta;
- open interest/volume;
- position limits;
- assignment/exercise mechanics;
- margin requirement;
- commission/fees;
- timestamp.

Production data caveat:

Nasdaq public JSON is research-use only. Production should use OPRA/vendor/broker data.

B. CME route

Use for:

- regulated futures/options overlay;
- institutional funds;
- miner/treasury clients preferring FCM route;
- basis/vol package comparison.

Possible sources:

- FCM;
- broker desk;
- CME DataMine/API/vendor;
- terminal export.

Required fields:

- futures contract reference;
- option contract;
- expiry;
- strike;
- bid/ask;
- futures price/basis;
- implied vol;
- contract multiplier;
- initial/maintenance margin;
- block liquidity;
- fees;
- position/accountability limits;
- settlement method;
- timestamp.

Caveat:

CME public pages are not enough for production quotes.

C. Deribit / crypto-native route

Use for:

- BTC options surface reference;
- crypto-native hedge/recycling;
- non-US sophisticated clients where legally permitted;
- internal screen monitor.

Possible sources:

- Deribit public API;
- Deribit RFQ/block if account eligible;
- crypto options market makers;
- prime/custody-integrated access provider.

Required fields:

- instrument name;

- expiry;
- strike;
- call/put;
- bid/ask/mark;
- mark IV;
- Greeks;
- open interest/volume;
- underlying/index price;
- settlement currency;
- margin requirement;
- collateral currency;
- liquidation/margin model;
- counterparty/venue restrictions;
- timestamp.

Caveat:

Deribit may not be usable for US clients. Treat as pricing/hedge venue only after legal review.

D. OTC/RFQ route

Use for:

- bespoke treasury/miner collars;
- block size;
- custom maturities;
- public-company friendly term sheets;
- structured-product hedges.

Possible sources:

- crypto OTC options desks;
- bank derivatives desks if available;
- prime brokers;
- market makers;
- structured-products issuers.

Required fields:

- counterparty;
- quote type: indicative vs firm;
- quote expiry time;
- structure legs;
- notional;

- premium/price;
- bid/offer spread;
- collateral/CSA terms;
- eligible collateral;
- haircuts;
- margin call timing;
- settlement method;
- documentation required;
- onboarding timeline;
- minimum ticket size;
- legal eligibility requirements;
- valuation agent;
- closeout/disruption terms.

Caveat:

OTC economics are hypothesis until quotes are received.

---

5. Standard RFQ template — treasury covered call

RFQ header

- RFQ ID:
- Date/time/timezone:
- Requestor:
- Counterparty:
- Quote type requested: indicative / firm
- Quote good-until:
- Legal capacity: internal research only / client-facing approved / principal
- Confidentiality level:

Client/exposure assumptions

- Exposure type: direct BTC / ETF shares / futures / mixed
- Total exposure:
- Hedge sleeve:
- Custody/venue restrictions:
- Settlement preference: cash / physical / ETF shares
- Collateral currency: USD / BTC / T-bills / stablecoin / other

Structure

- Product: covered call
- Underlying/reference:
- Notional:
- Expiry:
- Strike:
- Moneyness:
- American/European:
- Settlement index/source:
- Premium currency:

Quote response fields

- Bid premium:
- Ask premium:
- Mid/indicative:
- Implied vol:
- Delta:
- Gamma:
- Vega:
- Theta:
- Upfront premium:
- Fees/commission:
- Margin/collateral requirement:
- Minimum ticket:
- Size available:
- Quote timestamp:
- Quote expiry:
- Notes:

Verification notes

- Screen model value:
- Difference vs screen:
- All-in cost/spread:
- Execution confidence:
- Source confidence:
- Follow-up required:

---

6. Standard RFQ template — treasury put-spread collar

Structure

- Product: put-spread collar
- Underlying/reference:
- Notional:
- Expiry:
- Buy put strike:
- Sell put strike:
- Sell call strike:
- Buy call strike:
- Target net premium: zero-cost / debit / credit
- Settlement:
- Premium currency:

Quote response fields

- Net premium/debit/credit:
- Each leg bid/ask:
- Net implied vol by leg:
- Package delta/gamma/vega/theta:
- Max protection:
- Max upside give-up:
- Collateral/margin:
- Fees:
- Size available:
- Quote good-until:
- Valuation source:
- Disruption/fallback notes if OTC:

Verification notes

- Cost as % of protected notional:
- Floor level:
- Cap/sold band:
- Scenario P&L at BTC -20%, -30%, +20%, +50%:
- Fit for board package: yes/no:
- Main legal/operational blockers:

---

7. Standard RFQ template — miner production hedge

Client/exposure assumptions

- Miner:
- Monthly conservative production:
- Hedge horizon:
- Total forecast production:
- Hedge ratio:
- Hedge BTC:
- Existing BTC inventory:
- Pledged/restricted BTC:
- Power/capex/debt dates:
- Settlement preference:
- Approved venues/counterparties:

Structure options requested

Request quotes for three alternatives:

1. **Outright disaster put**

- Notional:
- Expiry:
- Strike: 80–85% of spot

2. **Production collar**

- Buy put: 80–90% of spot
- Sell call: 115–125% of spot

3. **Put-spread collar**

- Buy put: 90% of spot
- Sell put: 75–80% of spot
- Sell call: 115–120% of spot
- Buy call: 125–140% of spot

Quote response fields

- Net premium by structure:
- Cost as % of hedged notional:
- Initial/variation margin:
- Eligible collateral:
- Size available:
- Minimum ticket:
- Settlement/index:
- Quote validity:

- Documentation/onboarding required:
- Operational constraints:

Verification notes

- Runway protection at BTC -20%, -30%, -40%:
- Premium vs emergency equity/dilution cost:
- Lender/covenant compatibility:
- Overhedging risk:
- Board fit:

---

8. Standard RFQ template — cross-venue vol spread

Structure

- Product: cross-venue vol spread
- Dislocation type: wrapper / skew / calendar / basis / margin / flow
- Long leg:
- Short leg:
- Underlying conversion:
- Expiry matching:
- Strike/moneyness matching:
- Delta hedge:
- Basis hedge:
- Notional:

Quote response fields

For each leg:

- venue;
- instrument;
- bid/ask;
- IV;
- Greeks;
- size;
- fees;
- margin/collateral;
- settlement;
- execution timestamp.

Package fields:

- gross IV spread;
- premium spread;
- delta-neutral cost;
- basis adjustment;
- financing/collateral adjustment;
- expected slippage;
- net spread;
- capacity;
- stress/basis risk.

Verification notes

- Does the screen IV gap survive all-in costs?
- Which leg is hard to execute?
- Which venue creates margin/collateral drag?
- Does basis/wrapper risk dominate?
- Is the spread client-useful or only prop-tradable?

---

9. Quote storage schema

Every quote should be stored with this schema.

```
quote_id:
created_at:
requested_by:
counterparty:
quote_type: indicative | firm | executable_screen | trade
source_confidence:
execution_confidence:
legal_capacity: internal_research_only | counsel_review_required |
external_partner_possible_later
client_archetype: treasury | miner | fund | family_office | structured_product | internal
product_type: covered_call | collar | put_spread_collar | miner_hedge | vol_spread |
structured_note
underlying:
venue:
notional:
expiry:
legs:
  - side:
    type:
    strike:
    quantity:
    bid:
    ask:
    mid:
    iv:
    delta:
    gamma:
    vega:
    theta:
premium_currency:
net_premium:
fees:
margin_initial:
margin_maintenance:
collateral_currency:
```

```
eligible_collateral:
settlement_method:
quote_timestamp:
quote_good_until:
size_available:
minimum_ticket:
documentation_required:
legal_notes:
operational_notes:
model_value:
difference_vs_model:
all_in_cost:
scenario_pnl:
verification_status:
follow_up:
```

---

10. Quote comparison rules

Rule 1 — never compare IV alone

Compare:

- premium;
- IV;
- delta/vega;
- expiry;
- forward/basis;
- collateral cost;
- fees;
- settlement;
- size;
- legal usability.

Rule 2 — normalize to BTC-equivalent exposure

For ETF options:

- convert ETF shares to BTC equivalent using official ETF holdings/NAV data.
- account for ETF expense drag, tracking, and share settlement.

For CME:

- account for futures basis and contract multiplier.

For Deribit:

- account for index, settlement currency, margin/collateral, and jurisdiction.

For OTC:

- account for CSA, credit, valuation, disruption clauses, and documentation cost.

Rule 3 — all-in economics beat screen marks

A trade is attractive only if it survives:

- bid/ask;
- slippage;
- fees;
- financing;
- collateral drag;
- hedging cost;
- tax/accounting friction if relevant;
- legal/operational feasibility.

Rule 4 — quote age matters

For BTC options:

- <5 minutes: fresh;
- 5–30 minutes: stale risk;
- >30 minutes: market color only unless quiet market and counterparty confirms.

Rule 5 — separate client product from hedge venue

A treasury client may need OTC or CME governance, while the desk may observe cheaper hedge risk on Deribit. Do not assume client can use the hedge venue directly.

---

11. Verification workflow

Step 1 — screen candidate

From BTC Vol Desk Monitor:

- identify structure;
- record screen prices;
- calculate model economics;
- label screen-only.

Step 2 — quote request package

Prepare RFQ with:

- exact legs;
- notional;
- expiry;
- settlement;
- collateral preference;
- quote type requested;

- no client name unless legally approved.

Step 3 — receive quote

Record:

- price;
- size;
- quote age;
- margin/collateral;
- fees;
- constraints;
- whether firm or indicative.

Step 4 — compare to model

Calculate:

- gross edge;
- all-in edge;
- collateral impact;
- basis/wrapper adjustment;
- execution confidence.

Step 5 — decision label

Assign:

- reject — no edge;
- research — keep watching;
- quote-verified — worth investor evidence;
- client prototype — if legal and product fit;
- trade candidate — only if legal/risk approval.

Step 6 — archive

Store quote and reasoning. Do not rely on memory.

---

12. Minimum quote set before investor claims

Before any investor memo claims “there is executable edge,” collect at least:

Treasury products

- 2 quotes for 30–45D covered calls.
- 2 quotes for 90–135D covered calls.

- 2 quotes for 90–135D put-spread collars.
- At least one ETF/options route and one OTC/crypto-native route if possible.

Miner products

- 2 quotes for 3-month production collars.
- 2 quotes for disaster puts.
- 2 quotes for put-spread collars.

Cross-venue RV

- 5–10 repeated observations where screen spread is followed by quote verification.
- At least 2 venues per package.
- Evidence that net edge survives all-in costs.
- Capacity estimate.

CME integration

- At least one licensed/vendor/broker source for CME chain and margin.
- Do not use CME in return claims from public-page data alone.

---

13. Counterparty due diligence checklist

For each potential counterparty:

- legal entity;
- regulatory status;
- jurisdiction;
- approved products;
- minimum ticket size;
- supported collateral;
- custody/settlement process;
- ISDA/CSA availability;
- FCM/broker relationship;
- credit support/parent guarantee;
- valuation agent process;
- dispute process;
- fees and markups;
- quote response speed;
- stress-market performance;
- conflicts;
- sanctions/AML/KYC process.

---

14. What not to do

Do not:

- send client-specific RFQs before legal approval;
- accept transaction compensation before legal approval;
- call screen-only data executable;
- use stale quotes as live pricing;
- compare Deribit IV to IBIT IV without wrapper/basis/margin adjustment;
- imply Deribit is available to US clients without legal review;
- quote a public company structure without board/accounting/covenant diligence;
- build investor return claims from one-day screen snapshots.

---

15. First 10 RFQs to design internally

These are internal templates only until counsel approves outreach.

1. 500 BTC, 45D, 15% OTM covered call.
2. 500 BTC, 45D, 25% OTM covered call.
3. 1,000 BTC, 135D, 20% OTM covered call.
4. 500 BTC, 135D, 90/80 put spread funded by 115/130 call spread.
5. 1,000 BTC, 135D, 90/80 put spread funded by 115/130 call spread.
6. Miner 225 BTC, 90D, 85% floor / 120% cap collar.
7. Miner 225 BTC, 90D, 80–85% outright disaster put.
8. Miner 225 BTC, 135D, 90/75 put spread funded by 115/130 call spread.
9. IBIT vs Deribit matched 30D ATM straddle package.
10. IBIT vs Deribit matched 25-delta risk reversal package.

---

16. Bottom line

The RFQ workflow is the bridge from story to proof.

Current status:

- Product thesis: developed.
- Data map: developed.
- Screen monitor: developed.
- Treasury/miner case studies: developed.

- Legal perimeter: developed.
- RFQ workflow: now defined.

Next proof milestone:

> Convert the first 10 internal RFQ templates into a quote-verification board, then collect quotes only after legal role/partner path is approved.

RFQ Verification Board

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-rfq-verification-board-v0.md`

Idea 1 — RFQ Verification Board v0

INTERNAL RESEARCH DRAFT ONLY · NOT FOR EXTERNAL DISTRIBUTION · NO RFQ SUBMISSION · NO EXECUTION CAPACITY

Purpose: Internal board for converting screen-only BTC treasury/miner/cross-venue structures into quote-verified evidence.

Status: Internal planning only. Do not send client-specific RFQs, solicit transactions, introduce orders, or accept transaction compensation until counsel approves the legal wrapper/partner model.

Confidence ladder: hypothesis → screen-only → quote-verified → trade-verified.

---

1. Board rules

1. Every row starts as **screen-only** unless a real quote is attached.
2. Every quote must include timestamp, good-until, quote type, counterparty, size, margin/collateral, fees, and legal capacity.
3. Two independent quotes are preferred before using a structure as investor evidence.
4. No return claim is allowed from screen-only data.
5. CME rows require licensed/vendor/broker source before production use.
6. Deribit rows require jurisdiction/legal review before client-use assumption.
7. ETF rows require OPRA/vendor/broker data before production use; Nasdaq public JSON is research-only.

---

2. Quote status legend

- **Not prepared:** structure not specified enough for RFQ.
- **Template ready:** exact RFQ terms drafted; no outreach.
- **Legal pending:** ready technically, but counsel/partner approval required before sending.
- **Sent:** RFQ sent under approved legal capacity.
- **Indicative quote received:** non-firm market color.
- **Firm quote received:** executable or time-limited firm quote.
- **Trade executed:** executed/confirmed trade.
- **Rejected:** quote did not survive all-in economics or legal/operational filter.

Current board starts at **Template ready / Legal pending**.

---

3. First 10 RFQ board

ID	Product	Client use case	Notional	Tenor
RFQ-001	Treasury covered call	Income sleeve	500 BTC	45D
RFQ-002	Treasury covered call	Upside-preserving income	500 BTC	45D
RFQ-003	Treasury covered call	Quarterly income	1,000 BTC	135D
RFQ-004	Treasury put-spread collar	Board downside policy	500 BTC	135D
RFQ-005	Treasury put-spread collar	Larger treasury sleeve	1,000 BTC	135D
RFQ-006	Miner production collar	Runway hedge	225 BTC	90D
RFQ-007	Miner disaster put	Severe downside hedge	225 BTC	90D
RFQ-008	Miner put-spread collar	Quarterly production hedge	225 BTC	135D
RFQ-009	Cross-venue straddle	RV / hedge recycling	BTC-equivalent TBD	30D
RFQ-010	Cross-venue risk reversal	Skew comparison	BTC-equivalent TBD	30D

---

4. Detailed RFQ cards

RFQ-001 — 500 BTC, 45D, 15% OTM treasury covered call

Objective: Verify premium available for a board-approved income sleeve with moderate upside cap.

Client archetype: BTC treasury company.

Terms:

- Exposure: 500 BTC-equivalent.
- Tenor: approximately 45 days.
- Strike: 115% of spot.
- Structure: sell call, fully covered by BTC/ETF exposure.
- Settlement: request alternatives — cash-settled BTC/USD, ETF-share-settled, or OTC cash-settled.
- Quote type: indicative first; firm quote later.

Fields required:

- premium bid/offer;
- IV;
- delta/vega;
- size available;
- collateral/margin;
- fees;
- assignment/exercise mechanics;
- quote good-until;
- documentation/onboarding.

Decision rule:

Advance if annualized premium remains attractive after all costs and cap is acceptable under policy.

RFQ-002 — 500 BTC, 45D, 25% OTM treasury covered call

Objective: Verify upside-preserving income alternative.

Client archetype: BTC treasury company with high upside sensitivity.

Terms:

- Exposure: 500 BTC-equivalent.
- Tenor: approximately 45 days.
- Strike: 125% of spot.
- Structure: sell covered call.

Fields required: same as RFQ-001.

Decision rule:

Advance if premium is still meaningful while upside cap is board-friendly.

RFQ-003 — 1,000 BTC, 135D, 20% OTM treasury covered call

Objective: Verify longer-tenor premium and capacity for larger BTC treasury sleeve.

Client archetype: BTC treasury company with quarterly/semiannual liquidity objective.

Terms:

- Exposure: 1,000 BTC-equivalent.
- Tenor: approximately 135 days.
- Strike: 120% of spot.

- Structure: sell covered call.

Fields required:

- package price for full size;
- partial-fill/layered execution price;
- collateral/margin;
- OTC vs listed economics;
- capacity discount vs 500 BTC;
- quote good-until.

Decision rule:

Advance if size does not degrade economics enough to make smaller staggered clips clearly superior.

RFQ-004 — 500 BTC, 135D treasury put-spread collar

Objective: Verify board-friendly downside band with bounded upside give-up.

Client archetype: BTC treasury company needing downside floor but reluctant to fully cap upside.

Terms:

- Exposure: 500 BTC-equivalent.
- Tenor: approximately 135 days.
- Buy put: 90% of spot.
- Sell put: 80% of spot.
- Sell call: 115% of spot.
- Buy call: 130% of spot.
- Target: low debit, zero-cost, or modest credit.

Fields required:

- net package premium;
- leg-by-leg prices;
- package Greeks;
- max protected band;
- max upside band sold;
- collateral/margin;
- execution as package vs legs;
- quote good-until.

Decision rule:

Advance if the structure creates meaningful protection with acceptable upside give-up and collateral burden.

RFQ-005 — 1,000 BTC, 135D treasury put-spread collar

Objective: Verify whether larger notional can be priced without excessive slippage.

Client archetype: larger BTC treasury company.

Terms: same as RFQ-004 but 1,000 BTC-equivalent.

Additional required fields:

- size availability;
- market impact estimate;
- staged execution alternative;
- block/RFQ minimums;
- OTC vs listed package comparison.

Decision rule:

Advance if size economics remain close to 500 BTC quote or if staged execution solves capacity.

RFQ-006 — Miner 225 BTC, 90D, 85% floor / 120% cap collar

Objective: Verify cost of a practical quarterly production hedge.

Client archetype: miner with 450 BTC conservative 3-month production and 50% hedge ratio.

Terms:

- Hedge notional: 225 BTC.
- Tenor: approximately 90 days.
- Buy put: 85% of spot.
- Sell call: 120% of spot.
- Structure: cash-settled collar preferred.

Fields required:

- net premium;
- margin/collateral;
- settlement/index;
- size available;
- documentation;
- lender/collateral implications;
- stress payoff at BTC -20%, -30%, -40%.

Decision rule:

Advance if cost is lower than credible dilution/forced-sale risk and collateral is manageable.

RFQ-007 — Miner 225 BTC, 90D, 80–85% disaster put

Objective: Verify standalone downside insurance cost.

Client archetype: miner wanting no upside cap.

Terms:

- Hedge notional: 225 BTC.
- Tenor: approximately 90 days.
- Buy put: request both 80% and 85% strikes.
- Settlement: cash-settled preferred.

Fields required:

- premium for 80% and 85% puts;
- delta/vega;
- liquidity/size;
- margin/collateral;
- stress payoff;
- quote good-until.

Decision rule:

Advance if premium is affordable relative to runway protected and no collateral issue emerges.

RFQ-008 — Miner 225 BTC, 135D, put-spread collar

Objective: Verify longer-dated runway hedge with lower upfront cost.

Client archetype: miner with debt/capex/refi window.

Terms:

- Hedge notional: 225 BTC.
- Tenor: approximately 135 days.
- Buy put: 90% of spot.
- Sell put: 75% of spot.
- Sell call: 115% of spot.
- Buy call: 130% of spot.

Fields required: same as RFQ-006 plus leg-by-leg package price.

Decision rule:

Advance if protected band is meaningful and upside give-up is acceptable for a specific liability window.

RFQ-009 — IBIT vs Deribit 30D ATM straddle package

Objective: Test whether screen-level ETF/Deribit ATM IV differences survive all-in costs.

Client archetype: internal RV / hedge-recycling engine, not first client product.

Terms:

- Tenor: closest common 30D expiry.
- Long leg: cheaper ATM straddle.
- Short leg: richer ATM straddle.
- Notional: BTC-equivalent matched.
- Delta hedge: define method and cost.
- ETF normalization: use official BTC/share.

Fields required:

- IBIT ATM call/put bid/ask;
- Deribit ATM call/put bid/ask;
- IV by leg;
- size available;
- fees;
- ETF share/BTC conversion;
- delta hedge cost;
- margin/collateral;
- basis/settlement mismatch;
- borrow/assignment issues.

Decision rule:

Advance only if net vol/premium spread survives all-in cost and wrapper basis risk is acceptable.

RFQ-010 — IBIT vs Deribit 30D 25-delta risk reversal

Objective: Test whether skew differences are actionable after normalization.

Client archetype: internal RV / hedge-recycling engine; possibly useful for pricing collars.

Terms:

- Tenor: closest common 30D expiry.
- Structure: 25-delta put vs 25-delta call risk reversal.
- Compare IBIT and Deribit.
- Notional: BTC-equivalent matched.

Fields required:

- 25D put/call strikes by venue;
- bid/ask;
- IV;
- deltas;
- package price;
- size;
- fees;
- margin/collateral;
- settlement/basis risks;
- quote timestamp.

Decision rule:

Advance only if skew differential survives all-in cost and is repeatable across observations.

---

5. Quote capture fields

Every quote response should be entered using this structure:

```
rfq_id:
quote_id:
created_at_cst:
counterparty:
contact:
legal_capacity: internal_research_only | counsel_review_required |
external_partner_possible_later
quote_type: indicative | firm | executable_screen | trade
quote_status:
product:
client_archetype:
underlying_reference:
notional_btc_equivalent:
tenor_days:
expiry:
spot_reference:
legs:
  - side:
    option_type:
    strike_percent_of_spot:
    strike_absolute:
    quantity:
    bid:
    ask:
    mid:
    iv:
    delta:
    gamma:
    vega:
```

```
        theta:
    net_premium:
    premium_currency:
    fees:
    margin_initial:
    margin_maintenance:
    collateral_currency:
    eligible_collateral:
    settlement_method:
    quote_timestamp_cst:
    quote_good_until_cst:
    size_available:
    minimum_ticket:
    documentation_required:
    model_value:
    difference_vs_screen:
    all_in_cost:
    scenario_pnl:
    source_confidence:
    execution_confidence:
    legal_notes:
    operational_notes:
    decision:
    follow_up:
```

---

6. Current missing inputs before live quote collection

Legal / role

- Counsel-approved role for RFQ outreach.
- Compensation model decision.
- Whether outreach is internal market color, client-facing, principal, or partner-led.

Counterparties

- Approved ETF options broker route.
- Approved FCM/CME route.
- Approved OTC options desk route.
- Approved crypto-native/Deribit route.
- Approved structured-product issuer route if relevant.

Data

- Current spot reference source.
- Production ETF options source, ideally OPRA/vendor/broker.
- CME source.
- Deribit account/RFQ/block access status if used.
- BTC/share normalization feed.

Operations

- Quote repository location.
- Versioning/audit trail.
- Person responsible for quote capture.
- Standard timestamp/timezone.

- Archive of raw counterparty responses.

---

7. Decision matrix

Decision	Condition
Reject	quote worse than screen after costs, poor liquidity, legal issue, or unacceptable collateral
Watch	screen interest remains but quote weak/stale/incomplete
Requote	quote missing fields, stale, or inconsistent with market
Quote-verified	quote complete enough to evidence market economics
Client prototype	legal wrapper approved and product fits client policy
Trade candidate	risk/legal/counterparty approval plus executable quote

---

8. First milestone

The first quote-verification milestone is **not execution**.

It is:

> Complete indicative/firm quote records for RFQ-001 through RFQ-008 from at least two legally approved quote sources, plus RFQ-009 and RFQ-010 from at least one ETF-options source and one Deribit/crypto-native source.

After that, update:

- treasury case study from screen-only to quote-verified;
- miner case study from screen-only to quote-verified;
- investor deck evidence slide;
- business-model economics;
- capacity estimates.

---

9. Bottom line

This board keeps the business honest.

The next proof threshold is not another memo. It is quote evidence. Until quotes are collected under an approved legal role, the current economics remain **screen-only illustrative research**.

Launch Model Decision Matrix

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-launch-model-decision-matrix-v0.md`

Idea 1 Launch Model Decision Matrix v0

Concept: BTC Volatility Intermediation Desk

Purpose: Decide the legally and commercially sane launch model before outreach, RFQs, execution, or fundraising claims.

Status: Strategic decision tool. Not legal advice. Requires derivatives/securities counsel review before implementation.

---

1. Executive conclusion

The best launch model is likely a **sequenced hybrid**, not a single all-in structure.

Recommended path:

1. **Start as analytics/research + proof package** — lowest legal friction, builds evidence.
2. **Add counsel-approved advisory/structuring** — board-ready treasury/miner packages.
3. **Use partner-led execution/RFQ** — avoid premature broker/IB/FCM/BD burden.
4. **Consider separate prop/fund sleeve later** — only after data + quotes prove edge.
5. **Treat structured products as phase 2/3** — high complexity, partner-led if pursued.

Do **not** start by claiming to be a live execution desk, broker, OTC dealer, or structured-product issuer.

---

2. Launch options compared

Model	What it sells	Legal friction	Revenue potential	Speed to market
A. Analytics/research company	BTC Vol Desk Monitor, market research, generic case studies	Low/medium	Low/medium	Fast
B. Advisory/structuring shop	Board-ready treasury/miner hedging policy and structure design	Medium/high	Medium	Medium
C. Partner-led RFQ/execution	Quotes/trades through registered broker/FCM/OTC partner	Medium/high	Medium/high	Medium

Model	What it sells	Legal friction	Revenue potential	Speed to market
D. Principal prop/fund vehicle	Own-account RV/vol trading	High	High but volatile	Medium/slow
E. OTC derivatives dealer	Client-facing counterparty/market maker	Very high	High	Slow
F. Structured-product platform	Notes/autocallables/income products	Very high	High	Slow
G. Software/SaaS only	Dashboard/calculators/API	Low/medium	Medium	Fast

---

3. Option A — Analytics/research company

Description

Build and sell/release the BTC Vol Desk Monitor, data appendices, generic treasury/miner research, and screen-only dislocation reports.

Products

- daily/weekly BTC Vol Desk Monitor;
- cross-venue IV/skew/calendar reports;
- BTC treasury/miner generic hedge research;
- market-structure notes;
- dashboard/API later.

Pros

- fastest to launch;
- lowest regulatory burden if kept generic;
- builds credibility and historical evidence;
- supports all other models;
- useful to investors, treasuries, miners, funds, and counterparties.

Cons

- lower near-term revenue;
- can become “just research” if not tied to client pain;
- not enough moat if data becomes commoditized;
- cannot claim execution economics.

Legal gates

- Avoid tailored advice.
- Avoid transaction recommendations.
- Avoid performance/return claims from screen-only data.

- Review CTA/RIA implications if paid/subscription advice covers commodity interests or securities.

Revenue

- subscription;
- enterprise dashboard;
- custom research under legal review;
- sponsored/partner research only with conflict controls.

Decision

Yes — start here. This is the evidence spine and the safest MVP.

---

4. Option B — Advisory/structuring shop

Description

Provide company-specific treasury/miner hedge design, board policies, scenario analysis, and structure recommendations.

Products

- BTC treasury board policy;
- covered-call/collar program design;
- miner production hedge program;
- accounting/disclosure/covenant checklist;
- RFQ package preparation;
- board/investor education.

Pros

- strongest commercial wedge;
- maps directly to pain;
- higher revenue than research;
- creates sticky relationships;
- generates future RFQ/execution flow.

Cons

- legal perimeter is materially heavier;
- may trigger CTA/RIA issues depending instruments and compensation;
- public-company advice requires counsel/auditor/lender coordination;
- cannot casually recommend specific derivatives.

Legal gates

- CTA/RIA analysis.
- Public-company governance/disclosure controls.

- Compensation model review.
- Communications policy.
- No transaction compensation unless approved.

Revenue

- advisory retainer;
- fixed-fee board package;
- implementation planning fee;
- legal/counterparty partner referral only if counsel-approved.

Decision

Yes, but only after counsel-approved scope. This is the best commercial wedge.

---

5. Option C — Partner-led RFQ/execution model

Description

Use registered/approved brokers, FCMs, OTC desks, swap dealers, prime brokers, or structured-product issuers for quotes and execution while the platform provides analytics/structuring.

Products

- quote-verification board;
- best-efforts RFQ comparison under partner model;
- broker/FCM/OTC introductions if legally approved;
- execution support handled by registered counterparty.

Pros

- practical path to quote verification;
- avoids premature broker/FCM/BD buildout;
- gives clients real quotes;
- supports investor evidence;
- can evolve into revenue-sharing if legally approved.

Cons

- lower economics than owning execution;
- dependency on partners;
- potential conflict/disclosure issues;
- transaction compensation risk if not structured correctly.

Legal gates

- No order solicitation unless approved.
- No transaction-based comp unless counsel-approved.

- Partner agreements.
- Client disclosures.
- Role clarity: analytics provider, adviser, solicitor, broker, IB, or agent?

Revenue

- partner referral/revenue share only if approved;
- advisory fees separate from execution;
- data/subscription;
- eventually execution economics via registered partner.

Decision

Preferred execution route for phase 1/2. Do not build own execution desk first.

---

6. Option D — Principal prop/fund vehicle

Description

Trade own capital or pooled capital in cross-venue BTC vol, skew, basis, and calendar dislocations.

Products

- internal prop sleeve;
- private fund;
- managed account;
- risk-recycling vehicle;
- hedge book for client-flow residuals later.

Pros

- upside if edge is real;
- validates analytics with P&L;
- can monetize dislocations without client-advisory complexity;
- useful for hedge recycling later.

Cons

- requires capital, risk infrastructure, drawdown tolerance;
- fund launch triggers CPO/CTA/RIA/private fund issues;
- prop returns are volatile and capacity-constrained;
- can distract from stronger client-flow wedge.

Legal gates

- Own-account trading vs pooled capital distinction.
- CPO/CTA/RIA analysis for fund/managed accounts.

• Exchange/FCM/custody/margin setup.
• Risk policy and valuation controls.

Revenue

• prop P&L;
• fund management/performance fees;
• risk warehousing economics later.

Decision

Later and separate. Do not lead with this until quote/trade evidence exists.

---

7. Option E — OTC derivatives dealer / market maker

Description

Act as counterparty to clients in BTC options/swaps/collars and hedge risk across venues.

Pros

• highest strategic control;
• captures spread;
• direct risk intermediation;
• aligns with long-term desk vision.

Cons

• very high legal/regulatory burden;
• capital-intensive;
• swap dealer/dealer analysis;
• ISDA/CSA/collateral operations;
• public-company counterparty diligence;
• risk of catastrophic controls failure.

Legal gates

• swap dealer / broker-dealer / commodity option analysis;
• ECP/client eligibility;
• ISDA/CSA;
• margin/collateral/custody;
• counterparty credit;
• market conduct/compliance.

Revenue

• bid/ask spread;

- structuring margin;
- hedge-recycling P&L.

Decision

No for day one. Long-term optionality only.

---

8. Option F — Structured-product platform

Description

Create BTC-linked notes, reverse convertibles, autocallables, buffered notes, defined-outcome products, or yield products.

Pros

- large wealth/private-bank market;
- packaged product economics;
- scalable if distribution exists;
- natural buyer base for BTC yield/protection.

Cons

- securities law heavy;
- issuer credit/risk disclosure;
- product committee and suitability requirements;
- FINRA/Reg BI/RIA issues;
- tax and secondary valuation complexity.

Legal gates

- registered offering or exemption;
- broker-dealer/FINRA/RIA;
- product approval;
- disclosure;
- issuer/hedge documentation;
- suitability/client eligibility.

Revenue

- structuring margin;
- issuer/distributor economics;
- hedge spread;
- advisory fee.

Decision

Not day one. Partner-led later if quote-verified demand exists.

---

9. Option G — Software/SaaS only

Description

Pure dashboard/API/calculator company for BTC vol surfaces, treasury/miner scenario analysis, and RFQ recordkeeping.

Pros

- clean productization;
- scalable;
- lower legal risk than advice/execution;
- useful for clients and partners;
- creates data moat if history and quote evidence accumulate.

Cons

- may under-monetize high-value structuring;
- harder to sell without advisory context;
- requires engineering/product support;
- can be copied if not paired with workflow/relationships.

Revenue

- SaaS subscription;
- enterprise licenses;
- API access;
- dashboard seats.

Decision

Useful infrastructure layer, not the full business. Build as internal tool first.

---

10. Recommended sequenced launch

Phase 1 — Evidence company

Duration: 0–30 days

Model: analytics/research/internal proof package

Build:

- daily BTC Vol Desk Monitor;
- treasury/miner calculators;
- legal perimeter;

- RFQ board;
- investor deck;
- data history.

Revenue: none or research/pre-commercial.

Goal: prove the package is credible.

Phase 2 — Counsel-approved advisory wedge

Duration: 30–90 days

Model: advisory/structuring within approved scope

Build:

- board-ready BTC treasury package;
- miner hedge package;
- counsel-reviewed communications;
- pilot-client discovery;
- counterparty map.

Revenue:

- fixed-fee advisory;
- retainer;
- research subscription.

Goal: validate client demand without execution overreach.

Phase 3 — Partner-led quote verification

Duration: 60–120 days

Model: partner-led RFQ/execution evidence

Build:

- legally approved quote process;
- two independent quotes per key structure;
- quote repository;
- updated case studies;
- execution partner workflow.

Revenue:

- advisory fee;

- partner economics only if approved;
- enterprise subscription.

Goal: turn screen-only economics into quote-verified evidence.

Phase 4 — Risk/revenue expansion

Duration: 120+ days

Model: choose based on evidence

Options:

- prop sleeve;
- private fund;
- execution partnership;
- OTC desk partnership;
- structured-product issuer partnership;
- SaaS product.

Goal: scale only where quote/trade evidence and legal structure support it.

---

11. Decision criteria

Choose analytics-first if:

- legal budget is limited;
- data pipeline still needs hardening;
- no quote evidence yet;
- investor materials need credibility;
- client access is preliminary.

Add advisory if:

- counsel approves scope;
- treasury/miner packages are board-ready;
- clients want specific help;
- compensation avoids transaction-risk issues.

Add partner execution if:

- registered/approved counterparties are available;
- RFQ process is legally clean;
- quote evidence is needed;
- clients need implementation path.

Add fund/prop if:

- repeated quote/trade evidence exists;
- risk-adjusted edge is durable;
- capital and risk controls exist;
- strategy is capacity-aware.

Add structured products if:

- distribution partner exists;
- product approval is feasible;
- suitability/disclosure can be handled;
- hedge economics are quote-verified.

---

12. Recommended near-term answer

If forced to choose today:

> Launch as an analytics + board-level structuring platform, with legal review for advisory scope and partner-led RFQ/execution, while keeping prop/fund and structured-product paths as later optionality.

Do not raise or sell it as:

- a live arbitrage fund;
- a broker;
- a swap dealer;
- a structured-note issuer;
- a client execution desk.

Do raise/sell it as:

- evidence-backed BTC vol market intelligence;
- treasury/miner hedge design;
- quote-verification workflow;
- future risk-intermediation platform.

---

13. Next required decisions

1. Who is the initial legal counsel for CTA/RIA/BD/IB/FCM/swap/structured-product review?
2. Is the first commercial product paid research or advisory retainer?
3. Are there existing broker/FCM/OTC relationships that can provide legally clean quotes?
4. Is the first investor ask for operating budget, data/counsel budget, or risk capital?

5. Should the first client package target BTC treasury companies or miners?

Recommended answers:

1. Counsel first, before client-specific outreach.
2. Paid research/analytics first; advisory only after counsel.
3. Partner-led quotes, not self-execution.
4. Raise validation budget, not trading capital.
5. BTC treasury companies first, miners second.

---

14. Bottom line

The best business is a staged platform:

1. **Evidence engine** — data, monitor, screen board.
2. **Structuring wedge** — BTC treasury/miner packages.
3. **Quote verification** — partner-led RFQ evidence.
4. **Risk intermediation** — principal/fund/structured-products only after proof.

This sequence protects credibility, reduces legal risk, and keeps the concept institutionally serious.

Investor Deep Dive

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-investor-deep-dive-v0.md`

Idea 1 Investor Deep-Dive v0

Concept: BTC Volatility Intermediation Desk — ETF / Deribit / CME / OTC volatility analytics, treasury yield/protection, and miner hedging.

Prepared: 2026-05-14 19:10 CDT

Status: Investor-grade first draft, but not final. Data evidence is screen-only unless explicitly marked otherwise. Legal/regulatory structure requires counsel before any execution or client-facing activity.

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1. Executive thesis

Bitcoin has become institutionally accessible through multiple wrappers: spot ETFs, CME futures/options, crypto-native options, and OTC derivatives. These wrappers reference substantially similar economic exposure but trade through different legal, margin, collateral, custody, and client-flow regimes.

That fragmentation creates a business opportunity: build an institutional BTC risk-intermediation platform that helps treasuries, miners, funds, and allocators transform BTC volatility into yield, protection, and structured risk.

The desk should not be framed as a generic crypto-options prop fund. The higher-quality institutional wedge is:

> Board-ready BTC treasury and miner hedging programs, powered by a proprietary cross-venue BTC volatility monitor.

The analytics engine normalizes ETF options, CME futures/options, Deribit options, and OTC/RFQ quotes into BTC-equivalent terms. The client-facing business translates that intelligence into covered calls, collars, put spreads, production hedges, and eventually structured notes.

---

2. Why now

2.1 BTC moved into institutional wrappers

The BTC market is no longer only spot exchange flow. Institutions can now access BTC exposure through:

- US spot BTC ETFs and listed ETF options.
- CME Bitcoin futures and options.
- Crypto-native options venues such as Deribit.

- OTC bilateral derivatives and structured products.

Each wrapper attracts different users and constraints:

- ETF options: securities accounts, RIAs, family offices, public-company treasury comfort, OCC-listed options.
- CME: regulated futures accounts, FCM margin, institutional futures infrastructure.
- Deribit: deep crypto-native vol surface, broad strikes/expiries, 24/7 market, crypto collateral.
- OTC/RFQ: custom tenors, notional sizes, payoff shapes, legal/collateral terms.

2.2 BTC treasury companies and miners create natural client flow

BTC treasury companies and miners are structurally long BTC. But they also face fiat obligations: debt, preferred dividends, operating expenses, power costs, ASIC purchases, site capex, and public-market dilution pressure.

They need a framework that lets them:

- monetize volatility without selling core BTC;
- buy disaster protection without abandoning upside;
- protect capex/runway windows;
- reduce forced-equity or forced-BTC-sale risk;
- explain the policy to boards, auditors, lenders, and investors.

2.3 Volatility fragmentation creates pricing and hedging intelligence

A BTC option is not just a BTC option. The economic comparison depends on:

- ETF wrapper mechanics;
- futures basis;
- BTC/share conversion;
- contract multipliers;
- fees and expense drag;
- bid/ask and executable size;
- margin and collateral currency;
- custody/venue eligibility;
- assignment and settlement mechanics;
- regulatory/mandate restrictions.

Most simple IV comparisons ignore these. The desk’s edge is to normalize them.

---

3. Core product menu

3.1 BTC treasury covered-call program

Client: BTC treasury companies, operating companies with BTC reserves, wealth platforms holding BTC ETFs.

Structure: Sell calls against a defined minority tranche of BTC/ETF exposure.

Purpose: Generate recurring premium income without selling spot BTC.

Variants:

- monthly OTM covered-call ladder;
- quarterly covered-call program;
- call-spread overwrite to preserve extreme upside;
- BTC-denominated premium reinvestment;
- ETF-option version for securities-account clients.

Revenue model: Structuring fee, execution commission/spread, recurring monitoring/reporting retainer, later principal risk spread if permitted.

Key control: Do not cap too much upside. Keep notional modest and policy-driven.

3.2 BTC treasury collar / downside floor

Client: BTC treasury companies, family offices, miners with BTC inventory, firms with debt/preferred obligations.

Structure: Buy puts or put spreads; finance partly/fully by selling calls or call spreads.

Purpose: Create a board-approved downside floor while preserving strategic BTC ownership.

Variants:

- 3-month zero-cost collar;
- 6-month put-spread collar;
- debt/coupon/refi-window protection;
- LTV/collateral protection around BTC-backed loans;
- drawdown-budget hedge for public-company boards.

Revenue model: Structuring fee, RFQ/execution spread, advisory retainer, risk recycling.

Key control: Define maximum BTC encumbrance, acceptable upside cap, and collateral process before quoting.

3.3 Miner production hedge

Client: Public/private Bitcoin miners.

Structure: Hedge a conservative slice of expected production or unencumbered BTC inventory using puts, put spreads, collars, or limited forwards.

Purpose: Protect cash-flow runway for power, capex, debt service, and ASIC/site investment.

Variants:

- monthly production collar;
- 3–6 month put spread on conservative production forecast;
- treasury BTC collar on 10–30% of unencumbered inventory;
- BTC-backed loan collateral protection;
- power/hashprice stress-window hedge.

Revenue model: Advisory/structuring, recurring execution, monitoring/reporting retainer.

Key control: Do not overhedge production. Size to conservative output, not optimistic hashrate targets.

3.4 Cross-venue BTC vol spread package

Client: Hedge funds, macro/RV funds, market makers, internal prop sleeve.

Structure: Relative-value packages across ETF options, CME, Deribit, and OTC.

Examples:

- ETF-vs-Deribit ATM vol spread;
- skew/risk-reversal spread;
- calendar spread;
- CME futures/options basis-vol package;
- event-vol package;
- client-flow hedge recycling.

Revenue model: Analytics subscription, execution/RFQ commission, later principal spread.

Key control: Only promote as an investor return engine after quote-verified or trade-verified evidence.

3.5 Structured notes / defined-outcome products

Client: Private banks, structured-product distributors, family offices, wealth platforms.

Structure: BTC-linked reverse convertibles, autocallables, buffered notes, principal-protected notes, income notes.

Purpose: Package BTC volatility into defined yield/upside/downside profiles.

Revenue model: Structuring margin, hedge spread, distribution economics.

Key control: Phase 2 only. Legal, suitability, issuer, disclosure, and valuation complexity is materially higher.

---

4. Data and analytics engine

4.1 Verified and available sources

Deribit: Verified public API. Live BTC options book summary returned ~936 contracts across 12 expiries in the first monitor pass. Fields include bid/ask/mid/mark, mark IV, underlying price, OI, volume, and instrument metadata.

IBIT/Nasdaq: Public Nasdaq JSON returned IBIT option chain records and bid/ask data. This is usable for pilot research but semi-official/undocumented. Production-grade use requires OPRA/vendor data.

BlackRock/iShares: Official holdings source supports BTC/share normalization. Previously verified official values showed BTC/share around `0.000564212717166`, or roughly `1,772.38` IBIT shares per BTC, using observed BTC holdings and shares outstanding.

OCC: Official listed-series/OI source, useful for series validation but not quotes/IV.

CME: Official API/feed paths exist, but public pages should not be scraped. Production access requires credentials/licensing, DataMine, WebSocket/MDP feed, broker/vendor access, or terminal export.

OTC/RFQ: Not yet verified. Must be quote-verified before economics are treated as executable.

4.2 First quantified screen-only monitor

On the first v0 monitor, Deribit ATM IV term structure was roughly:

- DTE 1: ~31.3%
- DTE 4: ~30.1%
- DTE 8: ~34.9%
- DTE 15: ~36.3%
- DTE 43: ~38.0%
- DTE 134: ~40.8%
- DTE 225: ~43.7%

IBIT ATM IV estimates from Nasdaq mid prices and a rough Black-Scholes model were roughly:

- DTE 1: ~41.5%
- DTE 4: ~34.1%
- DTE 8: ~36.9%
- DTE 15: ~37.9%

Initial screen-only comparison:

- DTE 1: IBIT richer than Deribit by ~10 vol points.
- DTE 4: IBIT richer by ~4 vol points.

• DTE 8: IBIT richer by ~2 vol points.

• DTE 15: IBIT richer by ~1.5–2 vol points.

This does not prove arbitrage. It proves there are screen-level differences worth quote verification.

4.3 Monitor output standard

Every candidate must show:

• dislocation type: wrapper, skew, calendar, basis, margin/collateral, flow;

• exact legs;

• native prices;

• BTC-equivalent normalized prices;

• gross spread;

• estimated costs;

• net spread;

• size/capacity;

• client application;

• risk notes;

• source confidence;

• execution confidence.

Confidence labels:

• **Hypothesis:** theory only.

• **Screen-only:** visible from public/vendor screens.

• **Quote-verified:** indicative counterparty quote received.

• **Trade-verified:** executed trade or firm executable quote.

---

5. Natural client map

5.1 BTC treasury companies — first wedge

Examples: Strategy/MicroStrategy, Metaplanet, Twenty One Capital, Semler Scientific, KULR, The Blockchain Group/Capital B, BTC-heavy miners.

Pain: Capital raises, dilution optics, preferred/convert obligations, operating runway, BTC NAV volatility.

Product fit: Covered calls, zero-cost collars, put-spread collars, policy/reporting package.

Hook:

> Monetize a capped, board-approved slice of BTC convexity to fund obligations or reduce dilution, while preserving strategic BTC ownership and adding downside disaster protection.

5.2 Bitcoin miners — second wedge

Examples: MARA, Riot, CleanSpark, Cipher, TeraWulf, IREN, Core Scientific, Hut 8, Bitfarms, Bitdeer.

Pain: Power costs, capex, ASIC purchases, debt/refi needs, BTC-backed loans, production variability.

Product fit: Production collars, put spreads, inventory collars, runway protection.

Hook:

> Protect cash-flow runway for power, capex, and debt service by collaring a conservative slice of expected production or unencumbered BTC inventory, while leaving most upside intact.

5.3 RIAs, family offices, and wealth platforms

Pain: BTC drawdown tolerance, client income needs, custody restrictions, ETF workflow preference.

Product fit: ETF-option covered calls, collars, defined-loss call spreads, option-income SMA.

Hook:

> Add a volatility overlay to BTC ETF exposure without changing custody or introducing crypto exchange operations.

5.4 Hedge funds / macro / RV funds

Pain: Need normalized cross-venue data, execution proof, margin-adjusted edge.

Product fit: RV spread packages, analytics subscription, RFQ sourcing.

Hook:

> We normalize ETF, CME, Deribit, and OTC wrappers for basis, margin, collateral, bid/ask, and executable size, then label opportunities by quote/trade confidence.

5.5 Structured-product desks / private banks

Pain: Need defined BTC payoff products and hedge-source transparency.

Product fit: Autocallables, reverse convertibles, buffered notes, principal-protected notes.

Hook:

> Provide hedge-source transparency and BTC vol intelligence for defined-outcome BTC products.

---

6. Operating model

Phase 1: Research and analytics platform

- Build daily BTC Vol Desk Monitor.
- Ingest Deribit, IBIT, ETF holdings, OCC, and eventually CME/vendor data.
- Produce screen-only dislocation board.
- Create treasury/miner case studies.
- Maintain evidence appendix.

Revenue possibility: Research/analytics/advisory only, if legally structured.

Phase 2: Advisory and structuring

- Build board-ready treasury and miner hedging policies.
- Produce sample term sheets.
- Advise on permitted instruments, notional limits, collateral process, reporting, and risk governance.
- Partner with regulated execution counterparties.

Revenue possibility: Advisory retainer, structuring fee, consulting/project fee.

Phase 3: RFQ and execution workflow

- Source executable quotes from listed venues, brokers, OTC desks, or market makers.
- Compare quote-verified economics to screen economics.
- Provide best-execution-style analysis where legally permitted.

Revenue possibility: Commission, spread, referral/economic sharing where legally allowed.

Phase 4: Risk intermediation / principal sleeve

- Selectively warehouse/recycle risk across venues.
- Hedge client flow using ETF/CME/Deribit/OTC instruments.
- Operate inside proper fund, dealer, CTA/CPO, broker, or other legal wrapper as determined by counsel.

Revenue possibility: Principal spread, carry, management/performance fees.

Phase 5: Structured products

- Work with issuers/distributors to create BTC-linked defined-outcome products.
- Use analytics engine for hedge sourcing and pricing support.

Revenue possibility: Structuring margin, hedging spread, distribution economics.

---

7. Economics model

Potential revenue lines

1. **Advisory/structuring fees**

- Treasury policy design.
- Miner hedge design.
- Board memo and risk framework.

2. **Monitoring/reporting retainer**

- Daily/weekly hedge marks.
- Exposure reporting.
- Risk dashboard.
- Board/lender reporting package.

3. **Execution economics**

- Commission or spread through permitted broker/dealer/OTC relationships.
- RFQ sourcing fee if legally allowed.

4. **Principal/risk spread**

- Later-stage only.
- Requires legal wrapper, capital, risk limits, and controls.

5. **Analytics subscription**

- Hedge funds, allocators, internal desks.
- Requires production-grade data and history.

6. **Structured-product economics**

- Phase 2+.
- Requires issuer/distribution/legal framework.

Unit economics to model next

For each sample structure:

- notional BTC or ETF share exposure;
- premium generated or paid;
- annualized yield;
- option bid/ask spread;
- hedge cost;
- RFQ spread;
- collateral/margin cost;
- reporting/advisory fee;
- maximum loss/upside cap;
- expected client value vs unhedged BTC;
- desk gross and net revenue.

---

8. Legal and regulatory perimeter

This is the main gating issue. The memo must not overclaim.

Potential activities touch multiple regimes:

- securities/options advice;
- commodity/futures/options advice;
- broker-dealer or introducing-broker activity;
- CTA/CPO analysis;
- investment adviser issues;
- OTC derivatives/ISDA counterparty activity;
- offshore exchange access;
- structured-product issuance/distribution;
- custody/collateral/prime brokerage arrangements.

Correct phased stance

Start with:

- research;
- analytics;
- hypothetical/screen-only monitor;
- education;
- internal memo;
- sample term sheets clearly marked illustrative;
- legal analysis.

Do not start with:

- quoting derivatives as principal to public companies;
- taking discretion over treasury assets;
- acting as broker-dealer/FCM/IB without structure;
- implying executable RFQ economics before quotes;
- offering securities/commodity advice without counsel.

Legal workstream deliverables

- regulatory activity map;
- permitted MVP activities;
- required disclaimers;
- entity structure options;
- partner/counterparty model;
- CTA/CPO/RIA/broker-dealer/IB analysis;

- OTC documentation checklist;
- public-company derivatives-policy template.

---

9. Risk controls

Market risk

- BTC delta and gamma.
- Volatility shifts.
- Skew/correlation.
- Calendar mismatch.
- Gap risk and weekend/24-7 risk.

Basis/wrapper risk

- ETF vs BTC NAV/premium-discount.
- CME futures basis.
- Deribit index/settlement differences.
- ETF options assignment and share settlement.

Liquidity risk

- Screen liquidity vs executable size.
- Wide short-dated markets.
- Stress-day spread widening.
- Position limits.

Margin/collateral risk

- USD vs BTC collateral.
- FCM margin calls.
- Crypto exchange margin/liquidation.
- OTC CSA terms.
- Public-company collateral restrictions.

Counterparty/venue risk

- Exchange/clearing/custody risk.
- OTC counterparty credit.
- Prime broker/FCM dependency.
- Offshore venue restrictions.

Operational/legal risk

- Inappropriate advice or execution without proper wrapper.
- Board/auditor/lender non-approval.
- Disclosure/regulatory issues.
- Valuation/model governance failure.

Narrative/reputation risk

- Treasury companies may view hedging as anti-BTC.
- Covered calls can be attacked if BTC rallies through strikes.
- Public companies need careful disclosure and policy framing.

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10. 30-day pilot

Week 1 — data spine

Deliverables:

- Deribit BTC options ingestion.
- IBIT option-chain ingestion.
- iShares BTC/share parser hardened.
- OCC series/OI cross-check.
- CME data-source decision.
- Daily screen-only BTC Vol Desk Monitor v0.

Success criteria:

- Produce ATM IV term structure for Deribit and IBIT.
- Produce first normalized screen-only dislocation board.
- Every output labeled with source and execution confidence.

Week 2 — product prototypes

Deliverables:

- Treasury covered-call calculator.
- Zero-cost collar calculator.
- Put-spread collar calculator.
- Miner production hedge template.
- Sample term sheets.
- Board memo template.

Success criteria:

- One hypothetical BTC treasury case study.
- One hypothetical miner case study.
- All prices labeled illustrative/screen-only.

Week 3 — client mapping and commercial validation

Deliverables:

- 20–50 target accounts.
- Decision-maker/persona map.
- Outreach hooks.
- Qualification checklist.
- Counterparty/RFQ list.

Success criteria:

- At least 3 client archetype packages.
- At least 3 standard RFQ templates.
- Identify legal gating questions before any outreach.

Week 4 — investor memo and quote verification plan

Deliverables:

- Investor memo v1.
- Evidence appendix.
- Data architecture.
- Legal perimeter memo draft.
- RFQ verification plan.
- 90-day roadmap.

Success criteria:

- Clearly separate thesis, screen evidence, quote evidence, and execution requirements.
- No unsupported economics in investor-facing materials.

---

11. 90-day roadmap

Days 31–45

- Integrate CME via vendor/broker/terminal export or official credentialed path.
- Add OPRA-grade ETF option data if budget allows.
- Store daily surfaces historically.

- Add skew/risk-reversal and calendar analytics.
- Validate IBIT IV model against vendor or broker data.

Days 46–60

- Collect indicative RFQs for standard structures:
- 1-month covered call;
- 3-month zero-cost collar;
- 6-month miner put spread;
- ETF vs Deribit vol spread.
- Convert screen-only dislocation board into quote-verified board.
- Build first counterparty matrix.

Days 61–75

- Build treasury/miner reporting package.
- Draft legal operating model with counsel.
- Define entity/registration/partner structure.
- Create first client-ready board deck.

Days 76–90

- Run 3–5 live commercial conversations if legal perimeter permits.
- Finalize investor deck.
- Decide whether next vehicle is analytics/advisory company, fund, desk partnership, or hybrid.
- Define capital, hires, vendors, and counterparties required for launch.

---

12. Team and infrastructure needs

Team

- BTC/options structurer.
- Quant/data engineer.
- Derivatives legal counsel.
- Institutional sales / capital markets lead.
- Operations/risk person with FCM/prime/OTC experience.

Infrastructure

- Market data: Deribit, OPRA/vendor, CME/vendor.
- Data store for option surfaces and historical marks.
- Pricing library and validation process.

- RFQ/counterparty tracker.
- Risk dashboard.
- Compliance archive.
- Board/client report generator.

Counterparties

- FCM / CME access.
- OPRA/options data vendor.
- ETF/options broker.
- OTC crypto options desks.
- Custodian/prime broker.
- Legal/accounting advisers.

---

13. Investment case

What makes the idea attractive

- BTC volatility is structurally high and institutionally relevant.
- BTC exposure now exists across fragmented wrappers.
- Public BTC treasuries and miners create natural hedging/yield flows.
- ETF/CME/Deribit/OTC access constraints create intermediation value.
- The first product is understandable: treasury income and protection.
- The analytics engine can become a proprietary pricing and risk-recycling layer.

What must be proven

- Screen dislocations are executable after costs.
- Clients will accept capped upside on a defined tranche.
- Legal structure permits advisory/execution economics.
- Data pipeline can produce reliable normalized surfaces.
- Counterparties will quote size.
- Risk can be hedged across wrappers under stress.

Main red flags

- Legal/regulatory perimeter may be heavier than expected.
- Pure vol-arb edge may be competed away.
- Public companies may avoid derivatives due to optics.
- OTC/RFQ spreads may consume apparent screen edge.
- Margin/collateral costs may erase relative-value profits.

- Deribit access may be unusable for key US clients.

---

14. Recommended investor-facing positioning

Lead with:

> We help institutional BTC holders transform volatility into income, protection, and structured risk through board-ready treasury and miner hedging programs, powered by a proprietary cross-venue BTC volatility monitor.

Avoid leading with:

> We run a BTC options arbitrage fund.

The former is a business. The latter is a trade.

---

15. Immediate next deliverables

1. Harden the data pipeline:

- fix iShares BTC/share parser;
- store Deribit/IBIT snapshots;
- compute IV consistently;
- add bid/ask filters.

2. Build first treasury case study:

- 10,000 BTC hypothetical holder;
- 10% covered-call sleeve;
- 3-month collar;
- 6-month put-spread collar;
- board-policy summary.

3. Build first miner case study:

- 3-month conservative production forecast;
- 25–50% hedge ratio;
- put-spread/collar;
- cash runway impact.

4. Start legal perimeter memo:

- what can be done as research;
- what requires registration/partner;
- what cannot be done before counsel.

5. Start RFQ verification plan:

- list standard structures;
- identify counterparties;
- define quote fields;
- label quote-verified vs screen-only.

Bottom line

Idea 1 is viable enough to continue, but only if built in the disciplined order:

1. evidence spine;
2. screen-only monitor;
3. treasury/miner product prototypes;
4. legal perimeter;
5. quote verification;
6. investor memo;
7. execution vehicle.

The strongest launch product is not a prop-vol fund. It is a board-ready BTC treasury and miner hedging platform with a proprietary cross-venue volatility engine behind it.

Data Evidence Appendix

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-data-evidence-appendix.md`

Idea 1 — BTC Vol Desk Data & Evidence Appendix

Purpose: Establish the evidence spine for a BTC ETF / Deribit / CME / OTC volatility intermediation desk before writing investor narrative.

Standard: Every data source is classified as verified, credentialed/paid, or hypothesis. Screen-only data is not treated as executable.

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1. Source classification

A. Deribit BTC options — verified public API

Status: Verified public API access.

Official docs / endpoints:

- OpenAPI spec: `https://docs.deribit.com/specifications/deribit_openapi.json`
- Instruments: `GET https://www.deribit.com/api/v2/public/get_instruments?currency=BTC&kind=option&expired=false`
- Book summary: `GET https://www.deribit.com/api/v2/public/get_book_summary_by_currency?currency=BTC&kind=option`
- Order book: `GET https://www.deribit.com/api/v2/public/get_order_book?instrument_name=BTC-...&depth=1`
- Historical volatility: `GET https://www.deribit.com/api/v2/public/get_historical_volatility?currency=BTC`

Live verification from May 14, 2026 pass:

- Active BTC option instruments: ~936
- Expiries observed: 12
- Near-ATM IV samples:
- 15MAY26: ~31.1%
- 22MAY26: ~35.1%
- 29MAY26: ~36.5%
- 26JUN26: ~38.1%
- 25SEP26: ~40.8%

Useful fields:

- Metadata: `instrument_name`, `instrument_id`, `expiration_timestamp`, `strike`, `option_type`, `state`, `is_active`, `contract_size`, `settlement_currency`, `quote_currency`, `price_index`
- Market snapshot: `bid_price`, `ask_price`, `mid_price`, `mark_price`, `mark_iv`, `underlying_price`, `underlying_index`, `interest_rate`, `open_interest`, `volume`, `volume_usd`, `estimated_delivery_price`

- Top-of-book: `best_bid_price`, `best_bid_amount`, `best_ask_price`, `best_ask_amount`, `bid_iv`, `ask_iv`, `mark_iv`, `greeks.delta`, `greeks.gamma`, `greeks.vega`, `greeks.theta`, `greeks.rho`
- Historical vol: live payload returns `[timestamp, value]` pairs despite docs schema implying `timestamp` / `value` objects.

Caveats:

- Deribit prices are crypto-native and must be normalized into USD/BTC-equivalent terms.
- Use WebSocket subscriptions for real-time production instead of polling every instrument.
- Rate limits are credit-based; `get_instruments` has stricter limits than normal endpoints.

B. IBIT / listed ETF options — verified semi-official + official supporting sources

Nasdaq option chain

Status: Verified public JSON, but undocumented/semi-official rather than guaranteed production API.

Endpoint:

- `https://api.nasdaq.com/api/quote/IBIT/option-chain?assetclass=etf`
- Greeks endpoint: `https://api.nasdaq.com/api/quote/IBIT/option-chain/greeks?assetclass=etf`

Live verification from May 14, 2026 pass:

- IBIT last shown: `$46.17 as of May 14, 2026`
- `totalRecord`: 412
- Expiries returned included May 15, May 18, May 20, May 22, May 26, May 27, May 29.
- Retrieved chain sample showed:
- call OI: ~1.12M contracts
- put OI: ~640k contracts
- call volume: ~336k
- put volume: ~109k

Useful fields:

- Chain fields: `expiryDate`, `strike`, `c_Last`, `c_Bid`, `c_Ask`, `c_Volume`, `c_Openinterest`, `p_Last`, `p_Bid`, `p_Ask`, `p_Volume`, `p_Openinterest`
- Greeks endpoint: `cDelta`, `cGamma`, `cRho`, `cTheta`, `cVega`, `cIV`, `pDelta`, `pGamma`, `pRho`, `pTheta`, `pVega`, `pIV`

Caveats:

- Not documented as a stable API contract.
- `table.asOf` observed as null.
- Display-oriented records, not OPRA-grade records.
- No guaranteed history, timestamps, quote conditions, NBBO provenance, or depth.
- Greeks methodology/source not specified.

OCC series search

Status: Official public source for listed series and open interest; not a quote/IV source.

Endpoint tested:

- `https://marketdata.theocc.com/series-search?symbolType=U&symbol=IBIT`

Useful fields:

- Product symbol
- Expiration year/month/day
- Strike
- Call/put indicator
- Call OI
- Put OI
- Position limit
- Exchanges where product trades

Caveats:

- No bid/ask/last/volume/IV/Greeks.
- Useful as official series/OI reference, not as tradeable market data.

C. IBIT NAV / BTC-per-share normalization — verified official BlackRock/iShares source

Status: Verified official public iShares data.

Sources:

- Product page: `https://www.ishares.com/us/products/333011/ishares-bitcoin-trust-etf`
- Holdings CSV: `https://www.ishares.com/us/products/333011/fund/1467271812596.ajax?fileType=csv&fileName=IBIT_holdings&dataType=fund`

Observed official fields from May 13/14 pass:

- NAV: `$45.05` as of May 13, 2026
- Sponsor fee: `0.25%`
- Net assets: `$65,235,245,502`
- Shares outstanding: `1,448,200,000`
- BTC quantity in holdings CSV: `817,092.85700 BTC`
- Benchmark index: CME CF Bitcoin Reference Rate - New York Variant
- Bloomberg index ticker: BRRNY

Normalization calculation:

- BTC/share = `BTC quantity / shares outstanding`
- Using observed values: `817,092.857 / 1,448,200,000 = 0.000564212717166 BTC/share`
- Shares per BTC ≈ `1,772.38117748`

Use in vol monitor:

- Convert IBIT option strikes to BTC-equivalent strikes via BTC/share.
- Convert IBIT option premiums to BTC-equivalent or USD-per-BTC premium.
- Compare IV/Greeks only after wrapper and forward assumptions are normalized.

Caveats:

- Public holdings/NAV data is generally as-of/T-1, not real-time.
- Holdings CSV warns market value/weight/notional may use third-party vendor pricing.
- Use official quantity and shares outstanding for BTC/share; avoid inferring from market value.

D. CME BTC futures/options — verified official docs, credentialed for production data

Status: Official public web pages are reference-only; automated production data requires credentials/licensing/API/vendor.

Public product pages:

- Bitcoin futures: `https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.html`
- Bitcoin futures quotes: `https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.quotes.html`
- Bitcoin futures specs: `https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.contractSpecs.html`
- Bitcoin options: `https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.quotes.options.html`
- Bitcoin options specs: `https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.contractSpecs.options.html`

Automation caveat:

- CME public website returned HTTP 403 to scripted access and explicitly blocks scraping under its website data terms.
- Treat public pages as human reference/display only.

Official CME API/feed paths:

- Real-Time Futures and Options WebSocket API docs: `https://cmegroupclientsite.atlassian.net/wiki/spaces/EPICSANDBOX/pages/457414253/CME+Market+Data+Over+WebSocket+API`
- Message spec: `https://cmegroupclientsite.atlassian.net/wiki/display/EPICSANDBOX/CME+Market+Data+Over+WebSocket+API+-+Message+Specification`
- Production WebSocket: `wss://markets.api.cmegroup.com/marketdatastream/v1`
- OAuth token production: `https://auth.cmegroup.com/as/token.oauth2`
- MDP 3.0 market data docs: `https://cmegroupclientsite.atlassian.net/wiki/display/EPICSANDBOX/CME+MDP+3.0+Market+Data`
- DataMine: `https://www.cmegroup.com/datamine.html`
- DataMine API: `https://www.cmegroup.com/datamine/datamine-api.html`

Useful official data classes:

- Top of book: bid/ask price, quantity, order count, timestamps, instrument metadata.
- Trade summary: trade price/quantity, aggressor side, timestamps, instrument metadata.
- Statistics: cleared volume, open interest, settlement price, settlement flags/date/timestamp.
- Historical DataMine: end-of-market summary, BBO, volume/OI.
- CME options analytics product appears to include Greeks and implied volatility, but requires credentials/entitlements.

Practical path:

- MVP: use broker/vendor data for CME BTC futures/options if available.
- Institutional: CME WebSocket API + DataMine historical after entitlements.
- Low-latency institutional: MDP 3.0 direct feed only if justified.

E. OTC / RFQ layer — not yet verified; must be quote-verified

Status: Hypothesis until live counterparties provide indicative quotes or executed RFQs.

Target sources:

- Crypto OTC desks
- Options market makers
- Prime brokers
- FCMs / institutional crypto brokers
- RFQ venues
- Structured-product desks

Required fields:

- Indicative bid/ask for standard structures
- Minimum ticket size
- Max executable size
- Counterparty credit terms
- Collateral requirements
- Settlement method
- Legal documents required
- Quote expiry
- Venue hedge assumptions
- Fees/commission/spread

Confidence labels:

- Hypothesis: based on expected market behavior only.
- Screen-only: based on listed market data only.
- Quote-verified: at least one real indicative quote from a counterparty.
- Trade-verified: actual executed trade or executable firm market.

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2. Normalized BTC-equivalent schema

Every option record should be transformed into a common schema before comparison.

Required normalized fields

- `source`: Deribit / Nasdaq / OCC / CME / OTC / vendor
- `venue`: Deribit / ETF-listed / CME / OTC
- `instrument_id`
- `instrument_name`
- `underlying_wrapper`: BTC index / IBIT ETF / CME futures / OTC reference
- `underlying_price_native`
- `btc_reference_price`
- `expiry_date`
- `days_to_expiry`
- `strike_native`
- `strike_btc_equivalent`
- `option_type`: call / put
- `bid_native`
- `ask_native`
- `mid_native`
- `mark_native`
- `bid_usd_equivalent`
- `ask_usd_equivalent`
- `mid_usd_equivalent`
- `bid_btc_equivalent`
- `ask_btc_equivalent`
- `iv_bid`
- `iv_ask`
- `iv_mid`
- `iv_mark`
- `delta`
- `gamma`
- `vega`
- `theta`
- `rho`
- `open_interest_native`
- `volume_native`
- `bid_size`
- `ask_size`
- `bid_ask_width_pct`
- `forward_price_assumption`
- `rate_assumption`

- `funding_or_basis_assumption`
- `margin_model`
- `collateral_currency`
- `fee_assumption`
- `slippage_assumption`
- `data_timestamp`
- `source_confidence`: verified_public / semi_official / credentialed / paid_vendor / hypothesis
- `execution_confidence`: screen_only / quote_verified / trade_verified

Wrapper-specific normalization

Deribit:

- Native quote is often in BTC terms for options.
- Use Deribit `underlying_price` / `estimated_delivery_price` to convert to USD-equivalent.
- Confirm exact option contract settlement and PnL convention per instrument metadata.

IBIT:

- Compute BTC/share from official iShares holdings.
- Convert IBIT strike to BTC-equivalent: `strike_native / BTC_per_share` only after checking whether comparing to BTC-per-one-BTC notional or per-share notional.
- Convert option premium to BTC-equivalent by scaling share option contract multiplier and BTC/share.
- Account for ETF fee, NAV premium/discount, and T-1 holdings data.

CME:

- Normalize from futures option to BTC spot-equivalent using futures price/basis.
- Include contract multiplier and whether using standard or micro BTC contract.
- Use settlement/open-interest stats from official CME feed/vendor.

OTC/RFQ:

- Normalize quote terms manually from term sheet.
- Include collateral and credit terms; these may dominate apparent vol difference.

---

3. Daily BTC Vol Desk Monitor — MVP format

Section 1: Market state

- BTC spot/reference
- IBIT last / NAV / premium-discount
- IBIT BTC/share
- CME front futures and basis
- Deribit futures/perps basis
- realized/historical vol reference

Section 2: Venue surface summary

- Deribit ATM IV by expiry
- IBIT ATM IV by expiry
- CME ATM IV by expiry
- 25-delta put IV by venue
- 25-delta call IV by venue
- risk reversal by venue
- term-structure slope by venue
- volume/OI by venue
- bid/ask quality by venue

Section 3: Dislocation board

For each candidate:

- Dislocation type: wrapper / skew / calendar / basis / margin / flow
- Instruments/legs
- Native prices
- BTC-equivalent normalized prices
- Gross edge
- Estimated costs: bid/ask, fees, margin, collateral, slippage
- Net edge
- Capacity estimate
- Execution confidence: screen-only / quote-verified / trade-verified
- Client application: treasury / miner / fund / family office / structured product
- Risk notes

Section 4: Client structure examples

- Treasury covered call quote grid
- Zero-cost collar grid
- Miner hedge ladder
- Fund relative-value spread package

Section 5: Missing data / warnings

- Source gaps
- stale fields
- wide markets
- unverified quote assumptions
- regulatory/venue constraints

---

4. Data gaps and vendor needs

Immediate gaps

- CME credentialed/production data source.
- OPRA-grade IBIT quote/trade history and live NBBO.
- Historical IBIT option IV/Greeks.
- Historical Deribit full surface data unless stored going forward or sourced from vendor.
- OTC/RFQ quote verification.
- Legal/compliance mapping for advisory/execution/principal activity.

Candidate vendors / infrastructure

- CME official WebSocket + DataMine.
- OPRA-licensed feeds.
- Databento for OPRA and possibly CME data depending license/product coverage.
- Cboe DataShop / LiveVol.
- OptionMetrics IvyDB US.
- ORATS.
- Polygon/Massive options APIs.
- dxFeed / Interactive Brokers / Bloomberg / Refinitiv / FactSet.
- Crypto options vendors for Deribit history/analytics if cheaper than self-storing.

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5. Build sequence recommendation

Step 1: Deribit + IBIT daily snapshot

Build the first monitor using verified public/semi-official data:

- Deribit book summary + instruments.
- Nasdaq IBIT chain + Greeks.
- BlackRock iShares holdings CSV for BTC/share.
- OCC series search for official listed-series/OI cross-check.

Step 2: CME data-source decision

Choose one:

- Broker/vendor API for quick pilot.
- CME WebSocket/DataMine if institutional path and budget justify.
- Bloomberg/Refinitiv/manual export for investor memo proof only.

Step 3: Normalize and label confidence

Every result must show:

- source confidence

- execution confidence
- stale/as-of timestamp
- cost/margin assumptions

Step 4: Produce first daily report

Output should be suitable for two audiences:

- internal trader/structurer: exact instruments and dislocations
- investor/client: plain-English structures and why they exist

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6. Current conclusion

The data spine is feasible, but the honest MVP must be labeled correctly:

- Deribit: usable immediately from official public API.
- IBIT: usable immediately for pilot via Nasdaq + BlackRock + OCC, but production-grade data needs OPRA/vendor.
- CME: official automated access exists, but requires credentials/licensing or vendor/broker access; public web pages must not be scraped.
- OTC/RFQ: must be quote-verified before any economics are presented as executable.

This supports the strategic thesis: the opportunity is not just IV comparison. The edge is normalizing different wrappers, margin regimes, collateral regimes, venue access constraints, and client flows into structures that a treasury, miner, fund, or allocator can actually use.

Client Map / Commercial Wedge

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-client-map-commercial-wedge.md`

Idea 1 — Natural Client Map & Commercial Wedge

Purpose: Define who buys what, why they buy it, what constraints block them, and how the BTC Vol Desk should sequence go-to-market.

Strategic reminder: The business should not lead as a generic BTC options-arbitrage fund. It should lead as a treasury/miner/institutional risk-transformation platform, using the cross-venue vol monitor as pricing and hedge intelligence.

---

1. Client hierarchy

Priority 1: BTC treasury companies

Why first:

- Cleanest commercial story.
- Direct pain: large BTC holdings, public-market volatility, capital raises, preferred/convert obligations, dilution optics.
- Product is board-understandable: covered calls, collars, put spreads.
- Does not require clients to understand cross-venue vol arbitrage.

Core product:

- Board-approved covered-call program on a defined BTC/ETF tranche.
- Zero-cost or low-cost collar to protect treasury floor.
- Put-spread financed by call spread when outright puts are too expensive.

Likely decision makers:

- CEO/founder/executive chair
- CFO/treasurer
- board investment/risk/audit committee
- capital markets lead
- general counsel
- IR, for narrative framing

Pain points:

- Repeated equity/convert issuance.
- Preferred dividends, coupon/refi needs, operating runway.
- BTC NAV premium/discount volatility.

- Shareholder concern around dilution.
- Need to avoid selling core BTC while still funding operations.

Constraints:

- Many are explicitly bullish BTC and may resist selling upside.
- Derivatives need board policy, auditor comfort, valuation controls, disclosure discipline, and collateral/custody controls.
- Covered calls can conflict with “maximize BTC per share” narratives unless sized carefully.

Correct hook:

> Monetize a capped, board-approved slice of BTC convexity to fund obligations or reduce dilution, while preserving strategic BTC ownership and adding downside disaster protection.

Public examples / targets:

- Strategy / MicroStrategy
- Metaplanet
- Twenty One Capital / Cantor Equity Partners
- Semler Scientific
- KULR Technology
- The Blockchain Group / Capital B
- BTC-heavy miners with treasury strategies: MARA, Riot, CleanSpark, Hut 8

Do not pitch:

> You should hedge because BTC may go down.

Do pitch:

> A treasury policy that converts a small, defined slice of volatility into recurring income or a drawdown floor without compromising the long-term BTC thesis.

Priority 2: Bitcoin miners

Why second:

- Miners are structurally long BTC production but owe fiat liabilities.
- They have recurring power, lease, debt, ASIC, and site capex obligations.
- Their business risk is not just BTC price; it is BTC price minus power/hashprice/capex timing.

Core product:

- Rolling production collars.
- Put-spread protection on expected production.
- Treasury BTC collar on 10–30% of inventory.
- Capex-window downside floor.
- LTV/collateral protection around BTC-backed loans.

Likely decision makers:

- CFO/treasurer
- CEO
- head of energy/power strategy
- corporate development/project finance lead
- board risk/audit committee
- lenders or PE sponsor for private miners

Pain points:

- Power costs and curtailment volatility.
- ASIC/site capex timing.
- Halving/hashprice compression.
- Debt/refinancing/convert pressure.
- Need to preserve equity upside while avoiding forced spot sales.

Constraints:

- Production volume is uncertain; do not overhedge hashrate targets.
- BTC may already be pledged to lenders.
- Debt covenants may restrict derivatives/collateral.
- Shareholders often want upside BTC beta.
- Hedge accounting/disclosure complexity.

Correct hook:

> Protect the cash-flow runway for power, capex, and debt service by collaring a conservative slice of expected production or unencumbered BTC inventory, while leaving most upside intact.

Public examples / targets:

- MARA
- Riot
- CleanSpark
- Cipher
- TeraWulf
- IREN
- Core Scientific
- Hut 8
- Bitfarms
- Bitdeer

Most likely first product:

- 3–6 month production put spread or zero-cost collar sized to conservative production, not full expected production.

Priority 3: RIAs, family offices, and wealth platforms

Why third:

- They hold ETF exposure and need income/downside framing.
- ETF options fit existing brokerage/wealth workflows.
- They are natural buyers of simple defined-outcome products.

Core product:

- IBIT/spot BTC ETF covered-call overlays.
- Protective collars.
- Defined-loss call spreads.
- Put-spread collars.
- Option-income SMA/strategy model.

Likely decision makers:

- CIO
- investment committee
- alternatives/digital-assets lead
- portfolio manager
- platform due diligence / product committee

Pain points:

- BTC allocation is hard to hold through drawdowns.
- Clients want income and defined downside.
- Advisors need simple language and operationally familiar securities wrappers.
- Custody/wallet complexity is unacceptable for many.

Constraints:

- Options approval/suitability.
- Platform restrictions.
- Tax treatment.
- Position limits and liquidity.
- Need scenario analysis versus unhedged ETF.

Correct hook:

> Add a volatility overlay to BTC ETF exposure without changing custody or introducing crypto exchange operations.

Best venue:

- ETF options first.
- CME for more sophisticated overlay accounts.
- OTC notes if platform-approved.
- Deribit usually unsuitable for US wealth unless offshore/eligible and compliance-approved.

Priority 4: Hedge funds / macro / relative-value funds

Why fourth:

- They understand cross-venue vol and can underwrite relative-value trades.
- But they are competitive and less sticky than treasury/miner clients.

Core product:

- ETF vs Deribit vol spreads.
- CME vs Deribit basis/vol packages.
- Skew/risk-reversal trades.
- Calendar trades.
- Event-vol packages.
- RFQ sourcing and analytics.

Likely decision makers:

- PM
- volatility trader
- CIO
- COO/legal for venue approval

Pain points:

- Need clean normalized surfaces.
- Need execution/liquidity proof.
- Need margin/collateral-adjusted edge.
- Need cross-venue basis risk quantified.

Constraints:

- They can build part of this themselves.
- Need strong proof and executable size.
- Deribit/CME/ETF venue access may vary by mandate.

Correct hook:

> We normalize ETF, CME, Deribit, and OTC BTC vol into one margin- and wrapper-adjusted surface, then identify executable packages rather than naive IV gaps.

Priority 5: Structured-product desks / private banks / product issuers

Why later:

- Potentially large economics.
- Highest legal/product-approval complexity.

Core product:

- BTC-linked reverse convertibles.
- Autocallables.
- Buffered notes.
- Principal-protected notes with BTC call exposure.
- Income notes backed by covered-call strategy.

Likely decision makers:

- structured products desk
- product approval committee
- legal/compliance
- issuer risk management
- distributor due diligence

Pain points:

- Clients want BTC-linked yield/upside with defined terms.
- Distribution channels need productized payoffs.
- Issuers need hedge-source transparency.

Constraints:

- Suitability/distribution rules.
- Prospectus/KID/risk factors.
- Issuer credit/CVA.
- Secondary valuation.
- Complex hedge governance.

Correct hook:

> Provide hedge-source transparency and BTC vol intelligence for defined-outcome BTC products.

---

2. Venue fit by client

ETF options

Best for:

- RIAs
- family offices
- ETF holders
- public-company treasuries that hold ETF shares
- covered-call products

Why:

- Existing securities account workflow.

- OCC-cleared listed options.
- No wallet/custody change.
- Easier governance for US wealth and securities-oriented clients.

Main frictions:

- US market hours.
- Position limits.
- Assignment/exercise mechanics.
- ETF wrapper and tracking/NAV issues.
- Need OPRA-grade data for production.

CME

Best for:

- institutional funds
- pensions/endowments with futures mandates
- miners/treasuries that prefer regulated derivatives
- macro/RV funds

Why:

- CFTC-regulated, centrally cleared futures/options route.
- FCM margin/risk systems.
- Institutional comfort.

Main frictions:

- Futures basis/roll.
- FCM onboarding.
- Margin/collateral needs.
- Contract size.
- Licensed data access.

Deribit

Best for:

- crypto-native funds
- non-US sophisticated clients
- market makers
- internal hedge/recycle venue if legally/operationally permitted

Why:

- Deep BTC option surface.
- 24/7 market.
- Broad strikes/expiries.

- Often best crypto-native vol reference.

Main frictions:

- Jurisdiction restrictions.
- Offshore venue/counterparty risk.
- Collateral/custody governance.
- Not appropriate for many US public/wealth clients.

OTC/RFQ

Best for:

- treasuries
- miners
- family offices
- structured-product desks
- large bespoke trades

Why:

- Custom payoff, tenor, notional, collateral, and settlement.
- Can hide operational complexity from client.
- Allows board-friendly term sheets.

Main frictions:

- Counterparty credit.
- ISDA/CSA/legal onboarding.
- Valuation governance.
- Quote transparency.
- Must be quote-verified.

---

3. Product-client matrix

Treasury covered call

Best clients:

- BTC treasury companies
- operating companies with BTC reserves
- wealth platforms with ETF holdings

Revenue source:

- structuring fee
- execution spread/commission
- advisory retainer

- later: principal/risk spread if legally approved

Proof needed:

- Premium/yield scenarios.
- Upside opportunity-cost scenarios.
- Board policy language.
- Collateral/custody workflow.
- Independent mark methodology.

Treasury collar / downside floor

Best clients:

- BTC treasuries with debt/preferred obligations
- miners with capex/debt windows
- family offices with drawdown limits

Revenue source:

- structuring + execution fee
- RFQ spread
- risk recycling across venues

Proof needed:

- Protected level.
- Upside cap.
- Cost/premium.
- Stress test vs unhedged BTC.
- Covenant/runway impact.

Miner production hedge

Best clients:

- public/private miners
- capex-heavy miners
- debt-financed miners
- miners with BTC-backed loans

Revenue source:

- advisory/structuring fee
- recurring hedge execution
- monitoring/reporting retainer

Proof needed:

- Conservative production forecast.
- Hedge ratio policy.

- Power/hashprice sensitivity.
- Overhedging controls.
- Lender/covenant review.

Cross-venue vol spread package

Best clients:

- hedge funds
- macro/RV funds
- market makers
- internal prop sleeve

Revenue source:

- research/analytics subscription
- execution/RFQ commission
- principal spread if risk-taking

Proof needed:

- Trade-verified dislocations.
- Net edge after fees/margin/collateral.
- Capacity.
- Stress/basis risk.
- Historical recurrence.

Structured note / defined outcome

Best clients:

- private banks
- structured-product distributors
- family offices
- RIAs if platform-approved

Revenue source:

- structuring margin
- hedge spread
- distribution economics

Proof needed:

- issuer/legal wrapper.
- suitability and disclosure.
- valuation agent.
- secondary market process.
- hedge-source transparency.

---

4. Commercial sequencing

Wedge A — board-ready BTC treasury overlay

First 30-day target:

- produce sample board package for a hypothetical BTC treasury with 10,000 BTC.
- structures:
- 10% covered-call sleeve
- 3-month zero-cost collar
- 6-month put-spread financed by call spread
- use live screen data for indicative pricing but label screen-only.
- add legal/accounting/custody checklist.

Why this wedge wins:

- The buyer understands the problem.
- The product is commercially clear.
- It creates recurring monitoring and execution needs.
- It avoids leading with hard-to-prove arbitrage.

Wedge B — miner runway hedge

First 30-day target:

- produce sample hedge for a miner with expected 3-month production.
- structure:
- protect 25–50% of conservative expected production.
- use put spread/collar, not full forward sale.
- show cash-flow runway impact.

Why second:

- Strong natural risk need.
- More complex due to production uncertainty and power/hashprice variables.

Wedge C — RV fund dislocation monitor

First 30-day target:

- daily screen-only dislocation board.
- then collect quote verification for top candidates.
- only after quote verification should this become an investor-facing return thesis.

Why third:

- Useful as internal edge.

• But pure RV clients are harder to own and easier to compete with.

---

5. First outreach hooks

BTC treasury CEO / executive chair

> You can preserve the long-term BTC accumulation story while letting a board-approved minority tranche generate premium income or finance downside protection. The goal is not to become bearish; it is to reduce dilution and create a treasury policy around volatility monetization.

BTC treasury CFO / treasurer

> We map BTC holdings, obligations, and liquidity needs into a permitted-instruments policy: how much BTC can be encumbered, what maturities match coupons/capex/runway, how marks are sourced, how collateral is controlled, and what gets disclosed.

Miner CFO

> Instead of selling spot BTC reactively to fund power/capex, hedge a conservative slice of expected production with collars or put spreads. The goal is budget certainty without giving up the majority of upside.

RIA / family office CIO

> Keep BTC exposure inside the ETF/securities workflow but add a defined options overlay: income, downside budget, or capped-upside structures. No crypto exchange operations required.

Hedge fund PM

> We are not comparing naive IV screens. We normalize ETF, CME, Deribit, and OTC wrappers for basis, margin, collateral, bid/ask, and executable size, then label opportunities by quote/trade confidence.

---

6. Qualification checklist before showing terms

Exposure

• BTC held directly, through ETF, through futures, or through production?

• How much is unencumbered?

• Is BTC pledged to lenders or custody-restricted?

• What is expected monthly production, if miner?

Objective

• Income?

• Downside floor?

• Financing/capex runway?

- Debt/covenant protection?
- Defined-outcome product for investors?

Constraints

- Derivatives permitted?
- Venue restrictions: ETF options, CME, OTC, Deribit?
- Collateral currency and custody limits?
- Position limits?
- Board/auditor/lender approvals?

Risk tolerance

- Maximum upside cap acceptable?
- Maximum premium spend?
- Maximum BTC encumbrance percentage?
- Stress loss tolerance?
- Disclosure sensitivity?

Operations

- Broker/FCM/prime broker/OTC counterparty access?
- ISDA/CSA in place?
- Independent valuation source?
- Daily margin reporting capacity?
- Who approves trades?

---

7. Go-to-market conclusion

The first commercial wedge should be:

> Board-ready BTC treasury covered-call/collar policies for public BTC treasury companies and BTC-heavy miners.

The second should be:

> Miner production and runway hedging.

The analytics engine should remain the proprietary internal advantage:

> Cross-venue ETF / CME / Deribit / OTC vol normalization used to price, hedge, and risk-manage client structures.

Only after quote verification should the memo emphasize proprietary cross-venue arbitrage economics.

Treasury Case Study

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-treasury-case-study-10000btc-v0.md`

Idea 1 Treasury Case Study v0 — 10,000 BTC Holder

Purpose: Create the first board/investor proof artifact for the BTC treasury covered-call/collar wedge.

Prepared: 2026-05-14 CDT

Data basis: Screen-only Deribit BTC options snapshot from 2026-05-14 ~19:16 CDT.

Execution confidence: Screen-only, not quote-verified, not trade-verified.

Legal status: Illustrative research only. Not an offer, recommendation, or executable quote.

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1. Hypothetical client profile

Company: Public BTC treasury company

BTC holdings: 10,000 BTC

BTC reference price used in snapshot: approximately $81,600–$82,200

Gross BTC treasury value: approximately $816M–$822M

Program objective: Generate recurring premium income and/or define a downside floor while preserving the strategic long-term BTC thesis.

Core governance principle: Only a minority tranche is used.

Suggested initial program sizing:

- **Total BTC holdings:** 10,000 BTC
- **Program sleeve:** 1,000 BTC, or 10% of holdings
- **Uncovered/unencumbered strategic BTC:** 9,000 BTC, or 90% of holdings

This framing matters. The company is not “selling its BTC thesis.” It is monetizing or protecting a controlled treasury tranche.

---

2. Board-level framing

The board question should not be:

> Are we bearish on BTC?

The correct question is:

> Can a defined minority tranche of BTC volatility be converted into recurring income or downside protection without undermining the company’s long-term BTC accumulation strategy?

A treasury derivatives policy should define:

- maximum BTC encumbrance percentage;
- permitted instruments;
- maximum upside cap;
- minimum acceptable floor;
- tenor limits;
- counterparty/venue limits;
- margin/collateral rules;
- independent valuation process;
- reporting cadence;
- board/audit committee approvals;
- disclosure thresholds.

---

3. Structure A — 43-day covered-call sleeve

Snapshot tenor: Deribit 26JUN26, DTE 43

Underlying reference: ~$81,607

ATM IV: ~37.9%

Sleeve: 1,000 BTC

Call overwrite alternatives

10% OTM call — strike $90,000

- Moneyness: ~110.3% of spot
- Premium: ~0.01604 BTC per BTC
- Premium on 1,000 BTC: ~16.04 BTC
- USD equivalent: ~$1.31M
- Annualized premium vs BTC notional: ~13.6%
- Tradeoff: strongest income, but upside capped close to spot.

15% OTM call — strike $94,000

- Moneyness: ~115.2% of spot
- Premium: ~0.00856 BTC per BTC
- Premium on 1,000 BTC: ~8.56 BTC
- USD equivalent: ~$699k

- Annualized premium vs BTC notional: ~7.3%
- Tradeoff: more upside retained, lower income.

20%+ OTM call — strike $100,000

- Moneyness: ~122.5% of spot
- Premium: ~0.00334 BTC per BTC
- Premium on 1,000 BTC: ~3.34 BTC
- USD equivalent: ~$273k
- Annualized premium vs BTC notional: ~2.8%
- Tradeoff: preserves more convexity, but income is modest.

Interpretation

For a BTC treasury company, the 15% OTM version is the cleaner board candidate than the 10% OTM version. It generates visible income while reducing the risk that the company is perceived as selling too much upside.

Indicative board framing:

> Sell 43-day calls against 10% of BTC holdings, targeting 10–20% OTM strikes. Retain 90% of BTC fully uncapped. Use premium for operating expenses, debt service, preferred dividends, or BTC accumulation.

---

4. Structure B — 43-day downside floor / collar

Structure: Buy 74,000 put, sell 90,000 call on 1,000 BTC sleeve.

Tenor: 43 days.

Spot reference: ~$81,607.

Protection leg:

- Put strike: $74,000
- Floor: ~90.7% of spot
- Put premium: ~0.01939 BTC per BTC
- Cost on 1,000 BTC: ~19.39 BTC / ~$1.58M

Financing leg:

- Call strike: $90,000
- Cap: ~110.3% of spot
- Net collar cost: ~0.00335 BTC per BTC debit
- Net USD cost on 1,000 BTC: ~-$273k

Interpretation

This structure creates a short-term downside floor about 9% below spot and caps upside about 10% above spot on the 10% sleeve. It is close to zero-cost but still a small net debit in the screen snapshot.

Board fit: Moderate. Useful around a financing/refi/covenant window, but the cap is tight for a BTC-maximalist treasury.

Better use case:

- debt issuance window;
- preferred dividend/coupon period;
- expected market stress;
- temporary protection around a major corporate event.

---

5. Structure C — 43-day put-spread collar

Structure: Buy 74,000 put / sell 65,000 put, financed partly by selling 94,000 call / buying 105,000 call.

Tenor: 43 days.

Spot reference: ~$81,607.

Downside protection:

- Protection starts near $74,000 (~90.7% of spot)
- Protection band extends to $65,000 (~79.6% of spot)
- Protected band: ~11% of spot

Upside sold:

- Call spread starts near $94,000 (~115.2% of spot)
- Call spread ends near $105,000 (~128.7% of spot)
- Upside give-up band: ~13.5% of spot

Net screen cost:

- ~0.00678 BTC per BTC
- On 1,000 BTC: ~6.78 BTC
- USD equivalent: ~$553k

Interpretation

This is a cleaner BTC-treasury structure than a tight full collar. It protects a defined downside band while avoiding unlimited upside give-up above the call-spread cap. The company gives up some upside between ~$94k and ~$105k, but regains participation above the long call.

Board fit: Stronger than the full collar for high-conviction BTC treasuries.

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6. Structure D — 134-day covered-call sleeve

Snapshot tenor: Deribit 25SEP26, DTE 134

Underlying reference: ~$82,180

ATM IV: ~40.7%

Sleeve: 1,000 BTC

Call overwrite alternatives

~10% OTM call — strike $90,000

- Moneyness: ~109.5% of spot
- Premium: ~0.05936 BTC per BTC
- Premium on 1,000 BTC: ~59.36 BTC
- USD equivalent: ~$4.88M
- Annualized premium vs BTC notional: ~16.2%
- Tradeoff: large premium, tight cap for a 4.5-month tenor.

~15% OTM call — strike $95,000

- Moneyness: ~115.6% of spot
- Premium: ~0.04204 BTC per BTC
- Premium on 1,000 BTC: ~42.04 BTC
- USD equivalent: ~$3.45M
- Annualized premium vs BTC notional: ~11.5%
- Tradeoff: attractive income but still meaningful upside cap.

~20% OTM call — strike $98,000

- Moneyness: ~119.3% of spot
- Premium: ~0.03381 BTC per BTC
- Premium on 1,000 BTC: ~33.81 BTC
- USD equivalent: ~$2.78M
- Annualized premium vs BTC notional: ~9.2%
- Tradeoff: more palatable cap with still significant income.

Interpretation

The longer tenor creates more meaningful premium. But it also creates more headline risk if BTC rallies sharply. For a public BTC treasury, a 4.5-month overwrite should probably be implemented as a ladder, not a single block.

Preferred design:

- 3–4 staggered monthly/quarterly expiries;
- strikes 15–25% OTM;
- max 10% BTC encumbrance at launch;
- explicit board review before scaling.

---

7. Structure E — 134-day zero/low-cost collar

Structure: Buy 74,000 put, sell 90,000 call on 1,000 BTC sleeve.

Tenor: 134 days.

Spot reference: ~$82,180.

Protection leg:

- Put strike: $74,000
- Floor: ~90.0% of spot
- Put premium: ~0.05538 BTC per BTC
- Cost on 1,000 BTC: ~55.38 BTC / ~$4.55M

Financing leg:

- Call strike: $90,000
- Cap: ~109.5% of spot
- Net collar credit: ~0.00398 BTC per BTC
- Net USD credit on 1,000 BTC: ~$327k

Interpretation

This collar is approximately self-financing in the screen snapshot, but it caps upside tightly for 134 days. That is commercially dangerous for a BTC treasury company unless tied to a specific obligation or risk window.

Use only when:

- the company has a specific financing/covenant/refi deadline;
- the board prioritizes balance-sheet floor over upside on the sleeve;
- the sleeve is small and explicitly disclosed as risk management.

---

8. Structure F — 134-day put-spread collar

Structure: Buy 74,000 put / sell 66,000 put, financed by selling 95,000 call / buying 105,000 call.

Tenor: 134 days.

Spot reference: ~$82,180.

Downside protection:

- Protection starts near $74,000 (~90.0% of spot)
- Protection extends to $66,000 (~80.3% of spot)
- Protected band: ~9.7% of spot

Upside sold:

- Call-spread starts near $95,000 (~115.6% of spot)
- Call-spread ends near $105,000 (~127.8% of spot)
- Upside give-up band: ~12.2% of spot

Net screen cost:

- ~0.00432 BTC per BTC
- On 1,000 BTC: ~4.32 BTC
- USD equivalent: ~$355k

Interpretation

This is the best board-friendly structure in the screen set. It protects a meaningful downside band, avoids unlimited upside cap, and has a relatively modest net cost versus the protected notional.

Board fit: Strong.

Narrative:

> The company protects a defined drawdown band on 10% of BTC holdings while preserving most upside and retaining 90% of BTC entirely unencumbered.

---

9. Recommended launch policy

Initial policy limits

- Maximum program sleeve: 10% of BTC holdings.
- Maximum single-expiry sleeve: 3–5% of BTC holdings.
- Preferred tenors: 30–150 days.
- Preferred covered-call strikes: 15–25% OTM.
- Preferred collar floors: 80–90% of spot.
- Avoid full upside caps unless tied to specific liability/covenant windows.
- Prefer call-spread financing over naked call overwrites for longer tenors.

Recommended first live-style package

For a 10,000 BTC holder:

1. **Income sleeve:**
- 500 BTC in staggered covered calls.
- 15–25% OTM.
- 30–90 day maturities.

2. **Protection sleeve:**
- 500 BTC in put-spread collars.
- floor starts near 85–90% of spot.
- use call spread rather than outright call sale where possible.

3. **Unencumbered reserve:**

• 9,000 BTC untouched.

This creates proof of discipline without over-optimizing the first trade.

---

10. Economics summary from screen snapshot

For a 1,000 BTC sleeve:

43-day tenor

• 15% OTM covered call could generate roughly **8.56 BTC / ~$699k**.

• 20%+ OTM covered call could generate roughly **3.34 BTC / ~$273k**.

• 90% floor / 110% cap collar costs roughly **3.35 BTC / ~$273k**.

• 90/80 put-spread funded by 115/129 call-spread costs roughly **6.78 BTC / ~$553k**.

134-day tenor

• 15% OTM covered call could generate roughly **42.04 BTC / ~$3.45M**.

• 20% OTM covered call could generate roughly **33.81 BTC / ~$2.78M**.

• 90% floor / 109.5% cap collar produces roughly **3.98 BTC / ~$327k credit**.

• 90/80 put-spread funded by 116/128 call-spread costs roughly **4.32 BTC / ~$355k**.

Again: these are screen-only approximations, not executable quotes.

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11. Key risk disclosure for board package

Upside opportunity cost

Covered calls and collars may underperform unhedged BTC in sharp rallies.

Mark-to-market volatility

Even if held to expiry, option positions can show interim losses/gains.

Collateral and liquidity

Some structures may require collateral, margin, or asset encumbrance.

Counterparty/venue risk

Deribit, CME, ETF options, and OTC each carry different counterparty, clearing, custody, and settlement risks.

Basis/wrapper risk

BTC direct holdings, ETF shares, CME futures, and Deribit options are not identical under stress.

Disclosure and governance

A public company must consider materiality, risk disclosure, board authorization, auditor review, and insider/public communication controls.

Narrative risk

Investors may misunderstand hedging as bearishness. Communications must emphasize limited sleeve, treasury discipline, and preservation of long-term BTC ownership.

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12. What needs verification before client use

1. Quote-verify each structure with at least two counterparties or executable venues.
2. Confirm whether client holds direct BTC, ETF shares, or both.
3. Confirm unencumbered BTC and custody/collateral restrictions.
4. Review board-approved treasury policy.
5. Review debt/preferred/convert covenants.
6. Confirm accounting treatment with auditor.
7. Confirm legal authority to advise, structure, or execute.
8. Validate independent pricing/marks.
9. Stress test BTC +25%, +50%, -25%, -50% scenarios.
10. Produce disclosure-safe language.

---

13. Conclusion

This case study supports the commercial wedge.

A 10,000 BTC treasury can use a **10% sleeve** to generate meaningful premium income or create a downside floor without compromising the strategic ownership of the remaining 90%.

The most board-friendly first structures are not aggressive full covered calls. They are:

1. staggered 15–25% OTM covered calls on a small sleeve; and
2. put-spread collars that protect a drawdown band while avoiding unlimited upside give-up.

This is the product that can open client conversations. The cross-venue vol monitor is the pricing and hedge engine behind it.

Miner Production Hedge Case Study

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-miner-production-hedge-case-study-v0.md`

Idea 1 Miner Production Hedge Case Study v0

Purpose: Create a proof artifact for the Bitcoin miner hedging wedge: protect production/cash-flow runway without forcing spot sales or abandoning BTC upside.

Prepared: 2026-05-14 19:24 CDT

Data basis: Screen-only Deribit BTC options snapshot from 2026-05-14 ~19:24 CDT.

Execution confidence: Screen-only, not quote-verified, not trade-verified.

Legal status: Illustrative research only. Not an offer, recommendation, or executable quote.

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1. Hypothetical client profile

Company: Public Bitcoin miner

Monthly BTC production assumption: 150 BTC

3-month production forecast: 450 BTC

Hedge ratio: 50% of conservative 3-month production

Hedged BTC: 225 BTC

BTC reference price: approximately $81,600–$82,200

Hedged notional: approximately $18.4M

Illustrative operating context:

- quarterly power/opex burden: high and USD-denominated;
- ASIC/site/capex needs: ongoing;
- equity story: long BTC beta and operational leverage;
- board concern: avoid forced BTC sales or dilutive equity during BTC drawdowns;
- lender concern: protect liquidity/collateral if BTC falls.

The product should be framed as **runway protection**, not bearish BTC speculation.

---

2. Why miners need a different hedge than treasuries

BTC treasury companies primarily hedge **inventory/NAV volatility**.

Miners hedge a moving target:

- future production volume is uncertain;
- realized BTC revenue depends on hashprice, difficulty, uptime, and pool economics;
- USD liabilities are recurring and non-discretionary;
- power costs and curtailment can spike;
- capex timing can force liquidity decisions;
- BTC inventory may already be pledged to lenders.

Therefore the first rule is:

> Hedge conservative production, not optimistic hashrate targets.

A miner hedge should usually start at 25–50% of conservative expected production, not 100%.

---

3. Product objective

The objective is not to maximize option P&L.

The objective is to protect a minimum USD revenue band for a defined production window while preserving most upside.

A board-ready miner policy should answer:

- How much expected production can be hedged?
- Are existing BTC holdings pledged or unencumbered?
- Is the hedge tied to power/capex/debt-service dates?
- What downside price creates forced-sale or equity-raise risk?
- What upside can the company afford to cap on the hedged slice?
- What collateral/margin/counterparty path is allowed?

---

4. Structure A — 43-day protective puts

Tenor: Deribit 26JUN26, DTE 43

BTC reference: ~$81,580

ATM IV: ~37.9%

Hedged amount: 225 BTC

Protective put alternatives:

$74,000 put

- Floor: ~90.7% of spot
- Premium: ~0.01938 BTC per BTC
- Cost on 225 BTC: ~$356k
- Cost as % of hedged notional: ~1.94%
- Best for: near-term high-confidence protection around power/capex/lender window.

$70,000 put

- Floor: ~85.8% of spot
- Premium: ~0.01117 BTC per BTC
- Cost on 225 BTC: ~$205k
- Cost as % of hedged notional: ~1.12%
- Best for: budget protection with less premium drag.

$65,000 put

- Floor: ~79.7% of spot
- Premium: ~0.00583 BTC per BTC
- Cost on 225 BTC: ~$107k
- Cost as % of hedged notional: ~0.58%
- Best for: disaster insurance only.

Interpretation

Outright puts are easiest for a board to understand: pay premium, retain all upside, receive protection below the strike.

But miners are often cash-sensitive. A $356k premium on a 225 BTC hedge may be acceptable around a specific risk window, but recurring outright puts can become expensive.

---

5. Structure B — 43-day collars

Tenor: 43 days

Hedged amount: 225 BTC

Screen-only collar alternatives:

$74,000 put / $94,000 call

- Floor: ~90.7% of spot
- Cap: ~115.2% of spot
- Net cost: ~$201k debit
- Fit: strong floor, still reasonable upside cap.

$70,000 put / $100,000 call

- Floor: ~85.8% of spot

• Cap: ~122.6% of spot

• Net cost: ~$145k debit

• Fit: more balanced miner hedge.

$65,000 put / $100,000 call

• Floor: ~79.7% of spot

• Cap: ~122.6% of spot

• Net cost: ~$47k debit

• Fit: low-cost disaster hedge.

Interpretation

For miners, the **$70k put / $100k call** looks like the cleanest 43-day structure in this snapshot:

• protects below roughly 86% of spot;

• allows BTC to rally more than 20% before cap binds;

• costs materially less than outright put protection;

• matches a short runway/capex window.

---

6. Structure C — 43-day put-spread collar

Structure: Buy $74,000 put / sell $62,000 put, financed partly by selling $94,000 call / buying $100,000 call.

Tenor: 43 days

Hedged amount: 225 BTC

Economics:

• Protection starts: $74,000

• Protection ends: $62,000

• Protected price band: $12,000 per BTC

• Protected USD band on 225 BTC: ~$2.70M

• Upside sold from $94,000 to $100,000

• Upside band sold on 225 BTC: ~$1.35M

• Net cost: ~$186k debit

Interpretation

This is a useful miner structure because it protects a defined stress band and avoids unlimited upside give-up.

It is better than a tight full collar when the miner wants to preserve upside beta for equity investors.

---

7. Structure D — 134-day protective puts

Tenor: Deribit 25SEP26, DTE 134

BTC reference: ~$82,150

ATM IV: ~40.7%

Hedged amount: 225 BTC

Protective put alternatives:

$74,000 put

- Floor: ~90.1% of spot
- Cost on 225 BTC: ~$1.02M
- Cost as % of hedged notional: ~5.54%
- Fit: expensive but strong protection for a full quarter-plus risk window.

$70,000 put

- Floor: ~85.2% of spot
- Cost on 225 BTC: ~$754k
- Cost as % of hedged notional: ~4.08%
- Fit: balanced but still premium-heavy.

$66,000 put

- Floor: ~80.3% of spot
- Cost on 225 BTC: ~$545k
- Cost as % of hedged notional: ~2.95%
- Fit: disaster protection.

Interpretation

Outright 134-day puts are clean but expensive. They may be appropriate if the miner faces a specific debt maturity, site energization delay, ASIC payment, or lender review period.

For normal recurring hedging, collars or put spreads are likely more efficient.

---

8. Structure E — 134-day collars

Tenor: 134 days

Hedged amount: 225 BTC

Screen-only collar alternatives:

$74,000 put / $94,000 call

- Floor: ~90.1% of spot
- Cap: ~114.4% of spot
- Net cost: ~$192k debit
- Fit: strong protection but relatively tight cap over 4.5 months.

$70,000 put / $98,000 call

- Floor: ~85.2% of spot
- Cap: ~119.3% of spot
- Net cost: ~$132k debit
- Fit: balanced quarterly/capex hedge.

$66,000 put / $102,000 call

- Floor: ~80.3% of spot
- Cap: ~124.2% of spot
- Net cost: ~$78k debit
- Fit: low-cost disaster hedge with wide upside room.

Interpretation

The **$70k put / $98k call** is the best balanced quarterly hedge in this snapshot. It protects below ~85% of spot while leaving nearly 20% upside on the hedged production.

For a miner, this is commercially easier than outright puts because it limits premium outlay.

---

9. Structure F — 134-day put-spread collar

Structure: Buy $74,000 put / sell $62,000 put, financed partly by selling $94,000 call / buying $102,000 call.

Tenor: 134 days

Hedged amount: 225 BTC

Economics:

- Protection starts: $74,000
- Protection ends: $62,000
- Protected price band: $12,000 per BTC
- Protected USD band on 225 BTC: ~$2.70M
- Upside sold from $94,000 to $102,000
- Upside band sold on 225 BTC: ~$1.80M
- Net cost: ~$265k debit

Interpretation

This is the cleanest board/lender structure when the miner wants protection but also wants to avoid a full upside cap.

It creates a defined stress buffer for the next 4.5 months while preserving upside outside the sold call-spread band.

---

10. Downside scenario impact

Using the 43-day snapshot and a 225 BTC hedge:

If BTC falls **20%** from ~$81,580 to roughly ~$65,264:

- unhedged 225 BTC production value declines by roughly $3.67M versus spot reference;
- a $74,000 put floor pays approximately **$1.97M** before premium/friction;
- this can fund a meaningful portion of power, payroll, or capex during drawdown.

If BTC falls **30%** to roughly ~$57,106:

- unhedged 225 BTC production value declines by roughly $5.51M versus spot reference;
- a $74,000 put floor pays approximately **$3.80M** before premium/friction;
- this materially reduces forced-sale or emergency-equity risk.

Using the 134-day snapshot:

- a $74,000 put floor pays approximately **$1.86M** at -20%;
- approximately **$3.71M** at -30%;
- before premium/friction.

The point is not that every miner should buy 90% floors. The point is that the product directly translates BTC volatility into runway protection.

---

11. Recommended miner launch policy

Hedge-sizing policy

- Hedge only 25–50% of conservative expected production at launch.
- Do not hedge optimistic production targets.
- Exclude production already committed under financing or hosting arrangements.
- Separately classify treasury BTC as:
- pledged;
- restricted;
- operational liquidity;
- strategic reserve.

Product policy

Preferred first structures:

1. **Quarterly production collar**
• floor: 80–90% of spot;
• cap: 115–125% of spot;
• tenor: 60–150 days.

2. **Put-spread collar**
• protect a defined stress band;
• use call spread to avoid unlimited upside give-up;
• best for board/lender conversations.

3. **Outright puts**
• use around specific high-risk windows;
• debt maturity, lender review, site capex, power-price stress, or expected financing.

Avoid:

• 100% production hedges;
• full forward sales unless the company explicitly wants no upside;
• short naked optionality beyond board-approved BTC inventory/production;
• structures requiring collateral the company cannot post under stress.

---

12. Board/lender package language

A miner can present the program as:

> The company is implementing a limited production hedge program on a conservative portion of expected BTC production to protect power, capex, and debt-service runway during BTC drawdowns, while preserving the majority of upside exposure for shareholders.

For lenders:

> The hedge reduces downside revenue volatility and supports minimum liquidity planning during collateral stress periods.

For shareholders:

> The program does not eliminate BTC upside. It only hedges a minority of expected production and uses collars/put spreads to retain upside participation.

---

13. Qualification checklist before client terms

1. Actual trailing monthly BTC production.
2. Conservative 3–6 month production forecast.
3. Hashrate, uptime, pool, and difficulty assumptions.
4. Expected monthly power cost and curtailment plan.
5. ASIC/site/capex payment schedule.
6. Debt, convert, lease, or project-finance covenants.
7. BTC held in treasury and whether pledged/restricted.
8. Normal policy: sell production daily, hold BTC, or mixed?
9. Permitted derivatives and approved venues/counterparties.
10. Collateral and margin capacity.
11. Board/audit/lender approval process.
12. Required accounting/tax treatment.

---

14. Why this proof artifact matters

This case study makes the miner wedge concrete:

- It ties option structures to power/capex/debt runway.
- It avoids overhedging.
- It preserves the miner's upside equity narrative.
- It gives lenders and boards a practical risk framework.
- It creates recurring monitoring/reporting demand.

This is a real client-flow product, not a speculative vol-arb pitch.

---

15. Best next miner package

For the first client-ready prototype:

- assume 450 BTC conservative 3-month production;
- hedge 225 BTC, or 50%;
- show three options:

1. 85% floor / 120% cap quarterly collar;
2. 90/75 put-spread funded by 115/125 call-spread;
3. outright 80–85% disaster put;

- present the effect on cash runway under BTC -20%, -30%, and -40%.

The preferred institutional product is the **quarterly production collar or put-spread collar**, not a full forward sale.

BTC Treasury Board Policy Template

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-btc-treasury-board-policy-template-v0.md`

Idea 1 BTC Treasury Derivatives Policy Template v0

Purpose: Board-level policy template for a public or private company holding strategic BTC reserves and considering limited BTC option overlays for income generation, downside protection, or liability/runway management.

Status: Template for counsel/auditor/board review. Not legal, tax, accounting, investment, or derivatives advice.

Use case: BTC treasury covered-call, collar, and put-spread collar programs.

---

1. Board resolution summary

The board authorizes management to evaluate and, subject to final approvals, implement a limited BTC treasury derivatives program designed to:

1. preserve the company's long-term strategic BTC position;
2. generate recurring option premium on a defined minority sleeve of holdings;
3. provide partial downside protection against severe BTC drawdowns;
4. support liquidity planning for debt service, preferred dividends, operating runway, or capital expenditures;
5. avoid forced BTC sales during market stress;
6. operate within strict risk, collateral, accounting, disclosure, and counterparty controls.

The program is not intended to convert the company into a trading company, investment fund, commodity pool, or speculative derivatives vehicle.

---

2. Strategic principles

2.1 Preserve strategic BTC ownership

The company's core BTC holdings remain strategic treasury assets. Derivatives may only be applied to a defined minority sleeve of holdings or economically equivalent exposure.

2.2 No uncontrolled leverage

The program may not create open-ended leverage, uncapped losses, or collateral obligations that could impair the company's operating liquidity.

2.3 Board-governed optionality monetization

Covered-call sales, collars, and put-spread collars must be structured as disciplined treasury overlays, not discretionary trading.

2.4 Downside protection before yield maximization

The company may monetize volatility, but the program should prioritize liquidity resilience, risk control, and balance-sheet durability over maximum premium income.

2.5 Transparency and auditability

All trades must be documented, independently valued, reconciled, and reviewed under accounting, disclosure, and internal-control procedures.

---

3. Permitted objectives

The program may be used for the following objectives:

- generate option premium on a capped minority sleeve of BTC exposure;
- fund or partially offset operating expenses, debt service, preferred dividends, taxes, or capex;
- establish downside floors or drawdown bands;
- reduce probability of forced BTC liquidation;
- improve board/lender visibility into treasury liquidity under stress;
- evaluate institutional risk-transfer alternatives across ETF, CME, OTC, and other legally approved venues.

The program may not be used for:

- speculative directional trading;
- increasing net BTC exposure beyond approved treasury policy;
- naked short option exposures without fully approved risk limits;
- derivatives unrelated to the company's BTC treasury exposure or liquidity needs;
- trades that would materially alter the company's business classification without board/counsel approval.

---

4. Eligible exposure

Eligible exposure may include:

1. unencumbered BTC held by the company;
2. BTC held through approved custodians;
3. spot BTC ETF shares held by the company;
4. CME futures/options or OTC derivatives used solely as hedging/overlay instruments after counsel review;

5. other economically equivalent BTC exposure approved by the board.

Excluded exposure:

- BTC pledged to lenders unless lender consent is received;
- BTC subject to liens, lockups, custody restrictions, or contractual encumbrances;
- BTC needed for near-term operations or collateral;
- BTC for which derivative overlay would create disclosure, accounting, tax, or covenant issues not yet reviewed.

---

5. Approved instruments

5.1 Initially permitted, subject to execution approval

- covered calls on approved BTC exposure;
- protective puts;
- collars;
- put-spread collars;
- call-spread-financed put spreads;
- listed BTC ETF options;
- CME BTC futures/options where appropriate;
- OTC BTC options/swaps under approved ISDA/CSA or equivalent documentation.

5.2 Not permitted unless separately approved

- naked short puts;
- naked short calls beyond covered exposure;
- leveraged option packages with uncapped downside;
- exotic options with path-dependent margin or valuation complexity;
- autocallables/reverse convertibles issued by the company;
- transactions with affiliated or conflicted counterparties;
- offshore exchange transactions unless counsel specifically approves.

---

6. Program limits

6.1 Total program sleeve

Initial maximum program sleeve:

- **10% of total BTC-equivalent holdings**, unless board approves otherwise.

Hard maximum without renewed board approval:

- **20% of total BTC-equivalent holdings**.

6.2 Single-expiry limit

Maximum exposure in any single expiry:

- **3–5% of total BTC-equivalent holdings**.

6.3 Tenor limits

Permitted tenor range:

- **30–180 days**.

Requires special approval:

- less than 14 days;
- more than 180 days;
- event-specific tenors around earnings, financing, covenant tests, or major BTC market events.

6.4 Strike/floor/cap guidelines

Covered calls:

- initial guideline: **15–25% OTM**;
- calls inside 10% OTM require special approval unless tied to a liability window.

Protective puts/collars:

- floors typically between **80–90% of spot**;
- floors above 90% require premium/cost justification.

Put-spread collars:

- long put generally 85–90% of spot;
- short put generally 70–80% of spot;
- short call generally 115–125% of spot;
- long call generally 125–140% of spot where used to cap upside give-up.

6.5 Premium/cost limits

Maximum net premium paid for protection in a single program cycle:

- [X]% of covered notional, or
- board-approved dollar budget.

Minimum premium/yield hurdle for covered-call sales:

- [X]% annualized after estimated costs, unless trade is part of a broader collar.

6.6 Collateral and liquidity limits

The program may not require collateral or margin exceeding:

- [X]% of unrestricted cash;
- [X]% of unencumbered BTC;
- [X]% of projected 90-day operating liquidity.

Management must maintain stress liquidity for:

- BTC ±20%, ±30%, ±40% moves;
- volatility spike;
- margin haircut increase;
- counterparty downgrade;
- exchange or custodian outage.

---

7. Counterparty and venue policy

Permitted venues/counterparties may include:

- regulated broker-dealers/options brokers;
- FCMs for CME products;
- regulated banks or swap dealers;
- approved OTC derivatives counterparties;
- approved prime brokers/custodians;
- other venues approved by counsel and the board.

Counterparty requirements:

- KYC/AML/sanctions review;
- credit review;
- regulatory status review;
- documentation review;
- margin/collateral terms;
- valuation/dispute process;
- close-out/netting analysis;
- operational settlement review;
- no material conflicts without board disclosure.

Counterparty concentration limit:

- No more than [X]% of derivatives exposure or collateral with one counterparty unless approved.

---

8. Documentation requirements

Before execution, each transaction must have:

- approved trade ticket;
- structure description;
- objective and exposure being hedged/monetized;
- notional and BTC-equivalent exposure;
- expiry;
- strikes;
- premium/debit/credit;
- Greeks and scenario analysis;
- margin/collateral estimate;
- source of quote;
- execution-confidence label;
- accounting treatment memo or preliminary review;
- disclosure/covenant checklist;
- approval signatures.

For OTC transactions:

- ISDA Master Agreement;
- Schedule;
- Credit Support Annex/Deed;
- confirmations;
- digital asset definitions;
- reference price/source/fallback language;
- fork/airdrop/protocol event treatment;
- valuation agent and dispute mechanics;
- close-out provisions.

---

9. Approval authority

9.1 Board approval required

- initial program adoption;
- changes to total program sleeve;
- new instrument categories;
- new counterparty categories;
- OTC documentation templates;

- structured products;
- principal risk-taking beyond treasury overlay;
- exceptions to risk limits.

9.2 Committee approval required

- quarterly program plan;
- counterparty list;
- maximum notional per tenor;
- premium/cost budget;
- collateral allocation;
- disclosure approach.

9.3 Management approval permitted

Within approved limits, management may execute trades if authorized by:

- CFO / Treasurer;
- Chief Risk Officer or equivalent;
- General Counsel / outside counsel signoff where required;
- Controller/accounting signoff for treatment/valuation.

9.4 Emergency authority

Emergency actions may include:

- closing positions;
- reducing exposure;
- transferring collateral;
- suspending the program.

Emergency trades must be reported to the board or committee within [24/48] hours.

---

10. Accounting, valuation, and controls

The company must maintain procedures for:

- ASC 815 derivative accounting review;
- FASB crypto asset fair-value treatment review;
- hedge accounting analysis if applicable;
- independent valuation;
- daily/weekly mark-to-market;
- confirmation matching;

- collateral reconciliation;
- journal-entry review;
- fair-value hierarchy support;
- disclosure controls;
- internal control over financial reporting.

Valuation sources should be approved and ranked:

1. executed trade/clearing price;
2. exchange settlement/official mark;
3. independent broker/dealer quotes;
4. vendor marks;
5. internal model, with documented assumptions.

---

11. Disclosure and communications policy

All external communications regarding the program must be reviewed by legal counsel.

Potential disclosure areas:

- risk factors;
- MD&A liquidity/trends;
- market-risk disclosures;
- derivative accounting disclosures;
- material agreement / Form 8-K analysis;
- covenant/collateral impact;
- non-GAAP or performance metrics;
- investor presentations;
- social media or executive public statements.

Prohibited language:

- guaranteed yield;
- risk-free income;
- arbitrage profit;
- no downside risk;
- executable pricing based only on screen marks;
- statements implying the program eliminates BTC volatility.

Preferred language:

- partial premium monetization;
- defined downside band;

- minority sleeve;
- screen-only / quote-verified / trade-verified;
- board-approved risk limits;
- subject to market, liquidity, accounting, legal, and counterparty risk.

---

12. Reporting package

Management should provide the board/committee a report at least monthly, or more frequently during market stress.

Report fields:

- total BTC-equivalent holdings;
- program sleeve used;
- open positions by expiry/strike;
- premiums received/paid;
- mark-to-market P&L;
- realized P&L;
- collateral posted;
- available liquidity;
- counterparty exposure;
- Greeks by position and aggregate;
- stress scenarios;
- upcoming expiries;
- exceptions/breaches;
- disclosure/accounting updates.

Stress scenarios:

- BTC -20%, -30%, -40%;
- BTC +20%, +50%, +100%;
- volatility +25/+50 vol points;
- liquidity gap / no bid;
- counterparty default;
- custodian outage;
- collateral haircut increase.

---

13. Risk limits and escalation

Immediate escalation required if:

- collateral usage exceeds [X]% of limit;
- BTC price breaches defined stress level;
- counterparty fails margin/payment obligation;
- valuation dispute exceeds [X];
- position exceeds approved notional/tenor/strike limits;
- covenant headroom changes materially;
- auditor flags accounting/control issue;
- legal counsel flags disclosure or regulatory concern;
- program loss exceeds [X] threshold;
- operational/custody issue affects settlement.

Possible escalation actions:

- suspend new trades;
- reduce or close positions;
- add collateral;
- novate or transfer exposure;
- change counterparty limits;
- notify lenders/auditors/board;
- update disclosure.

---

14. Example approved starter program

For a company holding **10,000 BTC**:

Program size

- total program sleeve: 1,000 BTC / 10%;
- max single expiry: 300–500 BTC;
- 9,000 BTC remains untouched.

Income sleeve

- 500 BTC;
- staggered 30–90D covered calls;
- 15–25% OTM strikes;
- no more than 250 BTC per expiry.

Protection sleeve

- 500 BTC;
- 90–150D put-spread collars;
- long put 85–90%;
- short put 70–80%;

- short call 115–125%;
- long call 130–140% where appropriate.

Program review

- monthly mark/report;
- quarterly board review;
- immediate review after BTC ±30% move or major volatility spike.

---

15. Pre-trade checklist

Before any transaction:

- [ ] Exposure is eligible and unencumbered.
- [ ] Trade fits permitted instrument list.
- [ ] Notional is within program sleeve.
- [ ] Expiry is within tenor limit.
- [ ] Strike/floor/cap meets guidelines or exception approved.
- [ ] Quote source and confidence label recorded.
- [ ] All-in cost/premium calculated.
- [ ] Margin/collateral stress tested.
- [ ] Accounting review completed.
- [ ] Disclosure review completed.
- [ ] Covenant/lender review completed.
- [ ] Counterparty approved.
- [ ] Trade ticket approved by authorized signers.
- [ ] Settlement/custody process confirmed.
- [ ] Post-trade reporting process ready.

---

16. Post-trade checklist

After execution:

- [ ] Confirm trade economics against ticket.
- [ ] Reconcile confirmation.
- [ ] Book accounting entry.
- [ ] Update position report.
- [ ] Update collateral report.
- [ ] Update Greeks/stress table.

- [ ] Store quote/execution evidence.
- [ ] Monitor disclosure/materiality triggers.
- [ ] Schedule expiry/roll review.

---

17. Annual review

At least annually, the board or committee should review:

- whether program objectives remain valid;
- performance vs objectives;
- realized and unrealized P&L;
- liquidity and collateral impact;
- accounting/audit findings;
- disclosure history;
- counterparty performance;
- changes in BTC market structure;
- regulatory developments;
- changes to risk limits or approved instruments.

---

18. Policy adoption language

The board adopts this BTC Treasury Derivatives Policy as of [date], subject to final legal, tax, accounting, auditor, lender, and counterparty review.

No transaction may be executed unless it complies with this policy or receives documented exception approval.

The board may amend, suspend, or terminate the program at any time.

---

19. Important disclaimer

This template is for institutional planning and policy design only. It is not legal, tax, accounting, investment, derivatives, or fiduciary advice. Any company considering BTC derivatives should consult qualified securities, commodities, tax, accounting, auditor, lender, and derivatives counsel before executing any transaction.

Miner Hedge Calculator Spec

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-miner-hedge-calculator-spec-v0.md`

Idea 1 Miner Hedge Calculator Specification v0

Purpose: Define the calculator needed to convert the miner hedging concept into a repeatable board/lender-ready analysis package.

Status: Product specification. Not investment, derivatives, legal, tax, or accounting advice. All pricing remains screen-only unless quote-verified.

---

1. Objective

The miner hedge calculator answers one question:

> How much BTC price protection should a miner buy or structure against conservative expected production so that power, debt service, capex, and runway are protected without overhedging or eliminating strategic upside?

The calculator should produce:

- conservative production exposure;
- hedge notional;
- structure economics;
- premium/cost;
- downside runway protection;
- upside cap/give-up;
- collateral/margin impact;
- lender/covenant checklist;
- board-ready recommendation.

---

2. Target users

Internal users

- structuring analyst;
- BTC Vol Desk Monitor operator;
- risk lead;

- sales/capital markets lead.

External users later

- miner CFO;
- treasurer;
- CEO;
- board/risk committee;
- lender/project-finance counterparty;
- auditor/counsel.

---

3. Core design principle

The calculator must prevent overhedging.

It should default to conservative assumptions and force the user to distinguish:

- current BTC inventory;
- unencumbered BTC inventory;
- pledged/restricted BTC;
- forecast production;
- conservative production;
- optimistic production;
- BTC price hedge;
- hashprice hedge;
- power cost hedge.

Initial calculator scope is **BTC price hedge only**, not full hashprice/power hedging.

---

4. Input sections

4.1 Company / scenario metadata

Required fields:

- company name;
- ticker, if public;
- date/time/timezone;
- analyst;
- scenario name;

- source confidence;
- execution confidence;
- legal status: internal / counsel reviewed / client approved;
- notes.

4.2 BTC market assumptions

Required fields:

- BTC spot/reference price;
- source: Deribit index / CME reference / Coinbase / composite / vendor;
- reference timestamp;
- implied volatility source;
- risk-free rate;
- funding/collateral rate;
- quote source: model / screen / indicative / firm / trade;
- bid/ask spread assumption;
- slippage assumption;
- fees assumption.

Derived fields:

- BTC shock scenarios: -10%, -20%, -30%, -40%, -50%, +20%, +50%, +100%;
- spot scenario prices.

4.3 Miner operating assumptions

Required fields:

- current deployed hashrate;
- expected deployed hashrate over hedge period;
- fleet efficiency;
- uptime assumption;
- pool fee;
- curtailment assumption;
- network difficulty assumption or production forecast source;
- monthly production forecast;
- conservative production haircut;
- conservative monthly production;
- hedge horizon in months;

- total conservative production.

Simplified v0 option:

- enter monthly BTC production directly;
- enter conservative haircut;
- calculator derives conservative production.

Example:

- monthly production: 150 BTC;
- conservative haircut: 0%;
- horizon: 3 months;
- conservative production: 450 BTC.

4.4 Treasury / balance sheet inputs

Required fields:

- current BTC inventory;
- unencumbered BTC inventory;
- pledged/restricted BTC;
- cash balance;
- debt service due in hedge window;
- power cost due in hedge window;
- capex due in hedge window;
- other fixed obligations;
- minimum liquidity reserve;
- BTC-backed loan exposure, if any;
- lender collateral thresholds.

Derived fields:

- total obligations in hedge window;
- unhedged runway at current BTC price;
- unhedged runway at BTC shock prices;
- required protected value.

4.5 Hedge policy inputs

Required fields:

- hedge ratio: default 25–50% of conservative production;

- maximum hedge ratio allowed;
- permitted instruments;
- max premium budget;
- max collateral usage;
- minimum acceptable floor;
- maximum acceptable upside cap;
- hedge horizon;
- approved venues/counterparties;
- whether BTC inventory can be used as collateral;
- whether cash collateral is available.

Derived fields:

- hedge notional in BTC;
- hedge notional in USD;
- hedge notional as % of conservative production;
- hedge notional as % of unencumbered BTC inventory;
- hedge notional as % of total enterprise BTC exposure.

---

5. Supported structures

5.1 Outright protective put

Inputs:

- strike percent of spot: default 80%, 85%, 90%;
- expiry;
- premium source;
- premium quote.

Outputs:

- premium BTC/USD;
- premium as % of hedged notional;
- floor price;
- payoff at shock scenarios;
- net protection after premium;
- no upside cap.

Use case:

- miner wants disaster protection and no upside give-up.

5.2 Production collar

Inputs:

- put strike percent: default 80%, 85%, 90%;
- call strike percent: default 115%, 120%, 125%;
- expiry;
- premium source.

Outputs:

- put premium;
- call premium;
- net debit/credit;
- floor price;
- cap price;
- downside protection;
- upside give-up above cap;
- net scenario P&L;
- collateral/margin impact.

Use case:

- miner wants low-cost protection and can accept upside cap on limited production.

5.3 Put-spread collar

Inputs:

- long put percent: default 90%;
- short put percent: default 75–80%;
- short call percent: default 115–120%;
- long call percent: default 130–140%;
- expiry;
- premium source.

Outputs:

- net premium;
- protected downside band;
- max protection;
- upside band sold;
- max upside give-up;
- residual tail risk below short put;

- scenario P&L;
- collateral/margin.

Use case:

- board/lender-friendly hedge that controls cost and avoids unlimited upside sale.

5.4 Forward/futures hedge — later version

Not in v0 unless explicitly enabled.

Reason:

- eliminates upside;
- creates margin/collateral needs;
- more sensitive to accounting/covenant concerns.

Future outputs:

- locked price;
- basis;
- margin;
- cash-flow impact;
- hedge accounting analysis.

---

6. Pricing engine requirements

6.1 v0 model pricing

For internal screen-only estimates:

- Black-Scholes style approximation for European options;
- use BTC spot/reference;
- risk-free/funding rate;
- IV from Deribit/IBIT/CME/quote source;
- interpolate IV by moneyness/expiry where needed;
- show model limitations.

All v0 model results must be labeled:

> screen-only model estimate — quote verification required.

6.2 Quote-verified pricing

When quotes are available, quote data overrides model pricing.

Quote fields:

- quote ID;
- counterparty;
- quote timestamp;
- quote good-until;
- firm/indicative;
- bid/ask;
- size;
- fees;
- margin/collateral;
- settlement;
- documentation required.

6.3 Price hierarchy

Use this hierarchy:

1. executed trade;
2. firm quote;
3. indicative quote;
4. executable broker/venue screen;
5. vendor screen;
6. public screen;
7. model estimate.

---

7. Scenario outputs

The calculator should produce a scenario table for BTC:

- -50%;
- -40%;
- -30%;
- -20%;
- -10%;
- unchanged;
- +20%;
- +50%;

- +100%.

For each scenario, show:

- BTC price;
- unhedged production value;
- hedge payoff;
- premium cost;
- net hedged value;
- incremental runway protection;
- cap/give-up if BTC rallies;
- collateral/margin stress;
- obligations covered;
- remaining liquidity gap.

---

8. Board output summary

The top of the output should contain a plain-English recommendation:

Example:

> Hedge 225 BTC, equal to 50% of conservative 3-month production, using a 90D 85% floor / 120% cap collar. Based on current screen-only pricing, the structure protects approximately $X of downside value in a 30% BTC drawdown while preserving upside up to $Y/BTC. Quote verification and lender/covenant review are required before execution.

Required board summary fields:

- recommended hedge ratio;
- hedge notional;
- structure;
- floor;
- cap;
- premium/debit/credit;
- downside protection at -20%, -30%, -40%;
- upside give-up at +20%, +50%, +100%;
- collateral impact;
- legal/accounting/covenant status;
- confidence label.

---

9. Lender/covenant output

The calculator should flag:

- hedge notional vs allowed hedging limits;
- whether hedge obligations may count as debt;
- whether collateral posting creates liens/restricted assets;
- BTC-backed loan LTV impact;
- cross-default/cross-acceleration risk;
- consent required;
- cash liquidity for margin/collateral;
- whether structure is speculative or tied to bona fide exposure.

Output labels:

- clear;
- review required;
- consent likely required;
- prohibited unless amended;
- unknown.

---

10. Accounting/disclosure output

The calculator should not decide accounting treatment, but it should produce a checklist:

- ASC 815 derivative accounting review required;
- hedge accounting feasibility review;
- FASB crypto fair-value impact;
- fair-value source hierarchy;
- collateral/restricted asset disclosure;
- market-risk disclosure impact;
- MD&A liquidity impact;
- risk-factor update;
- Form 8-K/materiality analysis if relevant.

---

11. Risk warnings

The calculator must warn if:

- hedge ratio exceeds conservative production;
- hedge ratio exceeds 50% without approval;
- pledged/restricted BTC is included;

- premium exceeds budget;
- collateral exceeds limit;
- upside cap is below policy threshold;
- tenor exceeds approved range;
- quote is stale;
- source is model-only or public-screen-only;
- venue/counterparty is not approved;
- legal status is not counsel-reviewed.

---

12. Data model

Suggested output schema:

```
scenario_id:
created_at_cst:
company:
analyst:
confidence:
  source: model | public_screen | vendor_screen | indicative_quote | firm_quote | trade
  execution: hypothesis | screen_only | quote_verified | trade_verified
market:
  btc_spot:
  spot_source:
  timestamp:
  iv_source:
  risk_free_rate:
  funding_rate:
miner:
  monthly_production_btc:
  conservative_haircut:
  conservative_monthly_production_btc:
  hedge_horizon_months:
  conservative_total_production_btc:
  current_inventory_btc:
  unencumbered_inventory_btc:
  pledged_inventory_btc:
obligations:
  power_cost_usd:
  debt_service_usd:
  capex_usd:
  other_usd:
  total_usd:
policy:
  hedge_ratio:
  hedge_notional_btc:
  max_premium_usd:
  max_collateral_usd:
  permitted_instruments:
structures:
  - type: protective_put | collar | put_spread_collar
    expiry:
    tenor_days:
    legs:
      - side:
        option_type:
        strike_pct:
        strike_abs:
        premium:
        iv:
        delta:
    net_premium_usd:
    margin_usd:
    collateral_usd:
    scenario_pnl:
```

```
recommendation:
  selected_structure:
  rationale:
  blockers:
  next_steps:
```

---

13. Interface v0

Simplest v0 implementation:

- spreadsheet or markdown-backed Python script;
- inputs in YAML/JSON;
- outputs markdown report and CSV scenario table;
- later upgrade to dashboard.

Recommended files:

- `miner_hedge_inputs.yaml`
- `miner_hedge_calculator.py`
- `miner_hedge_scenarios.csv`
- `miner_hedge_report.md`

---

14. Example default scenario

Default demo inputs:

- monthly production: 150 BTC;
- hedge horizon: 3 months;
- conservative total production: 450 BTC;
- hedge ratio: 50%;
- hedge notional: 225 BTC;
- BTC spot: latest monitor spot;
- structures:

1. 90D 85% put;
2. 90D 85% floor / 120% cap collar;
3. 135D 90/75/115/130 put-spread collar.

Default output:

- compare net cost;
- show BTC -20/-30/-40 protection;
- show upside give-up;
- recommend one structure;

- mark as screen-only unless quote data loaded.

---

15. Acceptance criteria

The calculator is usable when it can:

- accept miner production and balance sheet inputs;
- prevent hedge notional exceeding conservative production unless explicitly overridden;
- price three standard structures;
- generate downside/upside scenario table;
- flag collateral, covenant, legal, accounting issues;
- label confidence correctly;
- produce a board-ready one-page summary;
- store assumptions and outputs for audit trail.

---

16. Bottom line

The miner calculator should make one thing obvious:

> A miner hedge is not a bearish BTC bet. It is a disciplined runway-protection policy applied to conservative production and bounded by collateral, covenant, and governance limits.

If the calculator cannot express that clearly, it is not ready for client use.

Data Pipeline Hardening Spec

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-btc-vol-monitor-data-pipeline-hardening-spec-v0.md`

Idea 1 BTC Vol Desk Monitor Data Pipeline Hardening Spec v0

Purpose: Define the production-grade data pipeline needed to turn the current BTC Vol Desk Monitor from a one-off screen-only snapshot into a reliable daily evidence engine.

Status: Technical/product specification. Not trading advice. Monitor outputs remain screen-only unless linked to quote/trade records.

---

1. Objective

Build a repeatable daily monitor that normalizes BTC volatility across:

- Deribit BTC options;
- IBIT / spot BTC ETF options;
- iShares/BlackRock ETF holdings and BTC-per-share data;
- OCC series/open-interest validation;
- CME BTC futures/options once licensed/vendor/broker data is available;
- OTC/RFQ quotes once legally approved.

The pipeline should produce:

- daily normalized option surfaces;
- ATM IV term structures;
- skew/risk-reversal tables;
- calendar spreads;
- wrapper-relative dislocation board;
- bid/ask/liquidity filters;
- margin/collateral annotations;
- source and execution confidence labels;
- auditable raw-data archive.

---

2. Core design principle

The monitor must never confuse **screen evidence** with **execution evidence**.

Every row, chart, and dislocation must carry:

1. **Source confidence**
- model estimate;
- public screen;
- semi-official public JSON;
- vendor screen;
- official exchange/API;
- broker/FCM source;
- indicative quote;
- firm quote;
- trade.

2. **Execution confidence**
- hypothesis;
- screen-only;
- quote-verified;
- trade-verified.

---

3. Data source map

3.1 Deribit

Role: Crypto-native BTC volatility reference surface.

Source: Official public API.

Endpoints:

- get instruments;
- get book summary by currency;
- get order book by instrument;
- historical volatility if needed.

Fields:

- instrument name;
- expiry;
- strike;
- call/put;
- bid;

- ask;
- mark;
- mark IV;
- underlying/index price;
- open interest;
- volume;
- Greeks from order book endpoint;
- timestamp.

Hardening requirements:

- retry logic;
- schema validation;
- stale timestamp checks;
- expiry parser;
- strike parser;
- remove inactive/expired instruments;
- bid/ask sanity checks;
- mark IV unit normalization;
- OI/volume conversion consistency.

3.2 IBIT / spot BTC ETF options

Role: TradFi listed-options BTC volatility surface.

Research source: Nasdaq public JSON / other public chain source.

Production source needed: OPRA/vendor/broker data.

Fields:

- ticker;
- expiry;
- strike;
- call/put;
- bid;
- ask;
- last;
- volume;
- open interest;
- implied volatility if available;
- Greeks if available;

- timestamp;
- underlying ETF price.

Hardening requirements:

- source fallback;
- numeric parser for commas, percents, blanks, dashes;
- expiry normalization;
- stale quote check;
- bid/ask crossed/zero filters;
- American exercise/assignment annotation;
- option multiplier handling;
- underlying ETF price timestamp check.

Caveat:

Public Nasdaq JSON is research-only. Investor/client materials should call it semi-official/public unless replaced by OPRA/vendor/broker data.

3.3 iShares / BlackRock holdings and BTC-per-share

Role: Convert IBIT ETF option exposure into BTC-equivalent notional.

Fields:

- BTC holdings;
- shares outstanding;
- NAV;
- market price;
- cash/other holdings;
- expense ratio if needed;
- data timestamp.

Derived:

- BTC per ETF share;
- ETF shares per BTC;
- ETF option contract BTC-equivalent;
- ETF strike to BTC-equivalent strike;
- ETF option notional in BTC.

Hardening requirements:

- robust CSV parser;
- identify BTC row by multiple possible labels;
- parse shares outstanding even if format changes;

• fallback to official factsheet/holdings page if CSV changes;
• store raw file;
• compare day-over-day BTC/share drift;
• alert if BTC/share changes unexpectedly.

Known issue:

The first live parser missed the BTC row in one run. This is priority hardening.

3.4 OCC

Role: Official validation for listed option series/open interest where available.

Fields:

• underlying;
• expiry;
• strike;
• call/put;
• contract symbol;
• open interest;
• series status;
• deliverable/multiplier if relevant.

Hardening requirements:

• map OCC symbology to vendor/Nasdaq option chain;
• reconcile series count;
• reconcile OI totals where possible;
• flag missing or stale series.

3.5 CME

Role: Regulated futures/options BTC volatility and basis layer.

Production source needed: CME licensed API, DataMine, broker/FCM feed, vendor, or manual terminal export.

Fields:

• futures contract;
• futures bid/ask/settlement;
• basis vs spot;
• option contract;

- expiry;
- strike;
- call/put;
- bid;
- ask;
- settlement;
- implied vol;
- open interest;
- volume;
- contract multiplier;
- initial/maintenance margin;
- block liquidity;
- fees;
- position/accountability limits.

Hardening requirements:

- do not scrape blocked public pages;
- define production data contract/source;
- normalize futures basis;
- map options to futures expiry;
- convert contract multiplier to BTC-equivalent;
- add FCM margin data;
- label as missing until sourced.

3.6 OTC/RFQ quotes

Role: Move economics from screen-only to quote-verified.

Source: Legally approved quote process only.

Fields:

- RFQ ID;
- quote ID;
- counterparty;
- indicative/firm/trade;
- structure legs;
- notional;
- bid/offer/premium;
- IV/Greeks if provided;

- margin/collateral;
- settlement;
- documentation;
- quote timestamp;
- quote good-until;
- legal capacity;
- source/execution confidence.

Hardening requirements:

- strict audit trail;
- raw response archive;
- quote-age checks;
- quote-to-model comparison;
- legal role field required;
- no client name unless approved.

---

4. Normalized schema

Every normalized option row should include:

```
record_id:
run_id:
as_of_cst:
source_name:
source_confidence:
execution_confidence:
venue:
wrapper: spot_btc | etf_option | cme_future_option | deribit_option | otc_quote
underlying_reference:
native_symbol:
option_type:
expiry:
dte:
strike_native:
strike_btc_equivalent:
spot_native:
btc_spot_reference:
forward_reference:
moneyness_spot:
moneyness_forward:
delta:
gamma:
vega:
theta:
iv_bid:
iv_ask:
iv_mid:
iv_mark:
price_bid:
price_ask:
price_mid:
price_mark:
open_interest:
volume:
bid_ask_width_abs:
bid_ask_width_pct:
contract_multiplier:
btc_equivalent_per_contract:
```

```
notional_btc:
notional_usd:
settlement_method:
collateral_currency:
margin_initial:
margin_maintenance:
fee_estimate:
timestamp_source:
stale_flag:
quality_flags:
raw_pointer:
```

---

5. Derived analytics

5.1 ATM IV term structure

For each venue/wrapper:

- select nearest ATM by forward moneyness where available;
- fallback to spot moneyness;
- compute bid/mid/ask IV;
- report DTE/expiry;
- include liquidity filters.

Output:

- table;
- line chart;
- term slope metrics.

5.2 Skew / risk reversals

For each expiry:

- identify 25-delta put;
- identify 25-delta call;
- compute 25D risk reversal = call IV - put IV or put IV - call IV, explicitly define convention;
- compute 10D tails later.

If Greeks unavailable:

- approximate delta by model;
- label as model-estimated.

5.3 Calendar spreads

For ATM and selected moneyness:

- compare near vs far IV;
- compute term slopes;
- identify event humps;
- avoid comparing stale/illiquid expiries.

5.4 Wrapper dislocation board

Compare:

- IBIT vs Deribit;
- IBIT vs CME;
- CME vs Deribit;
- OTC vs screen;
- ETF option vs BTC-equivalent option.

Every candidate row includes:

- gross IV difference;
- premium difference;
- bid/ask cost;
- basis adjustment;
- collateral/margin adjustment;
- liquidity score;
- legal usability;
- confidence label;
- next action: watch, quote, reject, trade candidate.

---

6. Quality controls

6.1 Source checks

- endpoint status;
- response timestamp;
- schema version;
- record count;
- missing fields;
- duplicate instruments;
- expired instruments;
- unexpected zeroes.

6.2 Market sanity checks

- bid <= ask;
- IV within plausible bounds;
- price non-negative;
- call/put parity warning;
- monotonic strike sanity where possible;
- OI/volume non-negative;
- DTE positive;
- moneyness in expected range.

6.3 Cross-source checks

- BTC spot across sources within tolerance;
- IBIT price vs BTC/share implied BTC value;
- OCC series vs option-chain series;
- Deribit index vs external BTC spot;
- CME futures basis vs spot.

6.4 Alert conditions

- source failed;
- record count drops >X%;
- BTC/share changes unexpectedly;
- IV spike/drop >Y vol points;
- bid/ask width expands >threshold;
- large new wrapper dislocation;
- stale data;
- quote good-until expired.

---

7. Storage design

Minimum storage:

```
artifacts/institutional/data/
  raw/
    deribit/YYYY-MM-DD/run_id.json
    ibit/YYYY-MM-DD/run_id.json
    ishares/YYYY-MM-DD/run_id.csv
    occ/YYYY-MM-DD/run_id.csv
    cme/YYYY-MM-DD/run_id.*
    rfq/YYYY-MM-DD/quote_id.*
  normalized/
    options_surface/YYYY-MM-DD/run_id.parquet_or_csv
    atm_term_structure/YYYY-MM-DD/run_id.csv
    skew/YYYY-MM-DD/run_id.csv
    dislocations/YYYY-MM-DD/run_id.csv
  reports/
    btc-vol-desk-monitor-YYYY-MM-DD.md
```

Every run gets:

- run ID;
- CST timestamp;
- source versions;
- raw pointers;
- quality report;
- generated monitor report.

---

8. Report layout

Daily report sections:

1. Header

- date/time CST;
- run ID;
- source status;
- confidence legend.

2. BTC/ETF reference

- BTC spot;
- IBIT price;
- BTC/share;
- ETF shares per BTC;
- CME futures basis if available.

3. Deribit surface

- ATM IV by expiry;
- OI/volume;
- skew;
- liquidity notes.

4. IBIT/ETF surface

- ATM IV by expiry;
- OI/volume;
- skew;
- BTC-equivalent notional.

5. CME surface

- futures curve;
- options IV;
- margin notes;

• if unavailable, explicit missing-data status.

6. Cross-venue comparison

• wrapper IV differences;

• calendar/skew spreads;

• bid/ask filters.

7. Dislocation board

• candidate;

• venue pair;

• gross spread;

• all-in caveats;

• confidence;

• next action.

8. RFQ/quote updates

• new quotes;

• stale quotes;

• quote-verified changes.

9. Quality warnings

• source failures;

• parser issues;

• stale/missing fields.

10. Action list

• watch;

• quote;

• reject;

• fix data.

---

9. Implementation plan

Phase 1 — Reliable daily snapshot

• Deribit ingestion hardened.

• IBIT chain parser hardened.

• iShares BTC/share parser fixed.

• Normalized CSV output.

• Daily markdown monitor.

• Quality report.

Phase 2 — Surface analytics

- ATM term structures;
- skew/risk reversal;
- calendar spreads;
- bid/ask liquidity filters;
- dislocation board.

Phase 3 — CME integration

- choose source;
- ingest futures curve;
- ingest options chain;
- add margin/basis;
- cross-venue comparison.

Phase 4 — RFQ integration

- quote repository;
- quote-to-screen comparison;
- update confidence labels;
- quote expiration/staleness tracking.

Phase 5 — Dashboard/API

- interactive filters;
- historical charts;
- client-specific scenario export only after legal approval;
- API or SaaS layer later.

---

10. Acceptance criteria

The data pipeline is v1-ready when it can:

- run repeatedly without manual edits;
- store raw and normalized data;
- calculate BTC/share correctly;
- produce Deribit and IBIT ATM IV tables;
- compute BTC-equivalent ETF option notionals;
- flag stale/bad data;
- generate daily markdown report;
- preserve confidence labels;

- show CME as either integrated or explicitly missing;
- accept RFQ quote records later;
- produce an auditable trail from report row to raw source.

---

11. Immediate technical priorities

1. Fix and harden iShares BTC holdings/BTC-share parser.
2. Create normalized schema CSV output.
3. Store raw source files by run ID.
4. Add quality flags and source status.
5. Implement ATM IV selection consistently.
6. Implement ETF BTC-equivalent conversion.
7. Add dislocation board output.
8. Define CME source path.
9. Define RFQ quote record import.

---

12. Bottom line

The monitor is the evidence engine for the entire business.

If the monitor is sloppy, the fund/desk concept becomes narrative. If the monitor is reliable, timestamped, normalized, and confidence-labeled, it becomes the spine for:

- investor credibility;
- treasury/miner structuring;
- RFQ verification;
- quote/trade evidence;
- eventual risk-intermediation economics.

Legal Perimeter Memo

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-legal-perimeter-memo-v0.md`

Idea 1 Legal Perimeter Memo v0

Concept: BTC volatility intermediation desk: treasury/miner hedging, ETF/CME/Deribit/OTC vol analytics, RFQ workflow, principal sleeve, structured products.

Prepared: 2026-05-14 CDT

Status: Research memo for strategy/counsel review.

Not legal advice. This is a gating map to prevent overclaiming and to define what must be reviewed by derivatives/securities counsel before client-facing execution.

---

1. Executive conclusion

The legal perimeter is the main gating issue for Idea 1.

The concept is commercially strong, but the desk cannot safely jump from analytics to "we quote BTC derivative structures to public companies" without a defined regulatory wrapper.

The correct launch sequence is:

1. **Research and internal analytics** — lowest friction.
2. **Illustrative term sheets and education** — possible with careful disclaimers and no tailored advice.
3. **Bespoke advisory/structuring** — likely registration/partner/counsel analysis.
4. **RFQ/execution support** — high broker/IB/FCM/broker-dealer risk.
5. **Principal risk-taking / warehousing** — possible, but depends on whether spot, futures, swaps, securities, or notes are involved.
6. **Structured products/distribution** — highest legal/product-approval complexity.

The MVP should be framed as:

> Research, analytics, illustrative case studies, legal perimeter design, and quote-verification planning.

It should not be framed as:

> A live principal derivatives desk quoting public-company BTC collars.

---

2. Instrument classification spine

Every product must first be classified by instrument.

Instrument / wrapper	Main perimeter
Spot BTC	CFTC anti-fraud/anti-manipulation; state money transmission/custody issues may apply depending activity
BTC ETF shares / ETF options	SEC, securities/options, FINRA/broker-dealer/RIA issues
CME BTC futures/options	CFTC/NFA, FCM/IB/CTA/CPO issues
Deribit BTC options	Jurisdiction/offshore exchange/counterparty/custody; CFTC/NFA implications if US-facing advice/execution
OTC BTC swaps/options	CFTC swap/commodity-option/ECP/SD rules; ISDA/CSA documentation
BTC-linked notes	Securities Act, Exchange Act, broker-dealer/FINRA/RIA, structured-products governance
Pooled vehicle trading BTC derivatives	CPO/CTA; potentially Advisers Act / Investment Company Act issues

This is why the memo must separate research, advice, execution, principal trading, and product distribution.

---

3. Activity phase map

Phase 0 — Internal research / proprietary analytics

Activities:

- Build BTC Vol Desk Monitor.
- Analyze Deribit/IBIT/CME/OTC surfaces.
- Run internal case studies.
- Generate screen-only dislocation board.
- Trade own capital internally if separately approved.

Lower-risk boundaries:

- No client-specific advice.
- No order solicitation.
- No transaction-based compensation.
- No client asset custody.
- No claims of executable pricing unless verified.

Regulatory note:

Pure internal research is generally lowest friction. If internal proprietary trading uses listed futures/options, exchange/clearing/position/reporting rules still apply.

Go/no-go: Green for MVP.

Phase 1 — Generic published research / education

Activities:

- Market commentary.
- Generic BTC vol research.
- Educational treasury/miner hedging content.
- Illustrative term sheets marked hypothetical.
- Public white paper or investor concept memo.

Main gates:

- CTA risk if compensated advice about futures, commodity options, swaps, or commodity interests.
- RIA risk if compensated advice about securities, ETF options, BTC-linked notes, funds, or investment contracts.
- Broker-dealer/FINRA risk if research is tied to securities solicitation or transaction compensation.

Allowed MVP posture:

- General education only.
- No personalized recommendation.
- No “buy/sell this option” language.
- No performance claim from screen-only data.
- Prominent disclaimer: illustrative, not executable, not legal/tax/accounting advice.

Go/no-go: Green/yellow. Green for generic research; yellow if paid or tailored.

Phase 2 — Bespoke advisory / structuring

Activities:

- Designing a BTC treasury collar for a specific company.
- Advising a miner on production hedge ratios.
- Recommending ETF option overlays.
- Designing swap/OTC structures.
- Producing client-specific board packages.

Main gates:

- **CTA:** if advising for compensation on futures, commodity options, swaps, or commodity interests.
- **RIA:** if advising on securities, BTC ETF shares/options, BTC-linked notes, funds, or securities-based swaps.
- **CPO:** if operating/soliciting a pool that trades commodity interests.
- **Broker-dealer:** if structuring is tied to distribution, placement, transaction compensation, or securities sales.

Practical counsel questions:

1. Is the advice about securities, commodity interests, swaps, or both?
2. Is compensation advisory, subscription, transaction-based, or performance-based?
3. Is the client a public company, RIA client, fund, ECP, accredited investor, or retail?
4. Are we only delivering analysis, or inducing a transaction?
5. Are we using a registered partner or relying on registration/exemption?

Go/no-go: Yellow/red. Do not do client-specific advisory until counsel defines wrapper or partner model.

Phase 3 — RFQ and execution support

Activities:

- Sending RFQs to dealers.
- Introducing client to market maker or OTC desk.
- Soliciting/accepting orders.
- Negotiating option/swap/structured-note terms.
- Routing trades.
- Receiving spread, commission, referral fee, markup, or transaction compensation.

Main gates:

- **CFTC Introducing Broker / FCM:** if soliciting or accepting orders for futures/options/swaps and not accepting margin/funds.
- **FCM:** if accepting customer funds/margin/property for futures/options/swaps.
- **Broker-dealer/FINRA:** if effecting or inducing securities transactions, including ETF options, notes, funds, or securities-based swaps.
- **Swap dealer:** if regularly making markets/holding out as counterparty in swaps.

Hard rule:

Transaction-based compensation is a major risk factor. Avoid any RFQ/execution economics until counsel approves the role.

Permissible lower-risk planning:

- Build RFQ templates.
- Define fields required for quote verification.
- Identify categories of counterparties.
- Run purely internal mock RFQs.
- Do not solicit live trades for clients.

Go/no-go: Red until legal/registration/partner model is established.

Phase 4 — Principal risk-taking / warehousing

Activities:

- Desk trades own BTC/options/futures book.
- Warehouses client-flow risk.
- Makes markets in OTC derivatives.
- Hedges structured-product exposure.

Main gates by activity:

Principal activity	Gate
Own-account spot BTC trading	Generally lower registration risk, but anti-fraud/manipulation/custody/tax controls apply
Own-account CME futures/options	CFTC exchange/clearing/reporting/position rules
OTC swap market-making	Swap dealer analysis
Principal securities/notes trading	SEC dealer/broker-dealer analysis
Warehousing client-flow risk	Depends on whether acting as dealer, broker, adviser, counterparty, or fund

Risk-control requirement:

Principal risk cannot be a casual extension of advisory. It needs:

- entity structure;
- capital allocation;
- risk limits;
- compliance policy;
- counterparty documentation;
- valuation controls;
- market-conduct policy;
- custody/margin controls.

Go/no-go: Yellow/red. Possible later, not MVP.

Phase 5 — Structured products / distribution

Activities:

- BTC-linked notes.
- Reverse convertibles.
- Autocallables.
- Buffered notes.
- Principal-protected notes.

- BTC income products.
- Wealth-channel distribution.

Main gates:

- BTC-linked notes are generally securities.
- Public offerings require Securities Act registration unless an exemption applies.
- Distribution usually triggers broker-dealer/FINRA analysis.
- RIA use requires fiduciary/best-interest analysis.
- Underlying hedge may trigger CFTC/swap/FCM issues.
- Product approval, suitability, Reg BI, options disclosure, FINRA communications, and structured-product guidance apply.

Controls needed:

- registered issuer/shelf or private-placement exemption;
- product committee approval;
- target-market definition;
- client eligibility grid;
- payoff/scenario disclosure;
- issuer credit-risk disclosure;
- tax memo;
- secondary-market/liquidity policy;
- hedge-source transparency.

Go/no-go: Red for MVP. Phase 2+ only with issuer/distributor/legal framework.

---

4. Public-company BTC treasury program checklist

Before any BTC treasury derivatives program is client-ready:

Governance

- Board or committee approval.
- Written treasury derivatives policy.
- Permitted instruments.
- Maximum BTC encumbrance.
- Tenor/strike/floor limits.
- Counterparty/venue list.
- Authority matrix for trade approval and collateral movement.
- Escalation rules for margin/collateral stress.

SEC disclosure

Potential disclosure areas:

- Reg S-K Item 105 risk factors.
- Reg S-K Item 303 MD&A trends/liquidity.
- Reg S-K Item 305 market-risk disclosures.
- Reg S-X derivative accounting-policy disclosure.
- Form 8-K analysis for material agreements/events.
- Reg FD controls for material hedge information.
- SEC crypto exposure/counterparty/custody disclosure themes.

Accounting/auditor

- FASB ASU 2023-08 for crypto fair value.
- ASC 815 derivative accounting.
- Hedge accounting feasibility, if any.
- Independent valuation and fair-value hierarchy.
- Collateral/restricted asset treatment.
- Internal controls over trade approval, valuation, custody, collateral, and journal entries.

Debt/covenant/collateral

- Are derivatives permitted?
- Are hedge obligations permitted debt?
- Are BTC/cash collateral liens permitted?
- Are pledged BTC assets available?
- Cross-default/cross-acceleration with ISDA?
- Liquidity stress under BTC gap moves?

OTC/ISDA

- ISDA Master, Schedule, CSA/credit support.
- Digital asset definitions.
- ECP status.
- Reference price/source/fallbacks.
- Fork/airdrop/protocol disruption clauses.
- Eligible collateral/haircuts.
- Netting/close-out enforceability.
- Valuation agent and dispute process.

---

5. Miner-specific checklist

Miners need all public-company controls above, plus exposure-specific controls:

Exposure definition

- BTC inventory hedge vs future production hedge.

- Hashprice exposure vs BTC price exposure.
- Power-price exposure vs BTC revenue hedge.
- Production from owned mining vs hosting/customer exposure.

Hedge ratio

- Use conservative production estimates.
- Start with 25–50% of expected production.
- Avoid hedging machine deployment targets or optimistic hashrate.
- Stress test difficulty, curtailment, downtime, pool luck, and power prices.

Covenant/collateral

- Check equipment finance, project finance, power agreements, hosting agreements, and BTC-backed loans.
- Determine whether BTC is pledged/restricted.
- Confirm lender consent before posting collateral or entering secured hedges.

Accounting

- Forecasted production hedges raise probability and hedge-accounting questions.
- Cash-settled derivatives may simplify delivery/custody but create liquidity/margin needs.
- Physical settlement requires custody, wallet, sanctions, and transfer controls.

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6. Wealth/RIA/structured-product checklist

For ETF-option overlays, wealth products, and notes:

Client channel

- Broker-dealer, RIA, dual registrant, bank, private fund, or institutional counterparty?
- Retail, accredited, QP, QIB, ECP, or institutional?

Recommendation standard

- Reg BI for retail broker-dealer recommendations.
- FINRA suitability/KYC/supervision/communications.
- RIA fiduciary duty: care, loyalty, mandate fit, cost/conflicts, alternatives.
- Options account approval and OCC disclosure if clients directly trade options.

Product governance

- New product committee.
- Target market / negative target market.
- Training and sales scripts.
- Scenario analysis.
- Concentration limits.
- Post-sale surveillance.

• Conflict and compensation disclosures.

Structured notes

• Securities Act registration or exemption.

• Prospectus/pricing supplement or private placement memo.

• Issuer credit-risk disclosure.

• Secondary liquidity process.

• Embedded fees/hedging conflicts.

• BTC/ETF/options/futures risk factors.

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7. Red/yellow/green activity matrix

Activity	Status for MVP	Comment
Internal BTC vol monitor	Green	Build now
Screen-only public research	Green/yellow	Avoid tailored advice/performance claims
Illustrative treasury/miner case studies	Green/yellow	Mark hypothetical/screen-only/not executable
Client-specific hedge recommendation	Yellow/red	Counsel/registration/partner required
Sending client RFQs	Red	Execution/IB/BD risk
Receiving transaction compensation	Red	High broker/IB/BD risk
Principal OTC swap dealing	Red	Swap dealer/counterparty analysis
Own-account listed derivatives trading	Yellow	Possible with proper account/risk/compliance
Pooled vehicle trading derivatives	Red until structured	CPO/CTA/investment adviser analysis
BTC-linked structured notes	Red	Securities/product/distribution framework required
Wealth-platform ETF-option overlay	Yellow/red	BD/RIA/options suitability and supervision

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8. Required counsel workstream

Before any client-facing business launch, counsel should produce:

1. Activity classification memo.
2. Entity/wrapper recommendation.
3. Registration/exemption analysis:

• CTA/CPO;

• IB/FCM;

- RIA;
- broker-dealer/FINRA;
- swap dealer;
- securities offering exemptions.

4. Communications/disclaimer policy.
5. Research publication policy.
6. Client eligibility matrix.
7. RFQ/execution role policy.
8. Compensation model review.
9. Public-company treasury derivatives policy template.
10. ISDA/CSA/digital asset terms checklist.
11. Structured-product go/no-go memo.

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9. Source references

CFTC / NFA

- CFTC Commodity Exchange Act and Regulations: https://www.cftc.gov/LawRegulation/CommodityExchangeAct/index.htm
- Commodity Exchange Act definitions, 7 U.S.C. § 1a: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title7-section1a&num=0&edition=prelim
- CFTC Commodity Trading Advisors: https://www.cftc.gov/IndustryOversight/Intermediaries/CTAs/index.htm
- CFTC Commodity Pool Operators: https://www.cftc.gov/IndustryOversight/Intermediaries/CPOs/index.htm
- CFTC Introducing Brokers: https://www.cftc.gov/IndustryOversight/Intermediaries/IBs/index.htm
- NFA who has to register: https://www.nfa.futures.org/registration-membership/who-has-to-register/index.html
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- CFTC Coinflip Bitcoin commodity release: https://www.cftc.gov/PressRoom/PressReleases/7231-15

SEC / FINRA / OCC

- SEC broker-dealer overview: https://www.sec.gov/about/divisions-offices/division-trading-markets/broker-dealers
- Exchange Act broker-dealer definitions, 15 U.S.C. § 78c: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section78c&num=0&edition=prelim
- Exchange Act broker-dealer registration, 15 U.S.C. § 78o: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section78o&num=0&edition=prelim
- Securities Act definition of security, 15 U.S.C. § 77b: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section77b&num=0&edition=prelim
- Securities Act registration, 15 U.S.C. § 77e: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section77e&num=0&edition=prelim
- Advisers Act definitions, 15 U.S.C. § 80b-2: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section80b-2&num=0&edition=prelim

- Advisers Act registration, 15 U.S.C. § 80b-3: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section80b-3&num=0&edition=prelim
- SEC spot bitcoin ETP approval order: https://www.sec.gov/files/rules/sro/nysearca/2024/34-99306.pdf
- SEC ISE options on iShares Bitcoin Trust order: https://www.sec.gov/files/rules/sro/ise/2024/34-101128.pdf
- OCC options disclosure document: https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document
- FINRA broker-dealer registration: https://www.finra.org/registration-exams-ce/broker-dealers
- FINRA Rule 2360 Options: https://www.finra.org/rules-guidance/rulebooks/finra-rules/2360
- FINRA Rule 2111 Suitability: https://www.finra.org/rules-guidance/rulebooks/finra-rules/2111
- FINRA Rule 2090 KYC: https://www.finra.org/rules-guidance/rulebooks/finra-rules/2090
- FINRA Rule 3110 Supervision: https://www.finra.org/rules-guidance/rulebooks/finra-rules/3110
- FINRA Rule 2210 Communications: https://www.finra.org/rules-guidance/rulebooks/finra-rules/2210
- SEC Reg BI: https://www.sec.gov/rules-regulations/2019/06/regulation-best-interest-form-crs
- SEC structured notes bulletin: https://www.sec.gov/investor/pubs/structurednotes.htm
- FINRA Notice 05-59 structured products: https://www.finra.org/rules-guidance/notices/05-59
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Public-company / accounting / OTC

- Reg S-K Item 105: https://www.law.cornell.edu/cfr/text/17/229.105
- Reg S-K Item 303: https://www.law.cornell.edu/cfr/text/17/229.303
- Reg S-K Item 305: https://www.law.cornell.edu/cfr/text/17/229.305
- SEC Market Risk Disclosure FAQ: https://www.sec.gov/divisions/corpfin/guidance/derivfaq.htm
- Reg S-X Rule 4-08: https://www.law.cornell.edu/cfr/text/17/210.4-08
- SEC crypto sample letter: https://www.sec.gov/rules-regulations/staff-guidance/disclosure-guidance/sample-letter-companies-regarding-recent
- FASB ASU 2023-08: https://storage.fasb.org/ASU%202023-08.pdf
- SEC SAB 122: https://www.sec.gov/rules-regulations/staff-guidance/staff-accounting-bulletins/staff-accounting-bulletin-122
- CFTC swap disclosure rule: https://www.law.cornell.edu/cfr/text/17/23.431
- CFTC swap confirmation rule: https://www.law.cornell.edu/cfr/text/17/23.501
- CFTC swap documentation rule: https://www.law.cornell.edu/cfr/text/17/23.504
- ISDA Digital Asset Derivatives Definitions release: https://cdn.aws.isda.org/a/EEygE/ISDA-Launches-Standard-Definitions-for-Digital-Asset-Derivatives.pdf
- Delaware DGCL §141: https://delcode.delaware.gov/title8/c001/sc04/index.html#141

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10. Bottom line

The legally clean next step is not live execution.

The legally clean next step is:

1. keep outputs as research/analytics/illustrative;

2. produce a counsel-ready activity map;
3. build RFQ templates but do not send client RFQs;
4. decide whether the business launches as:

- analytics/research company;
- registered/partnered advisory/structuring shop;
- fund/prop vehicle;
- broker/FCM/OTC partner model;
- structured-product partnership.

Until counsel defines the wrapper, every market price and structure remains **screen-only illustrative research**.

Investor Deck Outline

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-investor-deck-outline-v0.md`

Idea 1 Investor Deck Outline v0

Concept: BTC Volatility Intermediation Desk

Audience: Seed investors, strategic partners, counsel/compliance reviewers, execution/prime/OTC counterparties

Status: Outline only. Not a .pptx yet. All market economics remain screen-only unless explicitly quote-verified.

Deck principle

This deck should not look or sound like a crypto trading pitch.

It should feel like an institutional commodity/derivatives market-structure deck:

- fragmentation;
- natural client flows;
- wrapper dislocations;
- treasury/risk-management need;
- disciplined legal perimeter;
- quote-verification roadmap;
- eventual risk-recycling economics.

The headline business:

> An institutional BTC risk-intermediation platform that transforms BTC volatility into treasury income, downside protection, miner runway hedges, and structured risk — powered by cross-venue ETF/CME/Deribit/OTC volatility analytics.

Design direction

Visual tone: institutional, dark, precise, not crypto-retail.

Palette:

- Primary: deep charcoal / near-black `#111827`
- Secondary: institutional navy `#1E2761`
- Accent: Bitcoin gold `#F7931A`

- Support: ice blue `#CADCFC`
- Neutral: off-white `#F8FAFC`

Motif: market plumbing / layered wrappers.

Suggested repeated visual elements:

- wrapper stack diagrams;
- cross-venue arrows;
- confidence labels: hypothesis / screen-only / quote-verified / trade-verified;
- red/yellow/green legal gates;
- client-flow to hedge-recycling loop.

Avoid:

- laser eyes;
- generic Bitcoin moon imagery;
- crypto meme language;
- overclaiming arbitrage.

Slide 1 — Title

Title: BTC Volatility Intermediation Desk

Subtitle: Institutional BTC risk transformation across ETF, CME, Deribit, and OTC markets.

Visual: Dark background, four-wrapper ring around BTC:

- ETF options
- CME futures/options
- Deribit BTC options
- OTC/RFQ

Speaker note: We are not pitching directional crypto exposure. We are building the risk-intermediation layer between fragmented BTC wrappers and natural institutional client flows.

Slide 2 — The problem: BTC exposure has institutionalized, but risk transfer is fragmented

Message: Institutions now hold BTC through many wrappers, but volatility risk is not centralized.

Content blocks:

- Spot BTC ETFs and ETF options

- CME futures/options
- Crypto-native options markets
- OTC/RFQ desks
- Public-company BTC treasuries
- Bitcoin miners

Visual: Fragmentation map: same BTC economic exposure split across different venues/legal/margin systems.

Key line: Same underlying risk; different wrappers, collateral rules, participants, and constraints.

Slide 3 — Why fragmentation creates a business opportunity

Message: Wrapper differences create recurring pricing, execution, and structuring needs.

Drivers:

- different participant bases;
- different collateral currencies;
- ETF share settlement vs BTC/index settlement;
- CME margin and regulated-account constraints;
- offshore crypto-native liquidity;
- public-company governance constraints;
- OTC custom terms and documentation.

Visual: Table or stacked cards: wrapper → constraint → opportunity.

Speaker note: The opportunity is not just “mispriced IV.” It is the service of translating risk between wrappers.

Slide 4 — The business: BTC risk intermediation, not generic crypto trading

Message: The desk turns BTC volatility into client-facing income/protection structures.

Four pillars:

1. Cross-venue analytics
2. Treasury/miner structuring
3. RFQ/quote verification
4. Hedge recycling / risk warehousing later

Visual: Flywheel:

Client exposure → structured hedge/yield product → RFQ/quote verification → hedge/recycle across venues → analytics improves → more client flow.

Key line: Client-flow wedge first; proprietary risk sleeve later.

Slide 5 — Initial wedge: BTC treasury companies

Message: BTC treasury holders need board-governed income and protection without abandoning the BTC thesis.

Pain points:

- dilution from equity/convert issuance;
- preferred/coupon/refi obligations;
- NAV volatility;
- need to avoid selling core BTC;
- board/auditor/covenant constraints.

Product:

- covered-call sleeve;
- zero-cost collar;
- put-spread collar;
- call-spread-financed protection.

Visual: 10,000 BTC holder: 90% strategic untouched / 10% program sleeve.

Key line: Monetize a controlled minority slice of convexity while preserving strategic BTC ownership.

Slide 6 — Treasury case study: 10,000 BTC holder

Message: A small sleeve can generate meaningful income/protection while leaving most BTC uncapped.

Case:

- 10,000 BTC treasury;
- 1,000 BTC program sleeve;
- 9,000 BTC untouched;
- 43D and 134D structures modeled from screen-only Deribit data.

Show examples:

- 15–25% OTM covered call sleeves;
- 90/80 put-spread funded by 115/130 call-spread;
- governance limits.

Visual: Scenario payoff chart: no hedge vs covered call vs collar.

Footer label: Screen-only illustrative research; quote verification required.

Slide 7 — Second wedge: Bitcoin miners

Message: Miners are structurally long BTC production but short USD operating certainty.

Pain points:

- power costs;
- debt service;
- ASIC/site capex;
- hashprice compression;
- BTC inventory drawdowns;
- lender/covenant pressure.

Product:

- 25–50% conservative production hedge;
- outright disaster puts;
- 85% floor / 120% cap collar;
- put-spread collar.

Visual: Miner cash-flow bridge: BTC production → USD obligations → hedge floor protects runway.

Key line: Hedge conservative production, not optimistic hashrate targets.

Slide 8 — Miner case study: production hedge

Message: A quarterly hedge can protect runway without eliminating BTC upside.

Case:

- 150 BTC/month production;
- 450 BTC conservative 3-month production;
- 225 BTC hedge at 50%;
- 45D and 134D structures modeled.

Show examples:

- 85% floor / 120% cap quarterly collar;
- 80–85% disaster put;
- 90/75 put-spread funded by 115/130 call-spread.

Visual: BTC -20%, -30%, -40% scenario table showing runway protection.

Footer label: Screen-only illustrative research; quote verification required.

Slide 9 — The proprietary engine: BTC Vol Desk Monitor

Message: The analytics engine normalizes BTC vol across wrappers.

Inputs:

- Deribit options;
- IBIT/ETF options;
- iShares BTC/share normalization;
- OCC validation;
- CME futures/options once licensed/vendor source is active;
- OTC/RFQ quotes after legal approval.

Outputs:

- ATM IV by expiry;
- skew/risk reversals;
- calendars;
- basis/futures curve;
- bid/ask width;
- OI/volume;
- margin/collateral notes;
- dislocation board.

Visual: Data pipeline diagram.

Slide 10 — Early evidence: screen-only dislocations exist

Message: The first monitor showed visible cross-wrapper differences worth quote-verifying.

Evidence from v0 monitor:

- Deribit BTC options API works and returned active surface.
- IBIT chain research source works.
- iShares holdings support BTC/share conversion.
- Initial sample showed near-dated IBIT ATM IV richer than Deribit.

Important caveat:

This is not executable arbitrage proof.

Visual: Bar chart: IBIT vs Deribit ATM IV by DTE with label “screen-only.”

Key line: The monitor justifies RFQ verification; it does not yet prove tradable edge.

Slide 11 — Evidence standard: no fake precision

Message: Every opportunity moves through a confidence ladder.

Ladder:

1. Hypothesis
2. Screen-only
3. Quote-verified
4. Trade-verified

Rules:

- no return claims from screen-only data;
- no IV-only comparisons;
- normalize for basis, margin, collateral, fees, settlement, and legal usability;
- quote age and size matter.

Visual: Four-stage horizontal pipeline with gates.

Key line: This is what makes the concept institutionally credible.

Slide 12 — RFQ verification plan

Message: The next proof milestone is quote verification of 10 standard structures.

First 10 internal RFQs:

- 500 BTC 45D covered calls;
- 1,000 BTC 135D covered calls;
- 90/80 vs 115/130 treasury put-spread collars;
- 225 BTC miner production collars;
- IBIT vs Deribit ATM straddle packages;
- IBIT vs Deribit 25-delta risk reversals.

Visual: RFQ board mockup with columns: product, notional, tenor, venue, quote status, confidence.

Legal caveat: No client-specific RFQs until counsel-approved wrapper.

Slide 13 — Legal perimeter and launch discipline

Message: The launch must be legally sequenced.

Green:

- internal research;
- screen-only analytics;
- illustrative education.

Yellow:

- bespoke advisory;
- own-account listed derivatives;
- wealth ETF overlays.

Red until counsel/partner model:

- client RFQs;
- transaction compensation;
- principal OTC swap dealing;
- structured notes;
- public-company derivatives advice.

Visual: Red/yellow/green activity matrix.

Key line: The clean MVP is research, analytics, case studies, legal design, and RFQ planning.

Slide 14 — Business model options

Message: Multiple revenue models exist; launch model depends on legal wrapper.

Models:

1. Analytics/research subscription
2. Advisory/structuring retainer
3. Execution/RFQ partner economics
4. Principal/risk warehousing P&L
5. Structured-product margin

Visual: Phased revenue stack, lower legal complexity to higher legal complexity.

Speaker note: Day one should not depend on taking principal risk or transaction compensation.

Slide 15 — Competitive advantage

Message: The edge is not one API call. It is the integration of client pain, cross-venue analytics, legal discipline, and quote evidence.

Advantages:

- BTC treasury/miner domain focus;
- cross-wrapper normalization;
- board-ready product design;
- RFQ verification discipline;
- legal perimeter awareness;
- potential client-flow feedback loop.

Visual: Moat stack: data → structuring → legal → relationships → flow → risk recycling.

Slide 16 — 30-day execution sprint

Message: The next month turns this from concept into launch decision package.

Week 1: Data spine / monitor hardening

Week 2: Product calculators / board templates

Week 3: Legal wrapper / RFQ board

Week 4: Investor package / launch decision

Visual: Four-week timeline with deliverables.

Slide 17 — 90-day roadmap

Message: After the 30-day sprint, advance only through gates.

Days 31–60:

- counsel memo;
- counterparty map;
- quote verification;
- pilot client conversations if approved.

Days 61–90:

- recurring monitor;
- quote-verified case studies;
- partner/execution model;

- investor package v1;
- go/no-go on entity/capital.

Visual: Gate-based roadmap: data → legal → quotes → commercial → capital.

Slide 18 — Team and infrastructure needs

Message: This requires quant/data, structuring, legal/compliance, capital markets, and risk/ops.

Roles:

- quant/data engineer;
- derivatives structurer;
- legal/compliance counsel;
- treasury/miner sales lead;
- risk/operations lead;
- execution/counterparty relationships.

Infrastructure:

- market data;
- normalized option surface store;
- quote repository;
- scenario calculator;
- document templates;
- compliance records.

Visual: Team/infrastructure operating stack.

Slide 19 — What we are asking for

Message: The immediate ask is not large trading capital. It is resources to complete validation.

Ask options:

- seed budget for 30–90 day validation;
- legal/compliance review;
- market-data/vendor access;
- counterparty introductions;
- pilot treasury/miner conversations;
- strategic partner for execution/RFQ.

Visual: Capital/resource allocation pie: data, legal, structuring, sales, ops.

Key line: Validate first; scale risk later.

Slide 20 — Closing thesis

Message: BTC volatility is now institutionally relevant, but institutional risk transformation is underbuilt.

Closing line:

> The opportunity is to become the desk that translates fragmented BTC volatility into board-ready treasury income, miner runway protection, allocator structures, and eventually quote-verified risk-recycling economics.

Visual: Same wrapper ring as Slide 1, now with client flows feeding into the desk.

Appendix slides

Appendix A — Data source map

- Deribit;
- IBIT/Nasdaq;
- iShares BTC/share;
- OCC;
- CME;
- OTC/RFQ.

Appendix B — Normalization schema

- BTC-equivalent notional;
- expiry/DTE;
- moneyness;
- IV;
- basis;
- margin/collateral;
- fees;
- confidence labels.

Appendix C — Treasury case details

- covered call;
- collar;
- put-spread collar;
- policy constraints.

Appendix D — Miner case details

- production forecast;
- hedge ratio;
- scenario table;
- lender/covenant checklist.

Appendix E — Legal red/yellow/green matrix

- research;
- advisory;
- RFQ/execution;
- principal;
- structured products.

Appendix F — RFQ templates

- treasury covered call;
- treasury collar;
- miner hedge;
- cross-venue vol spread.

Slides to build first if making a short deck

For a first 10-slide version:

1. Title
2. Problem: BTC risk transfer is fragmented
3. Business: BTC volatility intermediation
4. Initial wedge: treasury companies
5. Treasury case study
6. Second wedge: miners
7. BTC Vol Desk Monitor
8. Evidence standard / RFQ verification
9. Legal launch discipline
10. 30-day sprint / ask

Final note

The deck should use measured language. It should make the idea feel more like a commodity derivatives / treasury risk-management business than a crypto prop-trading deck.

The central claim is not:

> We found free arbitrage.

The central claim is:

> Institutional BTC volatility markets are fragmented, natural clients need risk transformation, and a disciplined desk can build the analytics, structuring, and quote-verification layer that connects them.

30-Day Execution Sprint

SCREEN-ONLY · NOT EXECUTABLE · INTERNAL/INVESTOR DILIGENCE ONLY · NOT A QUOTE · NOT AN OFFER

Source artifact: `artifacts/institutional/idea1-30day-execution-sprint-v0.md`

Idea 1 — 30-Day Execution Sprint v0

Concept: BTC Volatility Intermediation Desk

Goal: Convert current evidence package into a counsel-ready, investor-ready, and quote-verification-ready institutional concept.

Status: Internal execution plan. No client RFQs or client-specific advice until legal wrapper is approved.

Sprint objective

By day 30, produce a complete launch decision package answering:

1. Is the data pipeline credible enough to run a daily BTC Vol Desk Monitor?
2. Are treasury/miner structures commercially compelling under screen-only assumptions?
3. What quotes must be collected to prove economics?
4. What legal wrapper or partner model is required?
5. Is this best launched as analytics/advisory, fund/prop, execution partnership, or hybrid?

Non-negotiable rules

- Every market claim gets a confidence label: hypothesis, screen-only, quote-verified, trade-verified.
- No screen-only dislocation becomes a return claim.
- No client-specific RFQ is sent before legal approval.
- No transaction compensation before counsel-approved model.
- No Deribit use is assumed for US clients without legal review.
- Public-company programs require board/accounting/covenant framing.

Week 1 — Data spine and monitor hardening

Deliverables

1. **Daily BTC Vol Desk Monitor v1**

- Deribit BTC options ingestion.
- IBIT option-chain ingestion from research source.
- iShares BTC/share normalization parser hardened.
- OCC series/OI validation path.
- CME source decision: broker/vendor/API/manual terminal export.

2. **Normalized schema implementation**

- venue;
- wrapper;
- expiry/DTE;
- strike/moneyness;
- bid/ask/mid/mark;
- IV;
- Greeks where available;
- OI/volume;
- basis/forward assumption;
- margin/collateral notes;
- source confidence;
- execution confidence.

3. **First historical store**

- save daily snapshots;
- preserve raw API/source output;
- preserve computed monitor output;
- timestamp in CST.

Success criteria

- Monitor runs daily without manual scraping confusion.
- Deribit and IBIT surfaces are comparable by expiry/moneyness.
- iShares BTC/share conversion is reliable.
- Outputs distinguish screen-only from quote-verified.
- CME plan is explicit even if not integrated.

Failure criteria

- Public data is too unstable to produce consistent monitor.
- IV calculations cannot be reconciled to vendor/broker values.
- No viable CME data path is identified.

Week 2 — Product calculators and board packages

Deliverables

1. **Treasury calculator v0**
• Input: BTC holdings, sleeve %, tenor, floor, cap, call OTM target.
• Output: premium/yield, downside floor, upside cap, BTC encumbrance, scenario table.
• Structures:
• covered call;
• full collar;
• put-spread collar;
• call-spread-financed hedge.

2. **Miner hedge calculator v0**
• Input: monthly production, conservative forecast, hedge ratio, power/capex/debt dates.
• Output: hedge notional, premium, runway protection, BTC -20/-30/-40 scenarios.
• Structures:
• outright disaster put;
• production collar;
• put-spread collar.

3. **BTC treasury board policy template**
• objectives;
• permitted instruments;
• max sleeve;
• tenor/strike/floor limits;
• collateral/custody policy;
• counterparty limits;
• valuation/reporting;
• approval authority;
• disclosure triggers.

4. **Miner board/lender memo template**
• production forecast methodology;
• hedge ratio policy;
• runway/capex linkage;
• lender/covenant review;
• collateral plan;
• scenario stress table.

Success criteria

• Treasury and miner case studies become reusable calculators/templates.
• Board language is disciplined and not promotional.

- Outputs explicitly label all prices as screen-only unless quote-verified.

Week 3 — Legal wrapper and RFQ verification board

Deliverables

1. **Counsel-ready activity map**

- research;
- generic education;
- bespoke advisory;
- RFQ/execution;
- principal risk;
- structured products;
- compensation models.

2. **Legal questions memo**

- CTA/CPO;
- RIA;
- broker-dealer/FINRA;
- IB/FCM;
- swap dealer;
- public-company treasury;
- structured products;
- Deribit/offshore access.

3. **RFQ verification board v0**

- first 10 internal RFQ templates;
- quote fields;
- required counterparties;
- legal capacity status;
- quote priority;
- no client names.

4. **Counterparty map**

- ETF/options broker route;
- FCM/CME route;
- crypto-native/Deribit route;
- OTC options desk route;
- structured-product issuer route.

Success criteria

• Counsel can answer concrete questions, not vague “can we do this?”

• RFQ board is ready but not sent until legal role is approved.

• Compensation models are separated: subscription, advisory fee, execution fee, spread, principal P&L.

Week 4 — Investor package and launch decision

Deliverables

1. **Investor deck outline**

• problem;

• why now;

• market structure;

• product menu;

• data evidence;

• screen monitor;

• treasury case;

• miner case;

• legal path;

• RFQ plan;

• 90-day roadmap;

• capital/team needs.

2. **Investor memo v1 refresh**

• integrate one-page summary;

• tighten legal caveats;

• add sprint outputs;

• clarify quote-verification milestones.

3. **Launch model decision matrix**

• analytics/research company;

• advisory/structuring with counsel-approved registration/partner;

• fund/prop vehicle;

• execution/RFQ partner model;

• structured-product partnership;

• hybrid.

4. **Go/no-go decision package**

• data readiness;

• quote readiness;

- legal readiness;
- commercial readiness;
- capital/team readiness.

Success criteria

- The concept can be explained in one page and defended in appendix depth.
- Investor materials do not overclaim screen-only economics.
- Next 90 days are concrete: data, counsel, quotes, counterparties, pilot clients.

Workstream owners / roles needed

Quant/data

- Build monitor and data store.
- Normalize option surfaces.
- Validate IV calculations.

Structuring/product

- Treasury calculator.
- Miner hedge calculator.
- Board templates.
- RFQ templates.

Legal/compliance

- Activity map.
- Registration/wrapper analysis.
- Communications policy.
- Client eligibility.

Sales/capital markets

- Target account list.
- Persona map.
- Counterparty map.
- Outreach language after legal approval.

Risk/operations

- Margin/collateral model.
- Counterparty diligence.
- Valuation/marks process.
- Scenario/stress reporting.

30-day artifacts checklist

Already created:

- `idea1-package-index-v0.md`
- `idea1-btc-vol-desk-plan.md`
- `idea1-data-evidence-appendix.md`
- `idea1-btc-vol-monitor-v0.md`
- `idea1-client-map-commercial-wedge.md`
- `idea1-investor-deep-dive-v0.md`
- `idea1-treasury-case-study-10000btc-v0.md`
- `idea1-miner-production-hedge-case-study-v0.md`
- `idea1-legal-perimeter-memo-v0.md`
- `idea1-rfq-verification-plan-v0.md`
- `idea1-one-page-executive-summary-v0.md`

To create next:

- `idea1-investor-deck-outline-v0.md`
- `idea1-treasury-board-policy-template-v0.md`
- `idea1-miner-hedge-calculator-spec-v0.md`
- `idea1-rfq-verification-board-v0.md`
- `idea1-launch-model-decision-matrix-v0.md`

Decision gates

Gate 1 — Data gate

Proceed only if daily monitor can reliably compute normalized surfaces and store snapshots.

Gate 2 — Legal gate

Proceed to client-facing work only if counsel approves the activity, compensation, and wrapper.

Gate 3 — Quote gate

Proceed to investor return/economics claims only after quote verification.

Gate 4 — Commercial gate

Proceed to outreach only if treasury/miner packages are board-ready, simple, and legally reviewed.

Gate 5 — Capital gate

Proceed to principal risk only with risk capital, limits, counterparties, and compliance.

Bottom line

The next 30 days should not be “raise money on a deck.”

The next 30 days should produce a defensible launch package:

- daily data evidence;
- board-ready treasury/miner products;
- legal wrapper clarity;
- RFQ verification readiness;
- launch model decision.

That is how this becomes a serious institutional business rather than a clever trade idea.